HOW CAN IT BE THAT MATHEMATICS, BEING AFTER ALL A PRODUCT OF HUMAN THOUGHT WHICH IS INDEPENDENT OF EXPERIENCE, IS SO ADMIRABLY APPROPRIATE TO THE OBJECTS OF REALITY?

ALBERT EINSTEIN

“AND WHAT IS THE USE OF A BOOK,” THOUGHT ALICE, “WITHOUT PICTURES OR CONVERSATIONS?”

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Introduction

This file serves as a gentle introduction to the vast topic of quantum communication. The mathematical foundation is built, not assumed.

This is far from an original work. It is compilation of notes taken from various sources: including textbooks, lecture notes, and online resources. Often, the notes are taken verbatim, and I have made no attempt to rephrase or rewrite them. Textbooks and other resources that I used are cited as they're used, or in the *Further Resources* subsection at the end of each section.

It should be noted that the *Quantum Mechanics* section is taken from the lecture notes used by Prof. Ben Heidenreich when he taught PHYSICS 614: Quantum Mechanics I at the University of Massachusetts Amherst in the Spring of 2025.

Mathematics

Contents

- Eigenvalues and Eigenvectors
- Dual Spaces and Inner Product Spaces
- Further Resources

The overall goal of this chapter is to build our way up to vector spaces. Vector spaces are the mathematical structure that underpins quantum mechanics and, by extension, quantum communication. After the development of vector spaces, we briefly discuss some additional mathematical topics that are useful in quantum communication.

The Mathematician's Tool

Mathematics is the art of giving the same name to different things. — Henri Poincaré

Abstraction

This subsection is taken from the youtube video titled "The Matematician's Weapon | An Intro to Category Theory, Abstraction and Algebra' by the channel Eyesomorphic, linked here: youtu.be/FQYOpD7tv3o?si=R8CFesSwN92JlWuR.

Everyone you've ever met, despite being incredibly different from one another, shares the same label: a person. This concept of giving the same name to different things has been argued to be the fundamental principle of mathematics.

Let us see how we can justify this notion that two arbitrary people, call them Alice and Bob, have enough in common for us to give them the same name. We start by stripping them of any unnecessary details that aren't relevant to the discussion at hand. Having done this, we see that, while Alice and Bob are different in many ways, they share the same *internal structure*, whatever this structure may be. The label 'person' is then a name we give to this internal structure that Alice and Bob share.

This process of ignoring irrelevant details and focusing on the internal structure of objects is known as **abstraction**, and it is the most potent tool in a mathematician's arsenal. Abstraction allows us to identify commonalities between seemingly different objects and to study these commonalities in a general way.

The general structure of these lectures is therefore as follows: we start with a concrete example of a mathematical object, such as the integers, and we identify the internal structure that these objects share. We then abstract this structure to arrive at a more general definition, such as that of a group. This process allows us to study the properties of groups in a general way, without having to worry about the specific details of any particular group.

Axioms

This subsection is taken, almost directly, word for word, from Israel Nathan Herstein's book *Abstract Algebra* (3rd edition, 1996); a wonderful text that I highly recommend for anyone, especially those interested in learning more about the topics covered in this chapter.

When studying abstract algebra, the reader will encounter difficulties not so much in the "algebra" part, but rather in the "abstract" one. The common reaction is the feeling of being adrift, not having anything to hold on to. It often leads to the question, *What is the point of all this?*

To get around this feeling, it is helpful to have a concrete example in mind. This is true of all things, but especially true of abstract algebra. The concrete example serves as a guide, a point of reference, and a source of intuition. It is also helpful to have a concrete example in mind when trying to understand the definitions and theorems that we encounter in abstract algebra.

The path we take is as follows: we start with some well-defined collection of objects, S, and endow this collection with an algebraic structure by assuming that we can combine, in one or several ways, elements of S to obtain elements of S. These ways of combining elements of S are called *operations* on S. Then we try to regulate the nature of S by imposing certain rules on how these operations behave on S. These rules are called the *axioms* defining the particular structure on S. These axioms are for us to define the structure we want to study.

One could try many sets of axioms to define new structures. Time has shown that certain structures defined by axioms play an important role in mathematics (and other fields) and that certain others do not. The ones we study, namely groups, rings, fields, and vector spaces, have stood the test of time.

In everyday language, "axiom" means a self-evident truth. But we are not using everyday language; we are dealing with mathematics. An axiom is not a universal truth. Rather, it is one of several rules spelling a mathematical structure. The axiom is true in the system we are studying because we have forced it to be true by hypothesis. It is a license, in the particular structure, to do certain things.

Group Theory

To begin our adventure towards vector spaces, we start with, what I consider, the most fundamental mathematical structure: groups. Groups are *sets*[1] equipped with an *operation* that satisfies four axioms. Groups are ubiquitous in mathematics and physics, and they have applications in many fields, including cryptography, coding theory, and particle physics.

[1] Briefly, a **set** is a collection of well-defined, distinct objects.

Operations

On any nonempty set A, an operation is a mapping from two elements in A onto some other, not necessarily unique, element in A. More formally,

Definition 1 (Operation)**.** A **binary operation** on a nonempty set A (simply, an **operation** on a set A), is a function $f : A \times A \to A$.

By this definition, A is closed under the operation defined by f.

We can familiarize ourselves with Defn. 1 by considering standard multiplication, denoted here by $(\cdot)$. Limiting ourself to the integers, $\mathbb{Z}$, we define $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ as $f(a, b) = a \cdot b$. One may repeat this process with integer addition $(+)$, defining $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ as $f(a, b) = a + b$. As a last example, suppose $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are vectors in $\mathbb{R}^3$. One way to define multiplication is the *vector product*, denoted $(\times)$, the mapping from two vectors to another vector, *i.e.*, $f : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$ defined by $f\left(\vec{\mathbf{a}}, \vec{\mathbf{b}}\right) = \vec{\mathbf{a}} \times \vec{\mathbf{b}}$.

Later in the text we will drop the mapping label of f and simply use, *e.g.*, $(a, b) = \square$, where $\square$ is to be filled.

Question 2. *Given Defn. 1, why would we not consider the scalar product of vectors to be a binary operation?*

Groups

Consider the set of integers equipped with addition, often denoted $(\mathbb{Z}, +)$. From our past experience, we can list a couple of facts about this set:

1. the sum of two integers is still an integer, and
2. the order in which we add integers does not affect the outcome.

In symbols, the first fact is written as $\forall\, a, b \in \mathbb{Z}, a + b \in \mathbb{Z}$. This is known as as the *closure* property. Similarly, the second fact is written as $\forall\, a, b, c \in \mathbb{Z}$, $a + (b + c) = (a + b) + c$. This is known as *associativity*.

These two facts are things we have taken for granted since elementary school. If we now consider the integer 0, we can add two more facts to our list:

3. the sum of any integer and 0 is simply that integer, *i.e.*, $\forall\, a \in \mathbb{Z}$, $a + 0 = a = 0 + a$, and
4. there exists a *negative* for each integer such that the sum of an integer and its negative is 0, *i.e.*, $\forall\, a \in \mathbb{Z}$, $a + (-a) = 0 = (-a) + a$.

Mathematicians refer to any element for which property 3 holds as the *additive identity* (or the *identity element* in general) and $-a$ is known as the *additive inverse* of a.

Stripping the addition operator from the set and replacing it with the multiplication operator, we now have $(\mathbb{Z}, \cdot)$. We have enough experience to know that the first two facts still hold, *i.e.*, the product of two integers is still an integer and multiplication of integers is associative. However, the identity element in this case would be the integer 1 since, $\forall\, a \in \mathbb{Z}$, $a \cdot 1 = a = 1 \cdot a$. We encounter an issue when trying to satisfy criterion 4 from above. $\forall a \in \mathbb{Z}$, we seek an integer a^{-1} s.t. $a \cdot a^{-1} = 1 = a^{-1}a$. In the special case of $a = 1$, we simply choose $a^{-1} = 1$ and we satisfy the criterion. However, for $a = 2$, we would need to choose $a^{-1} = \frac{1}{2}$, not an integer! Thus, we cannot satisfy criterion 4 for all integers in $\mathbb{Z}$.

The fact that the *set* $(\mathbb{Z}, +)$ carries this additional property makes it *special*. When abstracting the four properties discussed above, we arrive at the following definition for the first mathematical structure we will study:

Definition 3 (Group)**.** A **group** is a nonempty set G equipped with a binary operation $(*)$ that satisfies the following axioms: $\forall\, a, b, c \in G$,

1. $a * b \in G$ (Closure)
2. $a * (b * c) = (a * b) * c$, (Associativity)
3. $\exists\, e \in G$ s.t. $a * e = a = e * a$, (Identity)
4. $\forall\, a \in G$, $\exists\, a^{-1} \in G$ s.t. $a * a^{-1} = e = a^{-1} * a$. (Inverse)

By this definition, the integers equipped with addition form a group. Note that in the previous definition we explicitly stated the closure property as an axiom even though this is already part of the definition of an operation.

Unlike the integers, the set of rational numbers excluding zero, denoted $\mathbb{Q} \setminus \{0\}$, do form a group when equipped with multiplication. To see this, consider generic $a, b, c, d \in \mathbb{Z}$ with $b \neq 0$ and $d \neq 0$. Then the values $\frac{a}{b}, \frac{c}{d} \in \mathbb{Q}$ (we assume the fractions here are in simplest form but this need not be true for what follows). Clearly, the product of these two values is still rational: $\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd} \in \mathbb{Q}$.[2] Associative multiplication in $\mathbb{Q}$ follows from the fact that multiplication of integers is associative. The identity element is the rational number 1 since $\frac{a}{b} \cdot 1 = \frac{a}{b} = 1 \cdot \frac{a}{b}$. Finally, every rational number except zero has a multiplicative inverse, namely its *reciprocal*, *e.g.* $\frac{b}{a}$, s.t. $\frac{a}{b} \cdot \frac{b}{a} = 1 = \frac{b}{a} \cdot \frac{a}{b}$. Thus, by Defn. 3, $(\mathbb{Q} \setminus \{0\}, \cdot)$ is a group. It can also be shown that $\mathbb{R} \setminus \{0\}$, $\mathbb{Q}^+$, and $\mathbb{R}^+$ are all groups under multiplication.

[2] The set of rational numbers requires that the denominator be nonzero. From previous experience, we know that the product of two nonzero integers is also nonzero. This property is consequential of the set of integers, $\mathbb{Z}$, being an integral domain. This is a topic we cover in the following section.

We take a slight detour in our development of vector spaces to introduce an interesting application of groups in geometry. We start with the following definition:

Definition 4 (Symmetry group)**.** The **symmetry group** G of a mathematical object O is the set of all transformations that preserve the object's structure.

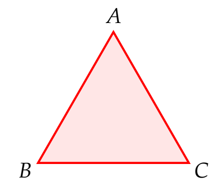


Figure 1: An equilateral triangle $\triangle ABC$. Note that the labels on the triangle's vertices are simply for us to visualize how the triangle is transformed. The labels do not imply that the triangle is oriented in any particular way.

For example, the symmetry group of an equilateral triangle $\triangle ABC$ (see Fig. 1) is the set of all transformations that preserve the triangle's shape, size, and *general* orientation. To build this group, we consider the following transformations:

1. Rotations: The triangle can be rotated (counterclockwise around its center) by $0°$, $120°$, or $240°$. Assigning names to each of these transformations, we define $\{e, r, r^2\}$, where e is the identity transformation (no rotation), r is the $120°$ rotation, and r^2 is the $240°$ rotation. We see each of these transformations in action below:

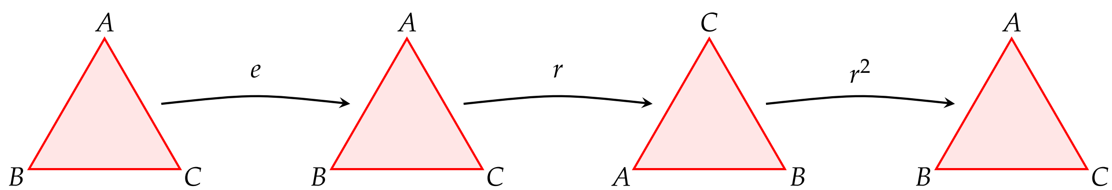


Figure 2: We see that the identity transformation e leaves the triangle unchanged, while the transformations r and r^2 rotate the triangle by $120°$ and $240°$ respectively. As one would expect, applying the r and r^2 rotations consecutively to the original triangle yields the original triangle again.

2. Reflections: Other than rotations around the centroid, we can also reflect the triangle about its axes of symmetry. The equilateral triangle has three axes of symmetry, each passing through one vertex and the midpoint of the opposite side. We can label these reflections as x, h, and v (see Fig. 4). We thus expand our set of transformations to $\{e, r, r^2, x, h, v\}$.

   Applying these new transformations to the original triangle in Fig. 1, we arrive at the results seen in Fig. 3.

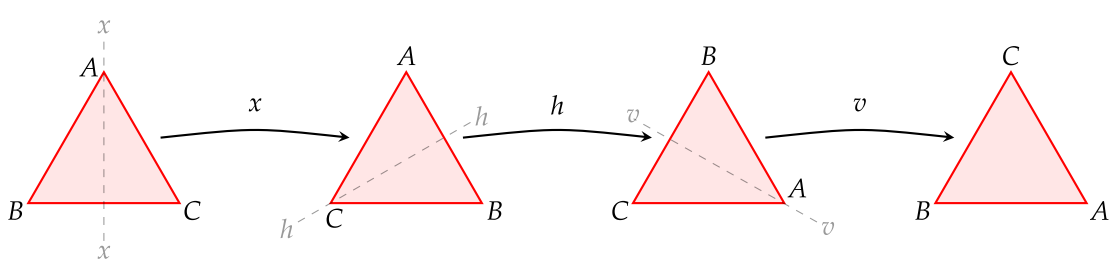


Figure 3: The axis of symmetry is shown (faintly) prior to reflection. Notice applying x, h, and v consecutively is equivalent to simply applying the h transformation to the original triangle.

As seen in Figs. 1–4, the transformations $\{e, r, r^2, x, h, v\}$ can be combined (*i.e.*, applied one after the other) to produce other, single transformations. If we think of a transformation as a function from the triangle onto itself, then the idea of combining transformations is equivalent to the composition of functions. In Fig. 3 (x followed by h and then followed by v), we can write $v \circ h \circ x$ (*i.e.*, first apply the x transformation, then the h, and finally the v) $= h$. In Fig. 2, we also saw that $r^2 \circ r = e$. Naming the set of transformations $D_3 \equiv \{e, r, r^2, x, h, v\}$[3] and equipping it with the composition operation, one can verify the Cayley table in Tab. 1.

[3] We reserve the symbol $\equiv$ to denote a definition. This is different from the symbol $\square \equiv \square \pmod{\square}$ used to denote congruence, which we will see in the next section.

Tab. 1 tells us that D_3 is closed under the composition operation, $(\circ)$, and composition of functions is known to be associative. Additionally, we've defined e as the identity element in this set and every element in D_3 has an inverse. For example, the inverse of r is r^2 since $r \circ r^2 = e = r^2 \circ r$. Similarly, the inverse of x is itself, *i.e.*, $x \circ x = e = x \circ x$. The same holds for h and v. Thus, we have shown that the set of transformations D_3 satisfies all four axioms in Defn. 3 and is therefore a group. In particular, it known as the **dihedral group** D_3, with the subscript of 3 indicating that our shape has 3 sides.

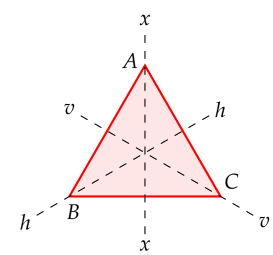


Figure 4: Showing all of the axes of symmetry on the equilateral triangle $\triangle ABC$.

In going through the above example, one may have noticed something rather peculiar about Tab. 1. Particularly, the fact the 3×3 block in the upper left corner is symmetric about the main diagonal, but no other 3×3 block in the table is symmetric. This symmetry tells us that, when applying rotations, the order in which we apply them does not matter. In other words, $r \circ r^2 = r^2 \circ r$, etc. This is not true for the reflections, however. For instance, $x \circ h \neq h \circ x$.

$\circ$	e	r	r^2	x	h	v
e	e	r	r^2	x	h	v
r	r	r^2	e	v	x	h
r^2	r^2	e	r	h	v	x
x	x	h	v	e	r	r^2
h	h	v	x	r^2	e	r
v	v	x	h	r	r^2	e

Table 1: The composition table for the symmetry group D_3 of the equilateral triangle. The way to read this table is as follows: the element in column i and row j is the result of applying the transformation i first and then the transformation j, *i.e.*, $j \circ i$. For example, the element in column r and row h is v, meaning that $h \circ r = v$.

Groups that satisfy this additional property, the property of *commutativity*: $a * b = b * a$ for all $a, b \in G$, are called *abelian* groups. The name is derived from the mathematician Niels Henrik Abel, who made significant contributions to mathematics in the 19th century.

Definition 5 (Abelian group)**.** A group is **abelian** if, $\forall\, a, b \in G$, it also satisfies the axiom

5. $a * b = b * a$. (Commutativity)

From what we mentioned in the previous paragraph, we see that the symmetry group D_3 is not abelian. However, the (proper) subgroup[4] of rotations, $\{e, r, r^2\}$, is abelian.

[4] A nonempty subset H of a group G is a **subgroup** provided that it is closed and their exists an inverse for every element in the subgroup. This is true of the subset $\{e, r, r^2\}$ of D_3. We use the word 'proper' to say that the subgroup is not equal to the whole group D_3.

We can now return to the list at the start of this subsection and append this additional property:

5. the addition of integers is independent of the order of the integers (*i.e.*, it is *commutative*: $\forall\, a, b \in \mathbb{Z}$, $a + b = b + a$),

stating that $(\mathbb{Z}, +)$ is an abelian group. By extension of the properties of $\mathbb{Z}$, $(\mathbb{Q}, +)$ is also an abelian group. Thus, it then follows that $(\mathbb{Z}, +)$ is a proper subgroup of $(\mathbb{Q}, +)$.

Question 6. *Produce the Cayley table of rotations, see Tab. 1, for the dihedral group D_4, the group of symmetries of the square.*

Further Resources

1. *Abstract Algebra: An Introduction* (3rd ed.) by Thomas W. Hungerford.

2. *Algebra: Abstract and Concrete* (ed. 2.6) by Frederick M. Goodman. https://homepage.divms.uiowa.edu/~goodman/algebrabook.dir/algebrabook.html

The material in this `\section{}` comes from Hungerford's text quite closely, though not in the same order. The structure is instead similar to Goodman's. Goodman does a thorough treatment of symmetries early in his text, which is why I have recommended it here. Goodman's text is also *free*[5] to download from his university webpage. Another (*free*) text that covers similar material is the following.

[5] See page ix of the text for its cost.

3. *A Crash Course on Group Theory* by Peter J. Cameron. https://cameroncounts.wordpress.com/wp-content/uploads/2016/11/groups.pdf

For a more traditional and advanced approach to group theory, consider the following text.

4. *Abstract Algebra* (3rd ed.) by I. N. Herstein.

Ring Theory

Rings are a natural extension of groups. Rings are sets equipped with two binary operations that generalize the arithmetic operations of addition and multiplication. Before we define rings formally, we first intrpoduce the concepts of *congruence* and *modular arithmetic*. These will not only be useful in our development of ring theory, but also in understanding the structure of rings themselves.

Congruence and Modular Arithmetic

Consider $a, b, c \in \mathbb{Z}$ with $c > 0$. We say ***a* is congruent to *b* modulo *c*** if c divides the difference of a and b. In symbols, $a \equiv b \pmod{c}$ if $c|(a-b)$.[6] For example, $15 \equiv 6 \pmod{3}$ since 3 divides $15 - 6 = 9$. Additionally, $23 \equiv -1 \pmod{6}$ since 6 divides $23 - (-1) = 24$.

[6] When expressing congruence, the symbol '$\equiv$' must accompanied by '$(\mathrm{mod}\, n)$' in order for it to actually *mean* anything. Without this, the same symbol of '$\equiv$' is used for definitions.

Now, let $a, c \in \mathbb{Z}$ with $c > 0$. The **congruence class of a modulo c**, denoted $[a]$, is the set of all integers congruent to a modulo c, *i.e.*,

$$[a] = \{n : n \in Z \text{ and } n \equiv a (\bmod c)\}. \tag{1}$$

For example, in congruence modulo 7, $[5] = \{\ldots, -9, -2, 5, 12, 19, \ldots\}$.[7] Restructuring this, equivalently $[5] = \{5, 5 \pm 7, 5 \pm 14, \ldots\}$. Thus, Eqn. (1) can be expressed instead as follows

$$[a] = \{a + kc : k \in Z\}, \tag{2}$$

a more digestible version.

[7] We must emphasize what the congruence class is with respect to since, notice, [5] by itself is insufficient information.
One may also arrive at Eqn. (2) by the definition of congruence in the previous paragraph. We know that $n \equiv a (\bmod c) \implies c|(n-a)$, which means $\exists k \in Z$ s.t. $n - a = kc$, hence $n = a + kc$.

Compactly, the set of all congruence classes modulo n is denoted $\mathbb{Z}_n$, read "$\mathbb{Z}$ mod n."[8] One may anticipate an issue in terms of representatives of these classes. For example, since $4 \equiv 7 (\bmod 3)$, $4 \equiv 16 (\bmod 3)$, and $4 \equiv -2 (\bmod 3)$, it can be argued that $[4] = [7] = [16] = [-2]$ should all be in $\mathbb{Z}_3$. However, since they are all equivalent by Eqn. (2), actually **none** of them will be in $\mathbb{Z}_3$. We avoid this redundancy by only including classes where the representative is less than the value of n. For example, $\mathbb{Z}_3 = \{[0], [1], [2]\}$. Every integer (class) can then be mapped onto (to) a class within $\mathbb{Z}_3$.

[8] It is important to emphasize that this is a set of congruence classes, not integers. Each class itself is a set of integers, as seen in the previous paragraph.

Finally, addition and multiplication in $\mathbb{Z}_n$ are defined by $[a] \oplus [b] = [a + b]$ and $[a] \odot [b] = [ab]$, respectively. Tables 2 and 3 show the addition and multiplication Cayley tables for all elements of $\mathbb{Z}_3$. Note that, *e.g.*, in Table 2 $[2] \oplus [2] = [1]$, replacing $[4]$.

$\oplus$	[0]	[1]	[2]
[0]	[0]	[1]	[2]
[1]	[1]	[2]	[0]
[2]	[2]	[0]	[1]

Table 2: Addition table for elements of $\mathbb{Z}_3$.

$\odot$	[0]	[1]	[2]
[0]	[0]	[0]	[0]
[1]	[0]	[1]	[2]
[2]	[0]	[2]	[1]

Table 3: Multiplication table for elements of $\mathbb{Z}_3$.

A further study can be done at this point and it is strongly encouraged that the reader explore congruence and modular arithmetic more deeply. However, for our purposes, this introduction suffices.

Question 7. *Make the Cayley tables for addition and multiplication in $\mathbb{Z}_4$. What do you notice about the column/row of* $[2]$ *under addition?*

Rings

Like groups, rings are best motivated by the generalization of properties of arithmetic in $\mathbb{Z}$ and $\mathbb{Z}_n$. Recall that in the Groups subsection we found that $(\mathbb{Z}, +)$ is an abelian group. Adding on to the 5 properties of abelian groups, we can list the following properties of the integers equipped with multiplication:

6. multiplication is closed, *i.e.*, the product of integers remains an integer;
7. multiplication is associative, *i.e.*, the *order in which we multiply* does not change the outcome;
8. the distributive properties hold;
9. multiplication is commutative, *i.e.*, the *order of the integers* being multiplied does not change the outcome;

Question 8. *Using Tables 2 and 3, verify that all of these properties hold for $\mathbb{Z}_3$.*

Question 9. *Do properties 6-11 hold for $\mathbb{Z}_n$, for any value of n? In particular, does property 11 hold for $\mathbb{Z}_4$? What about, $\mathbb{Z}_5$, $\mathbb{Z}_6$, and $\mathbb{Z}_7$? If you start to think the trend is parity-dependent, consider $\mathbb{Z}_2$. What must be true of n for properties 6-11 to hold for $\mathbb{Z}_n$?*

10. a multiplicative identity exists, namely the integer 1, which can be multiplied to every integer without altering their value; and lastly
11. if the product of two integers is zero, then at least one of the original integers was itself zero.

The last property in the list above is not something we often explore in elementary mathematics, maybe because it's trivial in the integers. This triviality will vanish as we move to more abstract settings. We begin the abstraction of these common properties with the following definition:

Because a ring is equipped with both addition and multiplication, we introduce the *distributive properties* to relate the two operations. These properties are axioms that must be satisfied in order for a set with two binary operations to be considered a ring.

Definition 10 (Ring)**.** A **ring** is an additive abelian group R equipped with multiplication, $(\cdot)$, s.t. $\forall\ a, b, c \in R$

6. $a \cdot b \in R$, (Closure under multiplication)
7. $a \cdot (b \cdot c) = (a \cdot b) \cdot c$, (Associative multiplication)
8. $a(b+c) = ab + ac$ and $(a+b)c = ac + bc$. (Distributive properties)

We use the symbol 0_R to denote the additive identity in a ring R. As of now, rings lack a multiplicative identity, denoted 1_R, and the existence of inverses for multiplication for them to form a group under multiplication as well. However, notice that rings introduce the distributive properties, which are not present in groups.

Rings which carry additional properties to the eight in Defn. 10 are given a different name. For example, a ring whose elements commute under multiplication is given a different name, though not as special as a commutative group:

Definition 11 (Commutative ring)**.** A ring R is a **commutative ring** if it satisfies the following axiom: $\forall\ a, b \in R$,

9. $ab = ba$. (Commutative multiplication)

If a ring forms both an additive abelian group and a multiplicative abelian group, we call it the following:

Definition 12 (Ring with identity)**.** A ring R which contains an element, say 1_R, satisfying the following axiom: $\forall\ a \in R$,

10. $a1_R = a = 1_R a$, (Multiplicative identity)

is a **ring with identity**.

Example 13. The set of integers $\mathbb{Z}$ equipped with the usual addition and multiplication operations, $(\mathbb{Z}, +, \cdot)$, is a commutative ring with identity. It satisfies all 10 properties listed in Defns. 10 through 12.

Example 14. Let E be the set of even integers. We claim that $(E, +, \cdot)$ is a commutative ring, but not a ring with identity. To see this, we verify that it satisfies properties 1-9 in Defns. 10 and 11.

1. If $a \in E$ and $b \in E$, then $a + b \in E$ since the sum of two even integers is even. That is, $a + b = 2m + 2n = 2(m + n)$ for some $m, n \in \mathbb{Z}$.
2. By the associativity of addition in $\mathbb{Z}$, $a + (b + c) = (a + b) + c$.
3. The additive identity in E is 0 since $\forall\, a \in E$, $a + 0 = a$. 0 is even since $0 = 2 \cdot 0$.
4. For any $a \in E$, the additive inverse is $-a$ since $a + (-a) = 0$. Note that $-a$ is even since $a = 2k$ for some $k \in \mathbb{Z} \implies -a = -2k = 2(-k)$.
5. By the commutativity of addition in $\mathbb{Z}$, $a + b = b + a$.
6. If $a \in E$ and $b \in E$, then $ab \in E$ since the product of two even integers is even. That is, $ab = (2m)(2n) = 2(2mn)$ for some $m, n \in \mathbb{Z}$.
7. By the associativity of multiplication in $\mathbb{Z}$, $a(bc) = (ab)c$.
8. By the distributive properties in $\mathbb{Z}$, $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$.
9. By the commutativity of multiplication in $\mathbb{Z}$, $ab = ba$.

There is no multiplicative identity in E since $1 \notin E$. Thus, $(E, +, \cdot)$ is not a ring with identity.

Question 15. *Verify that the set of odd integers with the usual addition and multiplication is not a ring.*

Example 16. Consider the set of congruence class modulo n in $\mathbb{Z}$, $\mathbb{Z}_n$.

We seek to show that $(\mathbb{Z}_n, \oplus, \odot)$ is a commutative ring with identity. To do so, we must verify that it satisfies all 10 properties listed in Defns. 10 through 12.

1. If $[a] \in \mathbb{Z}_n$ and $[b] \in \mathbb{Z}_n$, then $[a] \oplus [b] = [a+b] \in \mathbb{Z}_n$ by definition of addition in $\mathbb{Z}_n$.
2. By the definition of addition in $\mathbb{Z}_n$, $[a] \oplus ([b] \oplus [c]) = [a] \oplus [b+c] = [a+(b+c)] = [(a+b)+c] = [a+b] \oplus [c] = ([a] \oplus [b]) \oplus [c]$.
3. The additive identity in $\mathbb{Z}_n$ is $[0]$ since $\forall\ [a] \in \mathbb{Z}_n$, $[a] \oplus [0] = [a+0] = [a]$.
4. For any $[a] \in \mathbb{Z}_n$, the additive inverse is $[-a]$ since $[a] \oplus [-a] = [a+(-a)] = [a-a] = [0]$.
5. By definition of addition in $\mathbb{Z}_n$, $[a] \oplus [b] = [a+b] = [b+a] = [b] \oplus [a]$.
6. If $[a] \in \mathbb{Z}_n$ and $[b] \in \mathbb{Z}_n$, then $[a] \odot [b] = [ab] \in \mathbb{Z}_n$ by definition of multiplication in $\mathbb{Z}_n$.
7. By the definition of multiplication in $\mathbb{Z}_n$, $[a] \odot ([b] \odot [c]) = [a] \odot [bc] = [a(bc)] = [(ab)c] = [ab] \odot [c] = ([a] \odot [b]) \odot [c]$.
8. By the definition of addition and multiplication in $\mathbb{Z}_n$, $[a] \odot ([b] \oplus [c]) = [a] \odot [b+c] = [a(b+c)] = [ab+ac] = [ab] \oplus [ac] = ([a] \odot [b]) \oplus ([a] \odot [c])$. Similarly, $([a] \oplus [b]) \odot [c] = [ac] \oplus [bc]$.
9. By the definition of multiplication in $\mathbb{Z}_n$, $[a] \odot [b] = [ab] = [ba] = [b] \odot [a]$.
10. The multiplicative identity in $\mathbb{Z}_n$ is $[1]$ since $\forall\ [a] \in \mathbb{Z}_n$, $[a] \odot [1] = [a \cdot 1] = [a]$.

As an interesting note, suppose n is not prime. Then, there exists $a, b \in \mathbb{Z}$ s.t. $1 < a < n, 1 < b < n$, and $ab = n$. Thus, in $\mathbb{Z}_n$, $[a] \odot [b] = [ab] = [n] = [0]$, even though neither $[a]$ nor $[b]$ is equal to $[0]$. Therefore, when n is not prime, $\mathbb{Z}_n$ contains what we call **zero divisors**, elements x and y such that $x \neq 0_R$, $y \neq 0_R$, but $xy = 0_R$. This property will be important later on when we study fields.

The study of rings often involves studying polynomials with coefficients in a given ring. For example, one may study polynomials with integer coefficients. However, this is not something useful for our current development. As an alternative, we instead consider matrices.

Matrices

For the time being, we will not formally define matrices. Instead, we will provide an informal, very boring definition of what a matrix is, followed by some examples of matrix operations. A more formal definition will be provided later in the text when we study vector spaces.

Definition 17 (Matrix)**.** Let m and n be positive integers. An m-by-n **matrix A** is a rectagular array of elements arranged in m rows and n columns, denoted

$$\mathbf{A} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}.$$

The notation a_{ij} denotes the element in the i-th row and the j-th column of $\mathbf{A}$. In other words, the first index refers to the row and the second index refers to the column.

Notice that the definition says nothing about what the elements a_{ij} are. In theory, they can be elements from any set, it simply depends on what you want to achieve with the matrices. For our purposes, we will consider matrices with real number entries, denoted $\mathbf{M}(\mathbb{R})$.

Example 18. Let $\mathbf{M}(\mathbb{R})$ be the set of all 2×2 matrices with real number entries. That is, $\mathbf{M}(\mathbb{R})$ contains all arrays of the form

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

where a, b, c, and d are real numbers. Two matrices are equal if and only if their corresponding entries are equal, *i.e.*,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} e & f \\ g & h \end{pmatrix} \iff a = e,\ b = f,\ c = g,\ d = h.$$

For example,

$$\begin{pmatrix} 4 & 9 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1+3 & 4+5 \\ 7-7 & 8-6 \end{pmatrix}, \quad \text{but} \quad \begin{pmatrix} 4 & 9 \\ 0 & 2 \end{pmatrix} \neq \begin{pmatrix} 4 & 9 \\ 2 & 0 \end{pmatrix}.$$

Addition of matrices is done entry-wise:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}.$$

For example,

$$\begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix} = \begin{pmatrix} 1+5 & -2+6 \\ 3+7 & 4+8 \end{pmatrix} = \begin{pmatrix} 6 & 4 \\ 10 & 12 \end{pmatrix}.$$

Multiplication of matrices is a bit more involved. The product of two 2×2 matrices is given by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}.$$

For example,

$$\begin{pmatrix} 7 & 2 \\ 0 & 5 \end{pmatrix} \cdot \begin{pmatrix} 1 & 6 \\ 8 & 4 \end{pmatrix} = \begin{pmatrix} 7\cdot 1+2\cdot 8 & 7\cdot 6+2\cdot 4 \\ 0\cdot 1+5\cdot 8 & 0\cdot 6+5\cdot 4 \end{pmatrix} = \begin{pmatrix} 23 & 50 \\ 40 & 20 \end{pmatrix}.$$

Reversing the order of the factors in matrix multiplication *may* produce a different result, as is the case in our previous example:

$$\begin{pmatrix} 1 & 6 \\ 8 & 4 \end{pmatrix} \cdot \begin{pmatrix} 7 & 2 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} 1\cdot 7+6\cdot 0 & 1\cdot 2+6\cdot 5 \\ 8\cdot 7+4\cdot 0 & 8\cdot 2+4\cdot 5 \end{pmatrix} = \begin{pmatrix} 7 & 32 \\ 56 & 48 \end{pmatrix}.$$

Multiplication is therefore not commutative in general for matrices. One can verify that $(\mathbf{M}(\mathbb{R}), +, \cdot)$ satisfies all the properties of a ring as given in Defn. 10 and is thus a ring. The zero element of said ring is the zero matrix

$$\mathbf{0} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

and the matrix $\mathbf{X} = \left(\begin{smallmatrix} -a & -b \\ -c & -d \end{smallmatrix}\right)$ is a solution of

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \mathbf{X} = \mathbf{0},$$

thus $\mathbf{X}$ is the additive inverse of $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. The multiplicative identity element is the identity matrix

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

satisfying

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \mathbf{I} = \mathbf{I} \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Interestingly, two nonzero matrices can multiply to give the zero matrix. For example,

$$\begin{pmatrix} 4 & 6 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} -3 & -9 \\ 2 & 6 \end{pmatrix} = \begin{pmatrix} 4\cdot(-3)+6\cdot 2 & 4\cdot(-9)+6\cdot 6 \\ 2\cdot(-3)+3\cdot 2 & 2\cdot(-9)+3\cdot 6 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Using the same definitions of matrix addition and multiplication as above, we can construct other rings using matrices. Consider the following question:

Question 19. *Let $\mathbf{K}(\mathbb{R})$ be the set of all 2×2 matrices of the form*

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix},$$

where $a, b \in \mathbb{R}$. Show that $(\mathbf{K}(\mathbb{R}), +, \cdot)$ is a commutative ring with identity.

In the examples we've done so far, we've seen a couple of rings that have the property of two nonzero elements multiplying to give zero. In the following subsection, we explore rings that do not have this property. We also explore rings which have an additional, new property. Rings with these additional properties are given special names and are of incredible importance in the study of algebra.

Integral Domains and Fields

Definition 20 (Integral domain)**.** An **integral domain** is a commutative ring R with identity $1_R \neq 0_R$ that satisfies the following axiom: $\forall a, b \in R$,

11. if $ab = 0$, then $a = 0$ or $b = 0$.

The condition $1_R \neq 0_R$ is included here to avoid trivial cases where the ring contains only one element; namely, the ring $\{0_R\}$. Elements of a ring that are themselves nonzero but multiply to give zero are called **zero divisors**:

Definition 21 (Zero divisor)**.** An element a in a ring R is a **zero divisor** provided that $a \neq 0_R$ and $\exists\, b \in R$, with $b \neq 0_R$, s.t. $ab = 0_R$.

Looking back at Example 18, we see that $\mathbf{M}(\mathbb{R})$ is not an integral domain since it contains zero divisors; namely, the two nonzero matrices that we multiplied to get the zero matrix, $\left(\begin{smallmatrix} 4 & 6 \\ 2 & 3 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} -3 & -9 \\ 2 & 6 \end{smallmatrix}\right)$.

Recall our short dicussion in the section on groups, where we mentioned that $(\mathbb{Z}, \cdot)$ is not a group because not every integer has a multiplicative inverse. However, we pointed out that $(\mathbb{Q} \setminus \{0\}, \cdot)$ is a group because every nonzero rational number has a multiplicative inverse. Thus far in our development, we have not studied rings where every nonzero element has a multiplicative inverse. We do so now with the following definition:

Definition 22 (Field)**.** A **field** is a commutative ring R with identity $1_R \neq 0_R$ that satisfies the following axiom: $\forall\, a \in R$, with $a \neq 0_R$,

12. $\exists\, a^{-1} \in R$ s.t. $aa^{-1} = 1$.

Note axiom 11 is not mentioned explicitly in the definition of a field. However, it can be shown that every field is an integral domain.[9]

[9] The inverse of this is not true, however, as it is not generally true that an integral domain is a field. Take, for example, $\mathbb{Z}$. It is an integral domain but not a field. Interestingly, it is true that a *finite* integral domain is a field.

Example 23. The set of rational numbers $\mathbb{Q}$ equipped with the usual addition and multiplication operations, $(\mathbb{Q}, +, \cdot)$, is a field. It satisfies all 12 properties listed in Defns. 10 through 22. The set of real numbers $\mathbb{R}$ is also a field under the usual addition and multiplication.

Further Resources

1. *Abstract Algebra: An Introduction* (3rd ed.) by Thomas W. Hungerford.

The entirety of this `\section{}` is pulled from Hungerford's text. I strongly believe it is one of the best introductory texts on ring theory, and I highly recommend it to anyone looking to learn more about rings, integral domains, and fields.

Complex Numbers

Definitions and Algebraic Properties

Definition 24 (Complex numbers)**.** The set of **complex numbers**, denoted $\mathbb{C}$, is defined as pairs of real numbers,

$$\mathbb{C} \equiv \{(x, y) : x, y \in \mathbb{R}\},$$

equipped with addition

$$(x_1, y_1) + (x_2, y_2) \equiv (x_1 + x_2, y_1 + y_2),$$

and multiplication

$$(x_1, y_1) \cdot (x_2, y_2) \equiv (x_1x_2 - y_1y_2, x_1y_2 + y_1x_2).$$

By the definition above complex numbers of the form $(x, 0)$ behave like real numbers under addition and multiplication:

$$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0), \quad \text{and} \quad (x_1, 0) \cdot (x_2, 0) = (x_1 x_2, 0).$$

Furthermore, the complex numbers $(0, 0)$ and $(1, 0)$ serve as the additive and multiplicative identities, respectively. With all of this, we now show that $(\mathbb{C}, +, \cdot)$ is a field:

Proposition 25. *The set of complex numbers $\mathbb{C}$ equipped with the addition and multiplication operations defined above, $(\mathbb{C}, +, \cdot)$, is a field.*

Question 26. *Prove the above proposition. (Answer provided below.)*

Proof. We must verify that $(\mathbb{C}, +, \cdot)$ satisfies all 11 properties listed in Defns. 10 through 22. $\forall\ (x_1, y_1), (x_2, y_2), (x_3, y_3) \in \mathbb{C}$,

1. $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) \in \mathbb{C}$ by closure of addition in $\mathbb{R}$,
2. $(x_1, y_1) + [(x_2, y_2) + (x_3, y_3)] = (x_1, y_1) + (x_2 + x_3, y_2 + y_3) = (x_1 + (x_2 + x_3), y_1 + (y_2 + y_3)) = ((x_1 + x_2) + x_3, (y_1 + y_2) + y_3) = (x_1 + x_2, y_1 + y_2) + (x_3, y_3) = [(x_1, y_1) + (x_2, y_2)] + (x_3, y_3)$ by associativity of addition in $\mathbb{R}$,
3. $\exists 0_F \in \mathbb{C}$, namely $(0, 0)$, s.t. $(x_1, y_1) + (0, 0) = (x_1 + 0, y_1 + 0) = (x_1, y_1) = (0 + x_1, 0 + y_1) = (0, 0) + (x_1, y_1)$ by the additive identity property in $\mathbb{R}$,
4. $\forall\ (x_1, y_1) \in \mathbb{C}$, $\exists$ an additive inverse, namely $(-x_1, -y_1)$, s.t. $(x_1, y_1) + (-x_1, -y_1) = (x_1 - x_1, y_1 - y_1) = (0, 0) = (-x_1 + x_1, -y_1 + y_1) = (-x_1, -y_1) + (x_1, y_1)$ by the additive inverse property in $\mathbb{R}$,
5. $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2) = (x_2 + x_1, y_2 + y_1) = (x_2, y_2) + (x_1, y_1)$ by commutativity of addition in $\mathbb{R}$,
6. $(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2) \in \mathbb{C}$ by closure of multiplication (and addition) in $\mathbb{R}$,
7. $(x_1, y_1) \cdot [(x_2, y_2) \cdot (x_3, y_3)] = (x_1, y_1) \cdot (x_2 x_3 - y_2 y_3, x_2 y_3 + y_2 x_3) = (x_1(x_2 x_3 - y_2 y_3) - y_1(x_2 y_3 + y_2 x_3), x_1(x_2 y_3 + y_2 x_3) + y_1(x_2 x_3 - y_2 y_3)) = (x_1 x_2 x_3 - x_1 y_2 y_3 - y_1 x_2 y_3 - y_1 y_2 x_3, x_1 x_2 y_3 + x_1 y_2 x_3 + y_1 x_2 x_3 - y_1 y_2 y_3) = (x_1 x_2 x_3 - y_1 y_2 x_3 - x_1 y_2 y_3 - y_1 x_2 y_3, x_1 y_2 x_3 + y_1 x_2 x_3 + x_1 x_2 y_3 - y_1 y_2 y_3) = ((x_1 x_2 - y_1 y_2) x_3 - (x_1 y_2 + y_1 x_2) y_3, (x_1 y_2 + y_1 x_2) x_3 + (x_1 x_2 - y_1 y_2) y_3) = (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2) \cdot (x_3, y_3) = [(x_1, y_1) \cdot (x_2, y_2)] \cdot (x_3, y_3)$ by associativity of multiplication and addition in $\mathbb{R}$,
8. $(x_1, y_1) \cdot [(x_2, y_2) + (x_3, y_3)] = (x_1, y_1) \cdot (x_2 + x_3, y_2 + y_3) = (x_1(x_2 + x_3) - y_1(y_2 + y_3), x_1(y_2 + y_3) + y_1(x_2 + x_3)) = (x_1 x_2 + x_1 x_3 - y_1 y_2 - y_1 y_3, x_1 y_2 + x_1 y_3 + y_1 x_2 + y_1 x_3) = ((x_1 x_2 - y_1 y_2) + (x_1 x_3 - y_1 y_3), (x_1 y_2 + y_1 x_2) + (x_1 y_3 + y_1 x_3)) = (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2) + (x_1 x_3 - y_1 y_3, x_1 y_3 + y_1 x_3) = (x_1, y_1) \cdot (x_2, y_2) + (x_1, y_1) \cdot (x_3, y_3)$. Similarly, $[(x_1, y_1) + (x_2, y_2)] \cdot (x_3, y_3) = (x_1, y_1) \cdot (x_3, y_3) + (x_2, y_2) \cdot (x_3, y_3)$ by associativity of multiplication and addition in $\mathbb{R}$,
9. $(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + y_1 x_2) = (x_2 x_1 - y_2 y_1, x_2 y_1 + y_2 x_1) = (x_2, y_2) \cdot (x_1, y_1)$ by commutativity of multiplication and addition in $\mathbb{R}$,
10. $\exists 1_F \in \mathbb{C}$, namely $(1, 0)$, s.t. $(x_1, y_1) \cdot (1, 0) = (x_1 \cdot 1 - y_1 \cdot 0, x_1 \cdot 0 + y_1 \cdot 1) = (x_1, y_1) = (1 \cdot x_1 - 0 \cdot y_1, 1 \cdot y_1 + 0 \cdot x_1) = (1, 0) \cdot (x_1, y_1)$ by the multiplicative identity property in $\mathbb{R}$,

11. $\forall\ (x_1, y_1) \in \mathbb{C}$, with $(x_1, y_1) \neq (0, 0)$, $\exists$ a multiplicative inverse, namely $\left(\frac{x_1}{x_1^2+y_1^2}, \frac{-y_1}{x_1^2+y_1^2}\right)$, s.t.

$$(x_1, y_1) \cdot \left(\frac{x_1}{x_1^2+y_1^2}, \frac{-y_1}{x_1^2+y_1^2}\right) = \left(\frac{x_1^2+y_1^2}{x_1^2+y_1^2}, \frac{0}{x_1^2+y_1^2}\right) = (1, 0) = \left(\frac{x_1}{x_1^2+y_1^2}, \frac{-y_1}{x_1^2+y_1^2}\right) \cdot (x_1, y_1).$$

$\square$

The multiplicative inverse used in the proof of property 11 above is derived from the fact that if we want

$$(x_1, y_1) \cdot (a, b) = (1, 0),$$

then we must have

$$(x_1 a - y_1 b, x_1 b + y_1 a) = (1, 0),$$

which gives us the system of equations

$$\begin{aligned} x_1 a - y_1 b &= 1, \\ x_1 b + y_1 a &= 0. \end{aligned}$$

Solving this system for a and b,

$$x_1 b + y_1 a = 0 \quad \Longrightarrow \quad b = \frac{-y_1 a}{x_1} \quad \text{(assuming } x_1 \neq 0\text{)},$$

$$\therefore,\ x_1 a - y_1\left(\frac{-y_1 a}{x_1}\right) = 1 \quad \Longrightarrow \quad x_1 a + \frac{y_1^2 a}{x_1} = 1 \quad \Longrightarrow \quad \boxed{a = \frac{x_1}{x_1^2+y_1^2}} \quad \text{and} \quad b = \frac{-y_1 a}{x_1} = \boxed{\frac{-y_1}{x_1^2+y_1^2}}.$$

The definition of multiplication in $\mathbb{C}$ may seem innocent at first glance, but it has deep implications. In particular, notice that

$$(0, 1) \cdot (0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0). \tag{3}$$

This identity, together with the fact that

$$(a, 0) \cdot (x, y) = (ax, ay),$$

allows for an alternative notation for complex numbers. Specifically, we can write

$$(x, y) = (x, 0) + (0, y) = (x, 0) \cdot (1, 0) + (y, 0) \cdot (0, 1),$$

where since $(x, 0)$ and $(y, 0)$ behave like real numbers, we can denote them simply as x and y, respectively. That is, we can write

$$(x, y) = x \cdot (1, 0) + y \cdot (0, 1).$$

This means that we can write any complex number (x, y) as a linear combination $(1, 0)$ and $(0, 1)$ with coefficients x and y. $(1, 0)$, in turn, behaves like the real number 1, while $(0, 1)$ behaves like a new number, call it i. Then the complex number that we used to call (x, y) can now be written as

$$(x, y) = x \cdot 1 + y \cdot i = x + yi.$$

Eqn. (3) then tells us that $i^2 = -1$.[10]

[10] $i^2 = -1$ reveals that i is the square root of -1. This not only says that an equation such as $x^2 + 1 = 0$ has a solution (or that the polynomial $x^2 + 1$ is *reducible*), but every nonconstant polynomial has roots in $\mathbb{C}$. This is the content of the Fundamental Theorem of Algebra, a deep and important result in mathematics that we will, unfortunately, ignore here.

Definition 27. The number x is called the **real part** and y the **imaginary part** of the complex number $x + yi$, often denoted $\operatorname{Re}(x + yi) = x$ and $\operatorname{Im}(x + yi) = y$, respectively.

Question 28. *Find the real and imaginary parts of $(x + iy)^4$, where, of course, $x, y \in \mathbb{R}$.* ***Hint:*** *Recall the binomial theorem, which states that*

$$(a + b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k. \tag{4}$$

In particular, when $n = 4$, we have

$$(a + b)^4 = a^4 + 4a^3 b + 6a^2 b^2 + 4ab^3 + b^4.$$

Foreshadowing: some other useful expansions of Eqn. (4) are the $n = 2$ case:

$$(a + b)^2 = a^2 + 2ab + b^2,$$

and the $n = 3$ case:

$$(a + b)^3 = a^3 + 3a^2 b + 3ab^2 + b^3.$$

Geometric Representation

Complex numbers are usually denoted by a single letter, such as z. Thus, we often write $z = x + yi$, where $x, y \in \mathbb{R}$. Because the product is commutative, there is no difference between writing $x + yi$ and $x + iy$. Visually, complex numbers can be represented as points in the Euclidean plane $\mathbb{R}^2$ having Cartesian coordinates (x, y). In this context we refer to $\mathbb{R}^2$ as the **complex plane** (alternatively, the **Argand plane** or the **Gaussian plane**).[11] We use the horizontal axis (the x-axis) to represent the real part of a complex number and the vertical axis (the y-axis) to represent the imaginary part.

[11] Notice that the plane is defined as $\mathbb{R}^2$, not $\mathbb{C}^2$. The complex plane is a way of visualizing complex numbers, which are elements of $\mathbb{C}$, using points in $\mathbb{R}^2$.

In Fig. 5, we plot the complex numbers $z = 2 + i$ and $w = -1 + 2i$ in the complex plane. Notice that the sum of these two complex numbers, $u = z + w = 1 + 3i$, is represented by the point obtained by adding their corresponding coordinates in the plane. Going back to our original definition of complex numbers as pairs of real numbers, we see that this is precisely how addition is defined in $\mathbb{C}$:

$$(2, 1) + (-1, 2) = (2 + (-1), 1 + 2) = (1, 3).$$

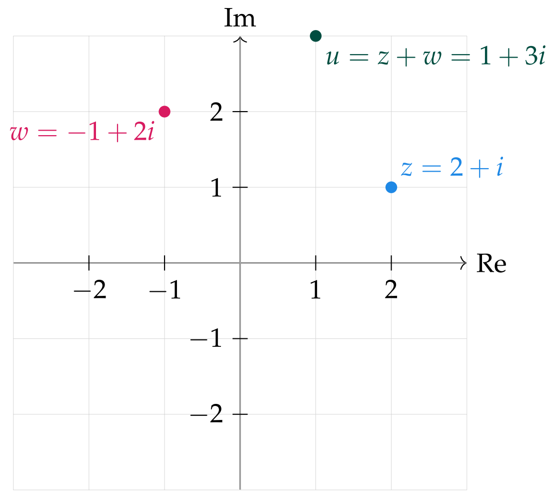


Figure 5: The complex plane. The horizontal axis represents the real part of a complex number, and the vertical axis represents the imaginary part. Here, we plot the complex numbers $z = 2 + i$, $w = -1 + 2i$, and their sum $u = z + w = 1 + 3i$. Notice that the sum of complex numbers is simply the sum of their corresponding coordinates in the plane.

One may prefer our original notation of complex numbers as pairs of real numbers when performing addition. Doing so is particularly useful when considering complex numbers as straight arrows starting from the origin $(0, 0)$ and ending at the point (x, y) in the complex plane. Fig. 5 is the as in Fig. 6.

These arrows are more formally known as **vectors**. A vector in $\mathbb{R}^2$ is an ordered pair of real numbers that can be represented as an arrow in the plane. Vectors have both a magnitude (or length) and a direction. We define these notions for complex numbers as follows:

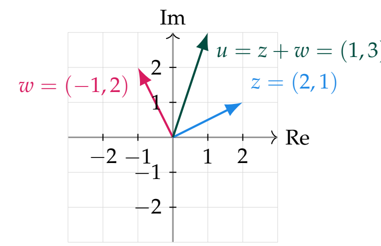


Figure 6: The complex plane with vectors representing complex numbers. Here, the complex numbers $z = (2, 1)$, $w = (-1, 2)$, and their sum $u = z + w = (1, 3)$ are represented as arrows starting from the origin. The sum of complex numbers corresponds to the vector addition of their respective arrows.

Definition 29. The **modulus** (or **magnitude**) of a complex number $z = x + yi$ is

$$r = |z| \equiv \sqrt{x^2 + y^2},$$

and an **argument** (or **angle**) of z is any real number ϕ s.t.

$$x = r\cos(\phi) \quad \text{and} \quad y = r\sin(\phi).$$

By the above definition we can represent any complex number $z = x + yi$ in terms of its modulus r and an argument ϕ as

$$z = r\cos(\phi) + ir\sin(\phi).$$

The periodic nature of the trigonometric functions implies that if ϕ is an argument of z, then so is $\phi + 2\pi k$ for any integer k. Thus, every nonzero complex number has infinitely many arguments differing by integer multiples of 2π. The number $0 = 0 + 0i$ has modulus zero and every real number ϕ is an argument.

Example 30. The modulus of the complex number $z = 1 + i\sqrt{3}$ is given by

$$r = |z| = \sqrt{1^2 + \left(\sqrt{3}\right)^2} = \sqrt{4} = 2.$$

To find an argument ϕ of z, we solve the system of equations

$$\begin{aligned} 1 &= 2\cos(\phi), \\ \sqrt{3} &= 2\sin(\phi). \end{aligned}$$

Solving the first equation for $\cos(\phi)$ tells us that $\phi = \frac{\pi}{3}, \frac{5\pi}{3}$. Similarly, solving the second equation for $\sin(\phi)$ tells us that $\phi = \frac{\pi}{3}, \frac{2\pi}{3}$. The only value of ϕ that satisfies both equations is $\phi = \frac{\pi}{3}$. Thus, one argument of z is $\frac{\pi}{3}$, and all other arguments are given by

$$\phi = \frac{\pi}{3} + 2\pi k, \quad k \in \mathbb{Z}.$$

When the argument ϕ of a complex number z is restricted to the interval $(-\pi, \pi]$, it is called the **principal argument** of z and is denoted by $\arg(z)$.

Question 31. *Consider the complex number $z = \frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}}$. Find its modulus and principal argument. Then, compute the modulus and smallest nonnegative argument of the complex numbers z^2, z^3, and z^4. What do you observe?* ***Hint:*** *Refer to Eqn. (4) and the handy expansions!*

Example 32. Given the modulus $r = 3$ and an argument $\phi = \frac{\pi}{4}$, we can find the corresponding complex number z as follows:

$$z = r\cos(\phi) + ir\sin(\phi) = 3\cos\left(\frac{\pi}{4}\right) + 3i\sin\left(\frac{\pi}{4}\right) = \frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2}i.$$

Prior to proceeding, recall the following trigonometric identities:

$$\cos(\alpha + \beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta), \tag{5}$$

$$\sin(\alpha + \beta) = \cos(\alpha)\sin(\beta) + \sin(\alpha)\cos(\beta). \tag{6}$$

Now, consider two complex numbers $z_1 = r_1\cos(\phi_1) + ir_1\sin(\phi_1)$ and $z_2 = r_2\cos(\phi_2) + ir_2\sin(\phi_2)$. Their product is given by

$$\begin{aligned} z_1 z_2 &= (r_1\cos(\phi_1) + ir_1\sin(\phi_1)) \cdot (r_2\cos(\phi_2) + ir_2\sin(\phi_2)) \\ &= r_1 r_2\Big[\cos(\phi_1)\cos(\phi_2) + i\cos(\phi_1)\sin(\phi_2) + i\sin(\phi_1)\cos(\phi_2) + i^2\sin(\phi_1)\sin(\phi_2)\Big] \\ &= r_1 r_2[\cos(\phi_1)\cos(\phi_2) - \sin(\phi_1)\sin(\phi_2) + i\cos(\phi_1)\sin(\phi_2) + i\sin(\phi_1)\cos(\phi_2)] \\ &= r_1 r_2[\cos(\phi_1)\cos(\phi_2) - \sin(\phi_1)\sin(\phi_2) + i(\cos(\phi_1)\sin(\phi_2) + \sin(\phi_1)\cos(\phi_2))] \\ &= r_1 r_2[\cos(\phi_1 + \phi_2) + i\sin(\phi_1 + \phi_2)], \quad \text{by Eqns. (5), (6).} \end{aligned} \tag{7}$$

Thus, the modulus of the product $z_1 z_2$ is the product of the moduli of z_1 and z_2, and an argument of $z_1 z_2$ is the sum of an argument of z_1

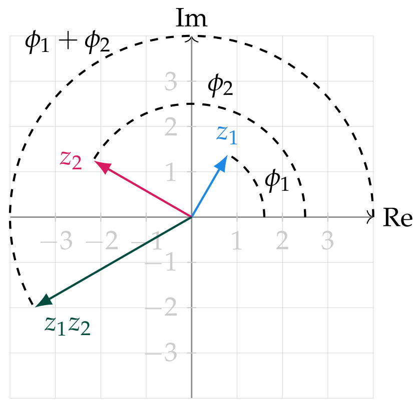


Figure 7: The complex plane with vectors representing complex numbers z_1, z_2, and their product $z_1 z_2$. The modulus of the product is the product of the moduli of z_1 and z_2, and an argument of the product is the sum of an argument of z_1 and an argument of z_2.

and an argument of z_2. In view of the above calculation, we will deal with complex numbers of the form $\cos(\phi) + i\sin(\phi)$, where ϕ is a real number. To simplify notation, we define

$$e^{i\phi} \equiv \cos(\phi) + i\sin(\phi). \tag{8}$$

At this point, this exponential notation is indeed purely a notation. However, later on we will see that this notation has deeper implications and connections to other areas of mathematics.

Fig. 8 shows some examples of complex numbers expressed in this new notation. We demonstrate the usefulness of this notation with

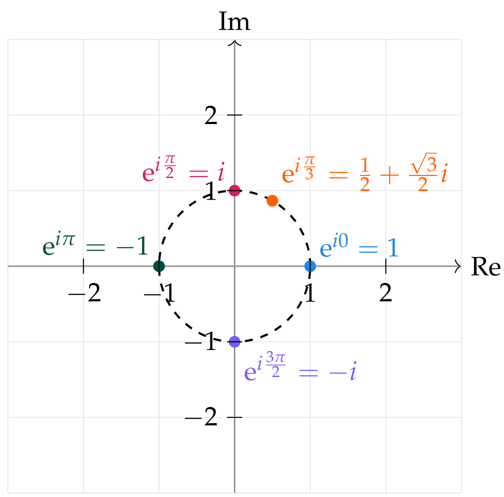


Figure 8: Examples of complex numbers in exponential notation on the complex plane. The resemblance to points on the unit circle is evident, as all these complex numbers have a modulus of 1.

the following proposition.

Proposition 33. *For any $\phi, \phi_1, \phi_2 \in \mathbb{R}$,*

1. $e^{i\phi_1} e^{i\phi_2} = e^{i(\phi_1+\phi_2)}$,
2. $e^{i0} = 1$,
3. $\frac{1}{e^{i\phi}} = e^{-i\phi}$,
4. $e^{i(\phi+2\pi)} = e^{i\phi}$,
5. $\left|e^{i\phi}\right| = 1$,
6. $\frac{d}{d\phi} e^{i\phi} = i\, e^{i\phi}$.

Question 34. *Prove the above proposition. (Answer below.)*

Proof. Going one by one:

1. Follows directly from Eqn. (7) with $r_1 = r_2 = 1$.
2. Follows directly from Eqn. (8) as
$$\mathrm{e}^{i0} = \cos(0) + i\sin(0) = 1 + i \cdot 0 = 1.$$
3. To show that $\frac{1}{\mathrm{e}^{i\phi}} = \mathrm{e}^{-i\phi}$, we need to show that $\mathrm{e}^{i\phi} \cdot \mathrm{e}^{-i\phi} = 1$. This follows directly from property 1 and property 2 as
$$\mathrm{e}^{i\phi} \cdot \mathrm{e}^{-i\phi} = \mathrm{e}^{i(\phi-\phi)} = \mathrm{e}^{i0} = 1.$$
4. Follows directly from Eqn. (8) as
$$\mathrm{e}^{i(\phi+2\pi)} = \cos(\phi+2\pi) + i\sin(\phi+2\pi) = \cos(\phi) + i\sin(\phi) = \mathrm{e}^{i\phi},$$
by the periodicity of the trigonometric functions.
5. Follows directly from Eqn. (8) as
$$\left|\mathrm{e}^{i\phi}\right| = \sqrt{\cos^2(\phi) + \sin^2(\phi)} = \sqrt{1} = 1,$$
by the definition of modulus in Defn. 29.
6. Follows directly from Eqn. (8) as
$$\frac{\mathrm{d}}{\mathrm{d}\phi}\mathrm{e}^{i\phi} = \frac{\mathrm{d}}{\mathrm{d}\phi}[\cos(\phi) + i\sin(\phi)] = -\sin(\phi) + i\cos(\phi) = i[\cos(\phi) + i\sin(\phi)] = i\,\mathrm{e}^{i\phi}.$$

□

With our new notation we can now express any complex number $z = x + yi$ in terms of its modulus r and an argument ϕ as
$$z = x + yi = r\,\mathrm{e}^{i\phi}.$$

With this new representation we now have five different ways of representing a complex number z:

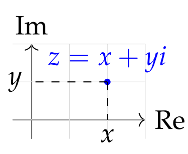
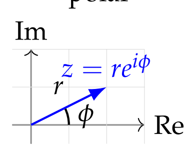
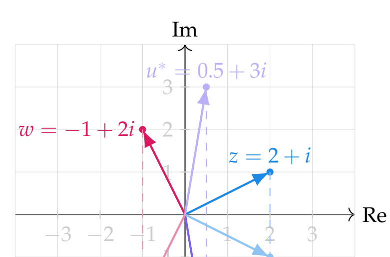
<table>
<tr><td rowspan="2">Formal
(x, y)</td><td>Algebraic:</td><td>rectangular
$x + iy$</td><td>exponential
$re^{i\theta}$</td></tr>
<tr><td>Geometric:</td><td>cartesian
</td><td>polar
</td></tr>
</table>

Table 4: Different representations of a complex number z.

We conclude this `\section{}` with a proposition that follows the following definition:

Question 35. *Find the values of*
$$(1 + 2i)^3, \quad i^n, \quad (1 + i)^n + (1 - i)^n,$$
where $n \geqslant 0$.



Definition 36. The **(complex) conjugate** of a complex number $z = x + yi$ is the complex number $z^* = x - yi$. We denote the conjugate by

$$(x + yi)^* \equiv x - yi.$$

Algebraically, all this is doing is changing the sign of the imaginary term. Geometrically, this is a reflection w.r.t. the real axis. This is shown in Fig. 9. We then see that the square of the modulus of a complex number $z = x + yi$ can be expressed as the product of z and its conjugate z^*:

$$|z|^2 = x^2 + y^2 = (x + yi)(x - yi) = zz^*. \tag{9}$$

The conjugate also enables us to consider division of complex numbers. Given two complex numbers $z_1 = x_1 + y_1 i$ and $z_2 = x_2 + y_2 i$ with $z_2 \neq 0$, their quotient is

$$\frac{z_1}{z_2} = \frac{x_1 + y_1 i}{x_2 + y_2 i}.$$

Similar to rationalizing the denominator when dealing with real numbers, we can multiply the numerator and denominator by the conjugate of the denominator to obtain

$$\frac{z_1}{z_2} = \frac{(x_1 + y_1 i)(x_2 - y_2 i)}{(x_2 + y_2 i)(x_2 - y_2 i)} = \frac{(x_1 x_2 + y_1 y_2) + i(y_1 x_2 - x_1 y_2)}{x_2^2 + y_2^2}.$$

This is now in the standard form $a + bi$ with

$$a = \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2}, \quad b = \frac{y_1 x_2 - x_1 y_2}{x_2^2 + y_2^2}.$$

Equivalently, we can express the quotient in terms of moduli and arguments. If $z_1 = r_1 \, e^{i\phi_1}$ and $z_2 = r_2 \, e^{i\phi_2}$, then

$$\frac{z_1}{z_2} = \frac{r_1 \, e^{i\phi_1}}{r_2 \, e^{i\phi_2}} = \frac{r_1}{r_2} \, e^{i(\phi_1 - \phi_2)},$$

provided $r_2 \neq 0$. The following proposition collects additional useful properties of complex conjugation.

An alternative notation for the complex conjugate is $\bar{z}$. This notation is often used in various mathematical texts and contexts. Our use of the notation z^* is to hint towards its connection to the adjoint operation in linear algebra.

Proposition 37. *For any z, z_1, $z_2 \in \mathbb{C}$,*

1. $(z_1 \pm z_2)^* = {z_1}^* \pm {z_2}^*$,
2. $(z_1 \cdot z_2)^* = {z_1}^* \cdot {z_2}^*$,
3. $\left(\frac{z_1}{z_2}\right)^* = \frac{{z_1}^*}{{z_2}^*}$ *provided* $z_2 \neq 0$,
4. $(z^*)^* = z$,
5. $|z^*| = |z|$,
6. $|z|^2 = zz^*$,
7. $\mathrm{Re}\,(z) = \frac{1}{2}(z + z^*)$,
8. $\mathrm{Im}\,(z) = \frac{1}{2i}(z - z^*)$,
9. $\left(\mathrm{e}^{i\phi}\right)^* = \mathrm{e}^{-i\phi}$.

Question 38. *Prove the above proposition. (Answer below.)*

Proof. Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$. Going one by one:

1. Follows directly from the definition of complex conjugation as $(z_1 \pm z_2)^* = ((x_1 + iy_1) \pm (x_2 + iy_2))^* = ((x_1 \pm x_2) + i(y_1 \pm y_2))^* = (x_1 \pm x_2) - i(y_1 \pm y_2) = (x_1 - iy_1) \pm (x_2 - iy_2) = {z_1}^* \pm {z_2}^*$.
2. Follows directly from the definition of complex conjugation as $(z_1 \cdot z_2)^* = ((x_1 + iy_1) \cdot (x_2 + iy_2))^* = ((x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2))^* = (x_1x_2 - y_1y_2) - i(x_1y_2 + y_1x_2) = (x_1 - iy_1) \cdot (x_2 - iy_2) = {z_1}^* \cdot {z_2}^*$.
3. Follows directly from the definition of complex conjugation as $\left(\frac{z_1}{z_2}\right)^* = \left(\frac{x_1+iy_1}{x_2+iy_2}\right)^* = \left(\frac{(x_1+iy_1)(x_2-iy_2)}{(x_2+iy_2)(x_2-iy_2)}\right)^* = \left(\frac{(x_1x_2+y_1y_2)+i(y_1x_2-x_1y_2)}{x_2^2+y_2^2}\right)^* = \frac{(x_1x_2+y_1y_2)-i(y_1x_2-x_1y_2)}{x_2^2+y_2^2} = \frac{(x_1x_2+y_1y_2)+i(x_1y_2-y_1x_2)}{x_2^2+y_2^2} = \frac{(x_1-iy_1)(x_2+iy_2)}{(x_2-iy_2)(x_2+iy_2)} = \frac{x_1-iy_1}{x_2-iy_2} = \frac{{z_1}^*}{{z_2}^*}$.
4. Follows directly from the definition of complex conjugation as $(z^*)^* = (x - iy)^* = x + iy = z$.
5. Follows directly from the definition of complex conjugation as $|z^*| = |x - iy| = \sqrt{x^2 + (-y)^2} = \sqrt{x^2 + y^2} = |z|$.
6. Shown in Eqn. (9).
7. Follows directly from the definition of complex conjugation as $\frac{1}{2}(z + z^*) = \frac{1}{2}((x + iy) + (x - iy)) = \frac{1}{2}(2x) = x = \mathrm{Re}\,(z)$.
8. Follows directly from the definition of complex conjugation as $\frac{1}{2i}(z - z^*) = \frac{1}{2i}((x + iy) - (x - iy)) = \frac{1}{2i}(2iy) = y = \mathrm{Im}\,(z)$.
9. Follows directly from Eqn. (8) as $\left(\mathrm{e}^{i\phi}\right)^* = (\cos(\phi) + i\sin(\phi))^* = \cos(\phi) - i\sin(\phi) = \cos(-\phi) + i\sin(-\phi) = \mathrm{e}^{-i\phi}$.

□

Property 6. implicitly states the inverse of a nonzero complex number z as

$$z^{-1} = \frac{1}{z} = \frac{z^*}{|z|^2}.$$

The advanced reader may note that properties 1, 2, and 3 imply that complex conjugation is a field automorphism of $\mathbb{C}$. That is, complex conjugation is a bijective map from $\mathbb{C}$ to itself that preserves both addition and multiplication.

Further Resources

1. *A First Course in Complex Analysis* by Beck, M., et al. Version 1.54. https://matthbeck.github.io/complex.html

This entire `\section{}` comes from Beck's brilliantly intuitive text. It is highly recommended for anyone looking to deepen their understanding of complex analysis. One considering further study in complex analysis should definitely consider this text. For a more traditional approach, consider the following standard undergraduate texts:

2. *Complex Analysis* by Ahlfors, L. V. McGraw-Hill, 1979.
3. *Complex Variables and Applications* by Brown, J. W., and Churchill, R. V. McGraw-Hill, 2009.

If not directly taken from Ahlfors, the questions in this `\section{}` are inspired by exercises from these texts.

Linear Algebra

Vectors: Algebraic and Geometric Interpretations

Definition 39. A **list of length *n*** (or an ***n*-tuple**), where $n \in \mathbb{Z}_{\geqslant 0}$, is an ordered collection of n elements separated by commas and surrounded by parentheses:

$$(x_1, x_2, \ldots, x_n).$$

Two **lists are equal** iff they have the same length and the same elements are in the same order.

In general, the elements of a list can be numbers, other lists, or even more abstract objects. A list of length 0 is called the **empty list** and is denoted by $()$. Additionally, by definition, a list cannot have infinite length.

Definition 40. Let F be a field. $\boldsymbol{F^n}$ is the set of all lists of length n whose elements are in F:

$$F^n \equiv \{(x_1, x_2, \ldots, x_n) : x_i \in F \text{ for } i = 1, 2, \ldots, n\}.$$

For $(x_1, x_2, \ldots, x_n) \in F^n$ and $j \in \{1, 2, \ldots, n\}$, we say that x_j is the ***j*-th coordinate** of the list.

In particular, $\mathbb{R}^n$ is the set of all lists of length n whose elements are real numbers, and $\mathbb{C}^n$ is the set of all lists of length n whose elements are complex numbers. We saw in the previous section that we initially treated the elements of $\mathbb{C}$ as ordered pairs of real numbers. Addition of lists in F^n is defined exactly as addition of ordered pairs in $\mathbb{C}$: consider $(x_1, x_2, \ldots, x_n)$ and $(y_1, y_2, \ldots, y_n)$, with $x_i, y_i \in F$ for all $i = 1, 2, \ldots, n$, their sum is

$$(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) \equiv (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n). \quad (10)$$

We denote the 0 list in F^n by

$$0_{F^n} \equiv (0_F, 0_F, \ldots, 0_F),$$

where 0_F is the additive identity in F. Under this addition operation, we propose the following:

Proposition 41. *Let F be a field. The set F^n, equipped with addition as defined in Eqn. (10), is an additive abelian group.*

Question 42. *Prove the above proposition. (Answer below.)*

Proof. We are tasked with showing that F^n with addition satisfies the group axioms (Defn. 3) and is abelian (Defn. 5). Going one by one, for all $(x_1, x_2, \ldots, x_n)$, $(y_1, y_2, \ldots, y_n)$, and $(z_1, z_2, \ldots, z_n)$ in F^n:

1. The sum $(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$ is also in F^n since $x_i + y_i \in F$ for all $i = 1, 2, \ldots, n$ (by the closure property of fields).
2. We have

$$\begin{aligned}
(x_1, x_2, \ldots, x_n) + ((y_1, y_2, \ldots, y_n) + (z_1, z_2, \ldots, z_n)) &= (x_1, x_2, \ldots, x_n) + (y_1 + z_1, y_2 + z_2, \ldots, y_n + z_n) \\
&= (x_1 + (y_1 + z_1), x_2 + (y_2 + z_2), \ldots, x_n + (y_n + z_n)) \\
&= ((x_1 + y_1) + z_1, (x_2 + y_2) + z_2, \ldots, (x_n + y_n) + z_n) \\
&= (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n) + (z_1, z_2, \ldots, z_n) \\
&= ((x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n)) + (z_1, z_2, \ldots, z_n)
\end{aligned}$$

   where the third equality follows from the associativity property of fields.
3. The 0 list $0_{F^n} = (0_F, 0_F, \ldots, 0_F)$ satisfies

$$(x_1, x_2, \ldots, x_n) + 0_{F^n} = (x_1 + 0_F, x_2 + 0_F, \ldots, x_n + 0_F) = (x_1, x_2, \ldots, x_n)$$

   where the second equality follows from the identity property of fields. Similarly, $0_{F^n} + (x_1, x_2, \ldots, x_n) = (x_1, x_2, \ldots, x_n)$.
4. The additive inverse of $(x_1, x_2, \ldots, x_n)$ is $(-x_1, -x_2, \ldots, -x_n)$ since

$$(x_1, x_2, \ldots, x_n) + (-x_1, -x_2, \ldots, -x_n) = (x_1 + (-x_1), x_2 + (-x_2), \ldots, x_n + (-x_n)) = (0_F, 0_F, \ldots, 0_F) = 0_{F^n},$$

   where the second equality follows from the inverse property of fields. Similarly, $(-x_1, -x_2, \ldots, -x_n) + (x_1, x_2, \ldots, x_n) = 0_{F^n}$.

5. We have

$$\begin{aligned}(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) &= (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n) \\ &= (y_1 + x_1, y_2 + x_2, \dots, y_n + x_n) \\ &= (y_1, y_2, \dots, y_n) + (x_1, x_2, \dots, x_n),\end{aligned}$$

where the second equality follows from the commutativity property of fields.

Therefore, F^n with addition is an additive abelian group.

□

In proving property 4 in the above proposition, we see that the additive inverse of a list $(x_1, x_2, \dots, x_n)$ is obtained by negating each of its coordinates, giving $(-x_1, -x_2, \dots, -x_n)$. Thus, there is a *global* negative sign that can be "factored" out:

$$-(x_1, x_2, \dots, x_n) \equiv (-x_1, -x_2, \dots, -x_n).$$

However, there is nothing special about a negative one that makes it different from any other value. We can easily define multiplication of a list by *any* scalar[12] $a \in F$:

[12] A **scalar** is simply an element of the field F.

$$a(x_1, x_2, \dots, x_n) \equiv (ax_1, ax_2, \dots, ax_n).$$

More formally, we have the following definition:

Definition 43. The **product** of a value $a \in F$ and a list $(x_1, x_2, \dots, x_n) \in F^n$ is defined by multiplying each coordinate of the list by a:

$$a(x_1, x_2, \dots, x_n) \equiv (ax_1, ax_2, \dots, ax_n).$$

This is what we call **scalar multiplication**.

Loosely speaking, lists that satisfy the above addition and scalar multiplication operations are called (row) **vectors**.[13] We are ommitting a formal definition of vectors here, as we will soon introduce the more general concept of vector spaces. For the time being, we seek to get a better understanding of the geometry of lists in F^n under addition and scalar multiplication.

[13] Vectors are written in **boldface** with an arrow over them, $\vec{\mathbf{v}}$. The term **vector** has multiple meanings in mathematics. In this context, we are referring to lists that satisfy specific algebraic properties under addition and scalar multiplication.

We've seen before that the complex plane uses $\mathbb{R}^2$ to represent complex numbers. More generally, we can use F^n to represent points (or vectors) in an n-dimensional space over a field F. For example, $\mathbb{R}^3$ is used to represent points (or vectors) in three-dimensional Euclidean space. As before, when we think of a list as an arrow, we refer to it as a vector. For instance, the vector $(2, 3)$ in $\mathbb{R}^2$ can be represented as an arrow from the origin $(0,0)$ to the point $(2,3)$ in the Cartesian plane. Similarly, the vector $(1,4,2)$ in $\mathbb{R}^3$ can be represented as an arrow from the origin $(0,0,0)$ to the point $(1,4,2)$ in three-dimensional space. In the following figure we show examples of vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$ and illustrate how their components lie on the axes.

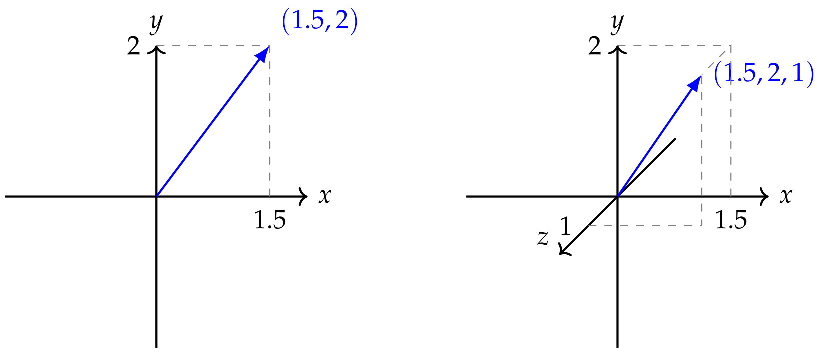


Figure 10: Showing vectors in $\mathbb{R}^2$ (left) and $\mathbb{R}^3$ (right) along with their components on each axis. There is an inherent order in the structure of lists, which is reflected in the labeling of the axes as x, y, and z. The first component of each vector corresponds to the x-axis, the second to the y-axis, and the third (if applicable) to the z-axis.

Visually, the addition of vectors corresponds to the *parallelogram law*. The parallelogram law for vector addition states that the sym of two vectors $\vec{\mathbf{x}}$ and $\vec{\mathbf{y}}$ that act at the same point P is the vector beginning at P that is represented by the diagonal of the parallelogram having $\vec{\mathbf{x}}$ and $\vec{\mathbf{y}}$ as adjacent sides. The point P is often taken to be the origin. This is illustrated in the following figure:

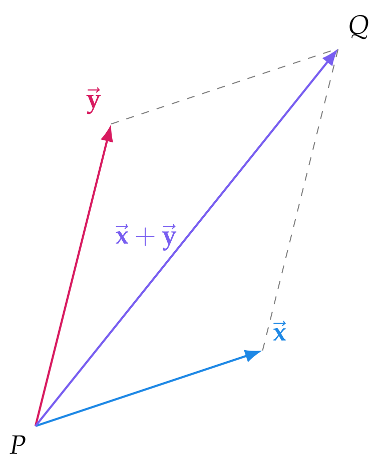


Figure 11: Since opposite sides of a parallelogram are parallel and of equal length, the endpoint Q of the arrow representing $\vec{\mathbf{x}} + \vec{\mathbf{y}}$ can be found by either allowing $\vec{\mathbf{x}}$ to act at P and then allowing $\vec{\mathbf{y}}$ to act at the endpoint of $\vec{\mathbf{x}}$, or by allowing $\vec{\mathbf{y}}$ to act at P and then allowing $\vec{\mathbf{x}}$ to act at the endpoint of $\vec{\mathbf{y}}$. Thus two vectors that both act at the point P may be added "tail-to-head," or "tip-to-tail."

Finally, scalar multiplication of a vector $\vec{\mathbf{v}}$ by a scalar a can be interpreted geometrically as stretching or compressing the vector $\vec{\mathbf{v}}$ by a factor of $|a|$ if $a > 0$, or reflecting the vector $\vec{\mathbf{v}}$ across the origin and then stretching or compressing it by a factor of $|a|$ if $a < 0$. If $a = 0$, the resulting vector is the zero vector, **not** the value 0. This is in accordance with Defn. 43. Each of these geometric interpretations can be visualized in the following figure:

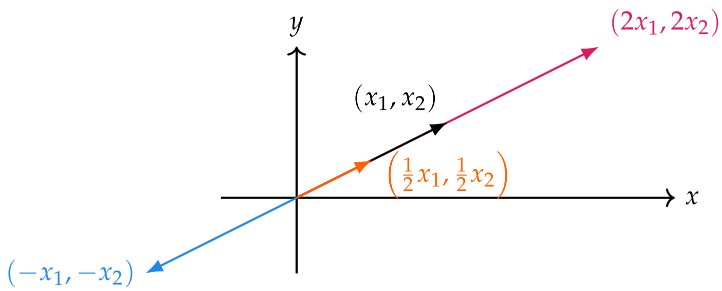


Figure 12: Scalar multiplication on a vector may affect its length or direction. The original vector is shown here in black. Multiplying by a scalar greater than 1 (red) stretches the vector, multiplying by a scalar between 0 and 1 (orange) compresses the vector, and multiplying by a negative scalar (blue) reflects the vector across the origin. Multiplying the vector by 0 would result in the zero vector at the origin (not shown).

Two nonzero vectors $\vec{\mathbf{x}}$ and $\vec{\mathbf{y}}$ are called **parallel** if $\vec{\mathbf{y}} = a\vec{\mathbf{x}}$ for some nonzero $a \in \mathbb{R}$. In particular, if $a < 0$, then $\vec{\mathbf{x}}$ and $\vec{\mathbf{y}}$ point in opposite directions and are then called **antiparallel**.

Question 44. *This question consists of four parts.*

1. *Find $x \in \mathbb{R}^4$ s.t.*

$$(4, -3, 1, 7) + 2x = (5, 9, -6, 8).$$

2. *Find the midpoint of the line segment joining the points (a, b) and (c, d) in $\mathbb{R}^2$.*

Let $\vec{\mathbf{v}}_1 = (a, 0)$ and $\vec{\mathbf{v}}_2 = (b, c)$ in $\mathbb{R}^2$. See Fig. 13 for the following questions.

3. *Find the area of the parallelogram formed by the vectors $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$. You may assume, without loss of generality, that a, b, and c are all positive.*

4. *We see in Fig. 13 that the vector $\vec{\mathbf{v}}_1 + \vec{\mathbf{v}}_2$ corresponds to one of the diagonals of the parallelogram formed by $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$. Find an expression, in terms of $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$, for the vector corresponding to the other diagonal of the parallelogram.*

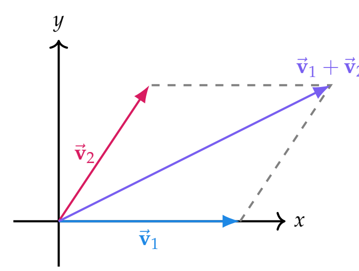


Figure 13: Reference figure for parts 3–4 of Quest. 44. The parallelogram is formed by the vectors $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$ in $\mathbb{R}^2$.

(Finite Dimensional) Vector Spaces

We've now arrived at the thing we've spent so long building up to:

Definition 45 (Vector space)**.** Let F be a field. A **vector space over F** is an additive abelian group (*i.e.*, an abelian group equipped with addition) V equipped with scalar multiplication s.t., $\forall\ a, a_1, a_2 \in F$ and $v, v_1, v_2 \in V$,

1. $a(v_1 + v_2) = av_1 + av_2$,

2. $(a_1 + a_2)v = a_1 v + a_2 v$,

3. $a_1(a_2 v) = (a_1 a_2)v$,

4. $1_F v = v$,

where 1_F is the multiplicative identity in F.

With this definition, we define our new use of the word **vector**:

Definition 46 (Vector)**.** Elements of a vector space are called **vectors** (or **points**).

The reader should not confuse this use of the word "vector" with the physical entity we discussed at the end of the previous `\subsection{}`.[14]

A vector space over $\mathbb{R}$ is called a **real vector space**, while a vector space over $\mathbb{C}$ is called a **complex vector space**. The set F^n, equipped with the addition and scalar multiplication operations defined in the previous `\subsection{}`, is a vector space over F. Though we were quite informal in our treatment of lists in the previous `\subsection{}`, vectors in F^n may be written as **column vectors**:

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

rather than as **row vectors**:

$$\begin{pmatrix} a_1 & a_2 & \cdots & a_n \end{pmatrix}.$$

Suppose V is a vector space over a field F and that w and v_1, v_2, ..., v_n are elements of V. We say that w is a **linear combination** of v_1, v_2, ..., v_n if w can be written in the form

$$w = a_1 v_1 + a_2 v_2 + \cdots + a_n v_n$$

for $a_i \in F$. If $a_1 = a_2 = \cdots = a_n = 0_F$, we have the **trivial representation** of the zero vector, $w = 0_V$, as a linear combination of v_1, v_2, ..., v_n. In general, all possible linear combinations of a set of vectors form the following definition:

[14] One should have also noticed by now that the notation for vectors in this section is different from that used in the previous subsection. We no longer use boldface with an arrow over it to denote vectors, as we are now working in a more abstract setting. In fact, vectors in a vector space need not even have a geometric interpretation at all!

Example 47. In $\mathbb{R}^3$, $(17, -4, 2)$ is a linear combination of $(2, 1, -3)$ and $(1, -2, 4)$ because

$$\begin{pmatrix} 17 \\ -4 \\ 2 \end{pmatrix} = 6\begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + 5\begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}.$$

Definition 48 (Span)**.** The set of all linear combinations of a set of vectors $v_1, v_2, \ldots, v_n$ in a vector space V over a field F is called the **span** of $v_1, v_2, \ldots, v_n$, denoted by $\operatorname{span}(v_1, v_2, \ldots, v_n)$. In symbols,

$$\operatorname{span}(v_1, v_2, \ldots, v_n) \equiv \{a_1 v_1 + a_2 v_2 + \cdots + a_n v_n : a_i \in F\}.$$

If every element of said vector space V is a linear combination of $v_1, v_2, \ldots, v_n$, we say that the set $\{v_1, v_2, \ldots, v_n\}$ **spans** $\boldsymbol{V}$ over F.

Example 49. In $\mathbb{R}^3$, the set

$$\left\{ \begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\1\end{pmatrix} \right\}$$

spans $\mathbb{R}^3$ because any vector

$$\begin{pmatrix}x\\y\\z\end{pmatrix} \in \mathbb{R}^3$$

can be written as a linear combination of these three vectors:

$$\begin{pmatrix}x\\y\\z\end{pmatrix} = x\begin{pmatrix}1\\0\\0\end{pmatrix} + y\begin{pmatrix}0\\1\\0\end{pmatrix} + z\begin{pmatrix}0\\0\\1\end{pmatrix}.$$

These three vectors are called the standard basis vectors of $\mathbb{R}^3$ and are often represented as simply $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ respectively, implying that each vector corresponds to an axis in $\mathbb{R}^3$.

Example 50. In $\mathbb{R}^2$, the set of a single vector

$$\left\{ \begin{pmatrix}a\\b\end{pmatrix} \right\}$$

does not span $\mathbb{R}^2$ since, *e.g.*, the vector

$$\begin{pmatrix}a+2\\b+1\end{pmatrix}$$

cannot **generally** be written as a linear combination of $\begin{pmatrix}a\\b\end{pmatrix}$. Linear combinations of $\begin{pmatrix}a\\b\end{pmatrix}$ are of the form

$$c\begin{pmatrix}a\\b\end{pmatrix} = \begin{pmatrix}ca\\cb\end{pmatrix},$$

for some $c \in \mathbb{R}$. That is, the set $\left\{ \begin{pmatrix}a\\b\end{pmatrix} \right\}$ only spans the line through the origin and the point (a, b) in $\mathbb{R}^2$.

Question 51. **TODO:** *NOT A GOOD QUESTION. REWRITE.*

In Exmp. 50 we said that the vector $\begin{pmatrix}a+2\\b+1\end{pmatrix}$ *cannot **generally** be written as a linear combination of* $\begin{pmatrix}a\\b\end{pmatrix}$. *Provide specific values of* a *and* b *s.t. this is this case, i.e.,*

$$\begin{pmatrix}a+2\\b+1\end{pmatrix} \neq c\begin{pmatrix}a\\b\end{pmatrix},$$

for any c $\in \mathbb{R}$. *On the other hand, provide expressions for* a *and* b *in terms of* c *s.t. the following holds:*

$$\begin{pmatrix}a+2\\b+1\end{pmatrix} = c\begin{pmatrix}a\\b\end{pmatrix}.$$

Definition 52 (Linear dependence)**.** A subset S of a vector space V over a field F is said to be **linearly dependent** over F if there exist a finite number of distinct vectors $v_1, v_2, \ldots, v_n$ in S and scalars $f_1, f_2, \ldots, f_n$ in F, not all zero, s.t.

$$f_1 v_1 + f_2 v_2 + \cdots + f_n v_n = 0_V,$$

with each $f_i \in F$, then, $\forall i$, $f_i = 0_F$. A set that is not linearly dependent is said to be **linearly independent**.

The following facts about linearly indpendent sets are true in any vector space:

1. The empty set is linearly independent, for linearly dependent sets must be nonempty.
2. A set containing of a single nonzero vector is linearly independent. If $\{v\}$ is linearly dependent, then there exists $f \in F, f \neq 0_F$, s.t. $fv = 0_V$. Multiplying both sides by f^{-1} gives $v = 0_V$. Thus, $\{v\}$ is linearly dependent only if $v = 0_V$.
3. A set is linearly independent iff the only representations of the zero vector as a linear combination of its elements is the trivial representation.

The third fact provides a useful method for determining whether a (finite) set is linearly independent.[15]

A linearly independent spanning set for a vector space possesses an incredibly useful property: every vector in said vector space can be uniquely expressed as a linear combination of the vectors in the set.[16] It is this property that makes linearly independent spanning sets the building blocks of vector spaces.

[15] This is done by setting up a system of linear equations and checking if the only solution is the trivial one. Traditionally, this involves writing the vectors as columns of a matrix and performing row reduction to echelon form. If this means nothing to you, don't worry; it **won't** be explained in more detail later because we won't need it for our purposes.

[16] The uniqueness of this representation is a direct consequence of linear independence and can be shown by assuming two different representations and subtracting them to obtain a nontrivial linear combination of the basis vectors that equals zero, which contradicts linear independence.

Definition 53 (Basis)**.** A subset $\{v_1, v_2, \ldots, v_n\}$ of a vector space V over a field F is said to be a **basis** of V if it spans V and is linearly independent over F.

Example 54. In F^n, let $e_1 = (1, 0, \ldots, 0)$, $e_2 = (0, 1, \ldots, 0)$, $\ldots$, $e_n = (0, 0, \ldots, 1)$. Then $\{e_1, e_2, \ldots, e_n\}$ is a linearly independent spanning set for F^n, and thus forms a basis for F^n. As we saw in Example 49, these vectors are the standard basis vectors for F^n.

Definition 55 (Dimensionality)**.** If a vector space V over a field F has a finite basis, then V is said to be **finite dimensional** over F. The **dimension of V over F** is the number of elements in *any* basis of V. If V does not have a finite basis, then V is said to be **infinite dimensional** over F.

Though we won't be proving it here, the following are true:

1. every finite-dimensional vector space has a basis, and
2. every linearly independent set of vectors in a finite-dimensional vector space can be extended to a basis of the vector space.

Any linearly independent set of vectors can be extended to a basis by adding more vectors to the set. In particular, if a vector space V has dimension n, then any linearly independent set of vectors in V can have at most n elements.

When attempting to find a basis for a vector space, the easiest basis to settle on is the standard basis. However, there are often many other bases for a vector space, some of which may be more useful than the standard basis for certain applications. Interestingly, although the choice of basis is not unique, the number of elements in any basis of a given vector space is always the same:

Theorem 56. *Any two bases of a finite-dimensional vector space V over a field F have the same number of elements.*

A light proof of this theorem relies on the last sentence of the preceding paragraph.

Proof. Suppose V is finite-dimensional. Let $B_1 \equiv \{v_1, v_2, \ldots, v_n\}$ and $B_2 \equiv \{u_1, u_2, \ldots, u_m\}$ be two bases of V. Then B_1 is linearly independent and B_2 spans V, so $n \leqslant m$. Similarly, B_2 is linearly independent and B_1 spans V, so $m \leqslant n$. Thus, $n = m$. □

Linear Maps

Definition 57 (Linear map)**.** A **linear map** (or **linear transformation**) from a vector space V over a field F to a vector space W over the same field F is a function $f : V \to W$ s.t., $\forall v_1, v_2 \in V$ and $a \in F$,

1. $f(v_1 + v_2) = f(v_1) + f(v_2)$,
2. $f(av_1) = af(v_1)$.

In other words, a linear map is a function that preserves vector addition and scalar multiplication.

Example 58. The identity map $I : V \to V$ defined by $I(v) = v$ for all $v \in V$ is a linear map. The zero map $Z : V \to W$ defined by $Z(v) = 0_W$ for all $v \in V$ is also a linear map.

Question 59. *Verify that $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x, y, z) = (2x - y + 3z, 7x + 5y - 6z)$ is a linear map.*

Example 60. We could also have a linear map $T : \mathbb{R}^3 \to \mathbb{R}^2$ defined by $T(x, y, z) = (2x - y, x + 3z)$. To check that T is a linear map, we need to verify that it preserves vector addition and scalar multiplication. For vector addition, let $u = (u_1, u_2, u_3)$ and $v = (v_1, v_2, v_3)$ be vectors in $\mathbb{R}^3$. Then,

$$\begin{aligned} T(u + v) &= T(u_1 + v_1, u_2 + v_2, u_3 + v_3) \\ &= (2(u_1 + v_1) - (u_2 + v_2), (u_1 + v_1) + 3(u_3 + v_3)) \\ &= ((2u_1 - u_2) + (2v_1 - v_2), (u_1 + 3u_3) + (v_1 + 3v_3)) \\ &= T(u) + T(v). \end{aligned}$$

Now for scalar multiplication, let $a \in \mathbb{R}$ and $u = (u_1, u_2, u_3)$ be a vector in $\mathbb{R}^3$. Then,

$$\begin{aligned} T(au) &= T(au_1, au_2, au_3) \\ &= (2(au_1) - (au_2), (au_1) + 3(au_3)) \\ &= a(2u_1 - u_2, u_1 + 3u_3) \\ &= aT(u). \end{aligned}$$

Thus, T is a linear map.

Now that we're somewhat comfortable with the definition of linear maps, we can start to explore some of their properties. The natural question to ask is how do two linear maps interact with each other?

Definition 61. Let V, W, and U be vector spaces over a field F. If $f : V \to W$, $g : V \to W$ and $h : W \to U$ are linear maps, and $a \in F$, then, for all $v \in V$,

1. the **sum** of f and g is the linear map $f + g : V \to W$ defined by $(f + g)(v) = f(v) + g(v)$,
2. the **product** of a and f is the linear map $af : V \to W$ defined by $(af)(v) = af(v)$,
3. the **composition** of h and f is the linear map $h \circ f : V \to U$ defined by $(h \circ f)(v) = h(f(v))$.

The composition of linear maps is also often simply called the product of linear maps, and is denoted by juxtaposition: hf instead of $h \circ f$. The following algebraic properties of linear maps are true, though we won't be proving them here:

1. $(T_1T_2)T_3 = T_1(T_2T_3)$ (associativity)

   whenever $T_1 : W \to X$, $T_2 : V \to W$, and $T_3 : U \to V$ are linear maps and U, V, W, and X are vector spaces over the same field F;

2. $TI = IT = T$ (identity)

   whenever $T : V \to W$ is a linear map, I is the identity map on the appropriate vector space (*i.e.*, $I : V \to V$ in the first term and $I : W \to W$ in the second term), and V and W are vector spaces over the same field F;

3. $(S_1 + S_2)T = S_1T + S_2T$ and $S(T_1 + T_2) = ST_1 + ST_2$ (distributivity)

   whenever T, T_1, $T_2 : U \to V$ and S, S_1, $S_2 : V \to W$ are linear maps and U, V, and W are vector spaces over the same field F.

It follows then that

Theorem 62. *Let $T : V \to W$ be a linear map between vector spaces V and W over a field F. Then $T(0) = 0$.*

Proof. By additivity, we have

$$T(0) = T(0 + 0) = T(0) + T(0).$$

Call $-T(0)$ the additive inverse of $T(0)$ in W. Then, adding $-T(0)$ to both sides of the above equation gives

$$0 = T(0) + (-T(0)) = (T(0) + T(0)) + (-T(0)) = T(0) + (T(0) + (-T(0))) = T(0) + 0 = T(0).$$

Therefore, $T(0) = 0$. □

We briefly discuss invertible linear maps, which are linear maps that have inverses.[17]

[17] Invertible linear maps are important because they allow us to establish isomorphisms between vector spaces, which are bijective linear maps that preserve the structure of the vector spaces.

Definition 63. A linear map $T : V \to W$ between vector spaces V and W over a field F is said to be **invertible** if there exists a linear map $S : W \to V$ such that $ST = I_V$ and $TS = I_W$, where I_V and I_W are the identity maps on V and W, respectively. In this case, we say that S is the **inverse** of T, and we denote it by T^{-1}.

Additionally, linear maps from a vector space to itself are so important they have their own name:

Definition 64. A linear map $T : V \to V$ from a vector space V over a field F to itself is called an **operator** on V.

Our study of linear maps allows us to finally be able to formally discuss matrices.

(Linear Maps as) Matrices

We start by repeating a defintion from before:

Definition 65 (Matrix)**.** Let m and n be positive integers. An m-by-n **matrix A** is a rectagular array of elements arranged in m rows and n columns, denoted

$$\mathbf{A} = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}.$$

The notation a_{ij} denotes the element in the i-th row and the j-th column of $\mathbf{A}$. In other words, the first index refers to the row and the second index refers to the column.

Example 66. If

$$A = \begin{pmatrix} 8 & 4 & 5 \\ 1 & 9 & 7 \end{pmatrix},$$

then $a_{11} = 8$, $a_{12} = 4$, $a_{13} = 5$, $a_{21} = 1$, $a_{22} = 9$, and $a_{23} = 7$.

We now come to an ideal definition, but the key definition in this section:

Definition 67. Let $T : F^n \to F^m$ be a linear map, and let

$$e_1, e_2, \ldots, e_n$$

be the standard basis of F^n. The standard matrix of T is the $m \times n$ matrix

$$A = \begin{pmatrix} | & | & & | \\ T(e_1) & T(e_2) & \cdots & T(e_n) \\ | & | & & | \end{pmatrix}.$$

That is, the j-th column of A is $T(e_j)$.

In the definition on the left, we use the notation e_i. The definition of e_i is as follows: e_i is the vector in F^n that has a 1 in the i-th position and 0's everywhere else. In symbols,

$$e_i = \begin{pmatrix} \delta_{i1} \\ \delta_{i2} \\ \vdots \\ \delta_{in} \end{pmatrix}$$

where δ_{ij} is the Kronecker delta.

In other words, the standard matrix of a linear map $T : F^n \to F^m$ is the matrix representation of T with respect to the standard bases of F^n and F^m. Though this is a very specific definition, it allows us to make the following important observation: every linear map from F^n to F^m can be represented by a matrix, and every matrix can be interpreted as a linear map from F^n to F^m.

Example 68. Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear map defined by

$$T(x, y) = (x + 3y, 2x + 5y, 7x + 9y).$$

To find the matrix representation of T with respect to the standard bases of $\mathbb{R}^2$ and $\mathbb{R}^3$, we first compute $T(e_1)$ and $T(e_2)$, where

$$e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

We have

$$T(e_1) = T(1, 0) = (1, 2, 7)$$

and

$$T(e_2) = T(0, 1) = (3, 5, 9).$$

Thus, the matrix representation of T with respect to the standard bases of $\mathbb{R}^2$ and $\mathbb{R}^3$ is

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 5 \\ 7 & 9 \end{pmatrix}.$$

We therefore see that matrices are linear maps in disguise. What this tells us is that we can use the properties of linear maps to derive properties of matrices, and vice versa.

We start by going over the algebraic properties of matrix addition and scalar multiplication, which are direct consequences of the algebraic properties of linear maps:

Definition 69 (Matrix addition)**.** The **sum of two matrices** A and B of the same dimensions is obtained by adding the corresponding entries of A and B:

$$\begin{aligned} A+b &= \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} + \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \cdots & b_{mn} \end{pmatrix} \\ &= \begin{pmatrix} a_{11}+b_{11} & \cdots & a_{1n}+b_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1}+b_{m1} & \cdots & a_{mn}+b_{mn} \end{pmatrix}. \end{aligned}$$

Definition 70. The **product of a scalar and a matrix** is the matrix obtained by multiplying each entry of the matrix by the scalar:

$$\lambda A = \lambda \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} = \begin{pmatrix} \lambda a_{11} & \cdots & \lambda a_{1n} \\ \vdots & \ddots & \vdots \\ \lambda a_{m1} & \cdots & \lambda a_{mn} \end{pmatrix}.$$

Matrix multiplication is a bit more complicated,

The product of a $1 \times n$ matrix and an $n \times 1$ matrix is a 1×1 matrix, which we identify with the single entry in the matrix.

Definition 71 (Matrix multiplication)**.** Let A be an m-by-n matrix and B be an n-by-p matrix. The **product of A and B**, denoted AB, is the m-by-p matrix whose entry in the i-th row and j-th column is given by

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

In words, the entry in the i-th row and j-th column of AB is computed by taking row i of A and column j of B, multiplying the corresponding entries together, and then summing those products.

Note that we define the product of two matrices only when the number of columns of the first matrix is equal to the number of rows of the second matrix. This goes back to the fact that the product of two linear maps $S : U \to V$ and $T : V \to W$ is only defined when the codomain of S is the same as the domain of T.

Example 72. Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 6 & 5 & 4 & 3 \\ 2 & 1 & 0 & -1 \end{pmatrix}.$$

Then,

$$\begin{aligned} AB &= \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 6 & 5 & 4 & 3 \\ 2 & 1 & 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 1 \cdot 6 + 2 \cdot 2 & 1 \cdot 5 + 2 \cdot 1 & 1 \cdot 4 + 2 \cdot 0 & 1 \cdot 3 + 2 \cdot (-1) \\ 3 \cdot 6 + 4 \cdot 2 & 3 \cdot 5 + 4 \cdot 1 & 3 \cdot 4 + 4 \cdot 0 & 3 \cdot 3 + 4 \cdot (-1) \\ 5 \cdot 6 + 6 \cdot 2 & 5 \cdot 5 + 6 \cdot 1 & 5 \cdot 4 + 6 \cdot 0 & 5 \cdot 3 + 6 \cdot (-1) \end{pmatrix} \\ &= \begin{pmatrix} 10 & 7 & 4 & 1 \\ 26 & 19 & 12 & 5 \\ 46 & 31 & 20 & 9 \end{pmatrix}. \end{aligned}$$

Example 73. Let

$$A = \begin{pmatrix} 3 & 4 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 6 \\ 2 \end{pmatrix}.$$

Then,

$$AB = \begin{pmatrix} 3 & 4 \end{pmatrix} \begin{pmatrix} 6 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \cdot 6 + 4 \cdot 2 \end{pmatrix} = \begin{pmatrix} 26 \end{pmatrix} = 26.$$

Matrix multiplication is not commutative, meaning that AB is not necessarily equal to BA. In fact, BA may not even be defined if the dimensions of A and B are such that the number of columns of B is not equal to the number of rows of A. It can be shown that matrix multiplication is associative and distributive over matrix addition, but we won't be proving those properties here. We conclude this subsection with the following corollary of the algebraic properties of linear maps:

Corollary 74. *Let A be an m-by-n matrix and c be an n-by-1 martrix (i.e., a column vector),*

$$c = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}.$$

Then,

$$Ac = c_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} + c_2 \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} + \cdots + c_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}.$$

In words, the product of a matrix A and a column vector c can be expressed as a linear combination of the columns of A, where the coefficients of the linear combination are given by the entries of c.

Lastly, though this is motivated by dual spaces, we introduce the notion of the transpose of a matrix, which is a matrix obtained by swapping the rows and columns of the original matrix:

Definition 75 (Transpose of a matrix)**.** Let A be an m-by-n matrix. The **transpose** of A, denoted $A^\intercal$, is the n-by-m matrix obtained by swapping the rows and columns of A:

$$A^\intercal = \begin{pmatrix} a_{11} & a_{21} & \cdots & a_{m1} \\ a_{12} & a_{22} & \cdots & a_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & \cdots & a_{mn} \end{pmatrix}.$$

The transpose of a product of matrices is the product of the transposes in reverse order:

Theorem 76. *Let A and B be matrices such that the product AB is defined. Then,*

$$(AB)^\intercal = B^\intercal A^\intercal.$$

Proof. Let A be an m-by-n matrix and B be an n-by-p matrix. Then, the product AB is an m-by-p matrix, and its transpose $(AB)^\intercal$ is a

p-by-m matrix. On the other hand, the transpose $B^\intercal$ of B is a p-by-n matrix, and the transpose $A^\intercal$ of A is an n-by-m matrix. Thus, the product $B^\intercal A^\intercal$ is also a p-by-m matrix. To show that $(AB)^\intercal = B^\intercal A^\intercal$, we need to show that their entries are equal. Let (i, j) be the indices of an arbitrary entry in $(AB)^\intercal$. Then,

$$\begin{aligned}(AB)^\intercal_{ij} &= (AB)_{ji} \\ &= \sum_{k=1}^{n} a_{jk} b_{ki} \\ &= \sum_{k=1}^{n} b_{ki} a_{jk} \\ &= (B^\intercal A^\intercal)_{ij}.\end{aligned}$$

Since the entries of $(AB)^\intercal$ and $B^\intercal A^\intercal$ are equal for all indices (i, j), we conclude that $(AB)^\intercal = B^\intercal A^\intercal$. □

Square matrices, which are matrices with the same number of rows and columns, have further properties that are worth studying. For example,

Invertible matrices are also called non-singular matrices, and non-invertible matrices are also called singular matrices.

Definition 77. A square matrix A is said to be **invertible** if there exists a square matrix B of the same dimensions such that $AB = BA = I$, where I is the identity matrix of the appropriate dimensions. In this case, we say that B is the **inverse** of A, and we denote it by A^{-1}.

A square matrix also carries information within its main diagonal,

Definition 78 (Trace of a matrix)**.** The **trace** of a square matrix A, denoted $\operatorname{tr} A$, $\operatorname{tr}(A)$, or $\operatorname{Tr}(A)$, is the sum of the entries on the main diagonal of A:

$$\operatorname{tr} A = a_{11} + a_{22} + \cdots + a_{nn}.$$

Question 79. *Show the following properties of the trace of a matrix:*

1. $\operatorname{tr}(A + B) = \operatorname{tr} A + \operatorname{tr} B$ *for all square matrices A and B of the same dimensions.*
2. $\operatorname{tr}(\lambda A) = \lambda \operatorname{tr} A$ *for all square matrices A and all scalars λ.*
3. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ *for all square matrices A and B of the same dimensions.*
4. $\operatorname{tr}(A^\intercal) = \operatorname{tr} A$ *for all square matrices A.*

For simplicity, you can assume that the matrices are 3-by-3 matrices when proving the third property.

More generally, it can be shown that the trace of a product of matrices is invariant under cyclic permutations of the factors:

$$\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB).$$

Eigenvalues and Eigenvectors

In the study of matrices, we are often interested in finding special vectors that are only scaled by the matrix, and not rotated or reflected. Consider a vector v in some vector space V. Let U be the set of all scalar multiples of v:

$$U = \{av : a \in F\} = \text{span}\,(v).$$

We call U a **1-dimensional subspace** of V because it is a subspace of V that is spanned by a single vector.[18] If we have a linear map $T : V \to V$, then we can ask whether T maps the subspace U to itself. In other words, we can ask whether there exists a scalar λ such that

$$T(av) = a\lambda v$$

for all $a \in F$.[19] This equation is important enough that vectors v and scalars λ that satisfy it have their own names:

Definition 80 (Eigenvalues and eigenvectors)**.** Let $T : V \to V$ be a linear map from a vector space V over a field F to itself. A nonzero vector $v \in V$ is called an **eigenvector** of T if there exists a scalar $\lambda \in F$ such that

$$T(v) = \lambda v.$$

In this case, we say that λ is the **eigenvalue** of T corresponding to the eigenvector v.

In this definition, the condition that v is nonzero is necessary because if v were the zero vector, then $T(v) = T(0) = 0$ for all linear maps T, and thus every scalar λ would satisfy the equation $T(v) = \lambda v$.

Question 81. *Let $T : F^2 \to F^2$ be the linear map defined by*

$$T(x, y) = (-y, x).$$

1. *If $F = \mathbb{R}$, find the matrix representation of T with respect to the standard basis of $\mathbb{R}^2$, and find all eigenvalues and eigenvectors of T.*

[18] A subspace of a vector space V is a subset of V that is closed under vector addition and scalar multiplication, and contains the zero vector. A subspace is called ***n*-dimensional** if it can be spanned by n vectors, but cannot be spanned by fewer than n vectors.

[19] Conversely, if there exists a scalar λ such that $T(av) = a\lambda v$ for all $a \in F$, then span (v) is a 1-dimensional subspace of V *invariant* under T.

2. *If $F = \mathbb{C}$, find all eigenvalues and eigenvectors of T.*

There are two theorems that are worth mentioning in this section, but we won't be proving them here. The ommission of the proofs is not due to the difficulty of the proofs, but rather due to the fact that the proofs require some additional lemmas and definitions that we haven't covered yet.

Theorem 82 (Linear independence of eigenvectors)**.** *Let $T : V \to V$ be a linear map from a vector space V over a field F to itself. Suppose λ_1, λ_2, ..., λ_k are distinct eigenvalues of T, and v_1, v_2, ..., v_k are corresponding eigenvectors. Then, the set $\{v_1, v_2, \ldots, v_k\}$ is linearly independent.*

The corollary below states that a linear map has at most n distinct eigenvalues, where n is the dimension of the vector space:

Theorem 83. *Let $T : V \to V$ be a linear map from a vector space V over a field F to itself. If V has dimension n, then T has at most n distinct eigenvalues.*

The implications of these theorems are profound. For example, if a linear map $T : V \to V$ has n distinct eigenvalues, where n is the dimension of V, then T has a basis of eigenvectors, and thus T can be represented by a diagonal matrix with respect to that basis. This process of finding a basis of eigenvectors and representing a linear map by a diagonal matrix is called diagonalization, and it is an important technique in linear algebra.

Dual Spaces and Inner Product Spaces

Linear maps into the field F are of incredible importance, and they have their own name:

Definition 84. Let V be a vector space over a field F. A **linear functional** on V is a linear map from V to F.

Example 85. We can define $T : \mathbb{R}^3 \to \mathbb{R}$ by $T(x, y, z) = 4x - 5y + 2z$. Then, T is a linear functional on $\mathbb{R}^3$.

The set of all linear functionals on a vector space V is called the dual space of V:

Definition 86. Let V be a vector space over a field F. The **dual space** of V, denoted V^*, is the set of all linear functionals on V:

$$V^* = \{f : V \to F \mid f \text{ is a linear functional}\}.$$

It can be shown that the dual space V^* of a vector space V has the same dimension as V, and if $\{v_1, v_2, \ldots, v_n\}$ is a basis of V, then there exists a (unique) basis $\{f_1, f_2, \ldots, f_n\}$ of V^* such that $f_i(v_j) = \delta_{ij}$ for all i and j. The basis $\{f_1, f_2, \ldots, f_n\}$ of V^* is called the dual basis of the basis $\{v_1, v_2, \ldots, v_n\}$ of V. We will get plenty of practice working with dual spaces and dual bases in the next chapter, so we won't be doing much more with them here. However, we will be introducing the notion of an inner product space.

To motivate the concept of inner product, we start by considering vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$. The length of a vector v in $\mathbb{R}^2$ or $\mathbb{R}^3$ is called the **norm** of v, and is denoted $\|v\|$. For $v \in \mathbb{R}^2$, we have

$$\|v\| = \sqrt{v_1^2 + v_2^2},$$

and for $v \in \mathbb{R}^3$, we have

$$\|v\| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

The pattern in these formulas is clear: the norm of a vector v in $\mathbb{R}^n$ is given by

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

The norm is not linear in the sense that $\|v + w\| \neq \|v\| + \|w\|$ in general. To get a linear function that gives us the norm, we can instead consider the square of the norm,

Definition 87 (Dot product)**.** Let v and w be vectors in $\mathbb{R}^n$. The **dot product** of v and w, denoted $v \cdot w$, is defined as

$$v \cdot w = v_1 w_1 + v_2 w_2 + \cdots + v_n w_n,$$

where v_i and w_i are the i-th entries of v and w, respectively.

Obviously, $v \cdot v = \|v\|^2$ for all $v \in \mathbb{R}^n$. The dot product on $\mathbb{R}^n$ has the following properties:

1. $v \cdot v \geqslant 0$ for all $v \in \mathbb{R}^n$,
2. $v \cdot v = 0$ if and only if $v = 0$,
3. $v \cdot w = w \cdot v$ for all $v, w \in \mathbb{R}^n$,
4. for a fixed $w \in \mathbb{R}^n$, the function $v \mapsto v \cdot w$ is linear in v.

Question 88. *Prove the properties of the dot product listed above. For property (4), use the function $f : \mathbb{R}^n \to \mathbb{R}$ defined by $f(v) = v \cdot w$ for a fixed $w \in \mathbb{R}^n$, and show that f is a linear functional on $\mathbb{R}^n$, i.e., show that f satisfies the following two properties:*

1. *$f(v + u) = f(v) + f(u)$ for all $v, u \in \mathbb{R}^n$,*
2. *$f(\lambda v) = \lambda f(v)$ for all $v \in \mathbb{R}^n$ and all scalars λ.*

An inner product is a generalization of the dot product to arbitrary vector spaces. This means that we must also consider complex vector spaces, and thus we need to modify the properties of the dot product to accommodate complex conjugation.

Recall that if $z = a + bi$ is a complex number, where a and b are real numbers, then

1. the modulus of z is given by $|z| = \sqrt{a^2 + b^2}$,
2. the complex conjugate of z is given by $z^* = a - bi$, and
3. $|z|^2 = zz^*$.

For vector $v = \begin{pmatrix} v_1 & \dots & v_n \end{pmatrix} \in \mathbb{C}^n$, we define the norm of v by

$$\|v\| = \sqrt{|v_1|^2 + |v_2|^2 + \cdots + |v_n|^2},$$

where the absolute values are necessary to ensure that the norm is a nonnegative real number. Note that

$$\|v\|^2 = v_1 v_1^* + v_2 v_2^* + \cdots + v_n v_n^*.$$

We want to think of $\|v\|^2$ as the inner product of v with itself, as we did with the dot product. The equation above suggests that we define the inner product of two vectors v and w in $\mathbb{C}^n$ by

$$w_1 v_1^* + w_2 v_2^* + \cdots + w_n v_n^*,$$

where $w = \begin{pmatrix} w_1 & \dots & w_n \end{pmatrix}$, $v = \begin{pmatrix} v_1 & \dots & v_n \end{pmatrix} \in \mathbb{C}^n$. Taking the complex conjugate of this expression gives us

$$w_1^* v_1 + w_2^* v_2 + \cdots + w_n^* v_n,$$

which is the inner product of v and w in the opposite order. This motivates the following definition of an inner product:

Definition 89 (Inner product)**.** Let V be a vector space over a field F. An **inner product** on V is a function $\langle\cdot,\cdot\rangle : V \times V \to F$ that satisfies the following properties for all $u, v, w \in V$ and all scalars λ:

1. $\langle v, v\rangle \geqslant 0$, (positivity)
2. $\langle v, v\rangle = 0$ if and only if $v = 0$, (definiteness)
3. $\langle u + v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$, (additivity in the first argument)
4. $\langle \lambda v, w\rangle = \lambda\langle v, w\rangle$, (homogeneity in the first argument)
5. $\langle v, w\rangle = \langle w, v\rangle^*$. (conjugate symmetry)

If the field we are working with is $\mathbb{R}$, then the complex conjugate vanishes in the above definition. The inner product of two vectors v and w is often denoted by $\langle v, w\rangle$, but it can also be denoted by (v, w) (we will use this notation to introduce the concept in the next chapter). Equipping a vector space V with an inner product yields a rich structure:

Definition 90 (Inner product space)**.** An **inner product space** is a vector space V over a field F equipped with an inner product $\langle\cdot,\cdot\rangle : V \times V \to F$.

Rigorously speaking, there are several different kinds of inner product, and simply stating that a vector space is equipped with an inner product does not necessarily tell us what the inner product is. However, in practice, the **Euclidean inner product** is taken to be the default inner product, defined by

$$\langle v, w\rangle = v_1 w_1^* + v_2 w_2^* + \cdots + v_n w_n^*$$

for $v = \begin{pmatrix} v_1 & \dots & v_n \end{pmatrix}$ and $w = \begin{pmatrix} w_1 & \dots & w_n \end{pmatrix} \in F^n$. This is the same inner product as the one we defined for $\mathbb{C}^n$ above, and it reduces to the dot product when $F = \mathbb{R}$. The following properties of the inner product are worth noting:

1. for each fixed $w \in V$, the function $v \mapsto \langle v, w\rangle$ is a linear functional on V,
2. $\langle 0, u\rangle = 0$ for all $u \in V$,
3. $\langle u, 0\rangle = 0$ for all $u \in V$,

4. $\langle u, v+w \rangle = \langle u, v \rangle + \langle u, w \rangle$ for all $u, v, w \in V$, and

5. $\langle v, \lambda w \rangle = \lambda^* \langle v, w \rangle$ for all $v, w \in V$ and all scalars $\lambda \in F$.

Question 91. *Prove the properties of the inner product listed above.*

Although we used norms to motivate the definition of an inner product, it is important to note that not all norms arise from inner products.

Definition 92 (Norm)**.** Let V be a vector space over a field F. For $v \in V$, the **norm** of v, denoted $\|v\|$, is defined by

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

The norm of a vector v in an inner product space V has the following properties:

1. $\|v\| = 0$ if and only if $v = 0$ and

2. $\|\lambda v\| = |\lambda| \|v\|$ for all $v \in V$ and all scalars λ.

Proof. 1. The desired result holds because $\|v\| = 0$ if and only if $\langle v, v \rangle = 0$, which is true if and only if $v = 0$ by the definiteness property of the inner product.

2. Suppose $\lambda \in F$. Then,

$$\|\lambda v\| = \sqrt{\langle \lambda v, \lambda v \rangle} = \lambda \sqrt{\langle v, \lambda v \rangle} = \lambda \lambda^* \sqrt{\langle v, v \rangle} = |\lambda| \sqrt{\langle v, v \rangle} = |\lambda| \|v\|.$$

□

To start concluding this chapter, we come to a crucial definition.

Definition 93 (Orthogonality)**.** Let V be an inner product space. Two vectors v and w in V are said to be **orthogonal** if $\langle v, w \rangle = 0$.

In the defintion above, the order of the vectors v and w does not matter because of the conjugate symmetry property of the inner product. Orthogonality is a generalization of the notion of perpendicularity in Euclidean geometry, and it comes with a couple of somewhat trivial properties:

1. the zero vector is orthogonal to every vector in V, and

2. the zero vector is the only vector in V that is orthogonal to itself.

Both of these properties follow from the definiteness property of the inner product, which states that $\langle v, v \rangle = 0$ if and only if $v = 0$.

There is so much more to say about linear algebra. The topics that we've covered in this section have only scratched the surface of the subject. We took what we needed from linear algebra, but even then we've not taken enough.

Resources

1. *Linear Algebra Done Right* (3rd ed.) by Sheldon Axler.

   Axler's book has been the main resource for this section. It is a great book that covers linear algebra in an elegant and intuitive way. It is also a great book for self-study, as it contains many exercises that are designed to help the reader develop a deep understanding of the material.

   Another book that was heavily used in this section is the following:

2. *Linear Algebra* (4th ed.) by Friedberg, Insel, and Spence.

   This book is more comprehensive than Axler's book, and it covers a wider range of topics in linear algebra. It is an excellent reference for those who have already studied linear algebra and want to learn more about the subject.

   In order to ground the material in the real world, I also consulted the following book:

3. *Linear Algebra and Its Applications* (4th ed.) by Gilbert Strang.

   Strang's book is more applied than the other two books, and it focuses on the applications of linear algebra in various fields such as engineering, computer science, and physics. It takes a different perspective on linear algebra than the one we've taken here, so if you're interested in seeing how linear algebra is used in practice, I highly recommend this book.

Quantum Mechanics

Contents

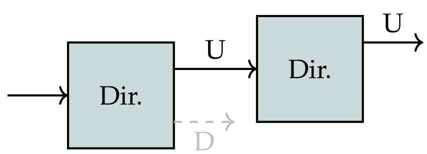
Quantum Interference

The following is taken (almost directly) from David Z. Albert's *Quantum Mechanics and Experience*. I still strongly recommend reading the original text, as it provides a lot of intuition for the concept of superposition.[20]

[20] Even further, I would **strongly** recommend that you watch the following video: youtu.be/lZ3bPUKo5zc?si=ZS7sjvBKoGGvToVM, which is a lecture given by (then) Prof. Allan Adams based on the same text. In fact, I would recommend the entire lecture series on MIT OCW: ocw.mit.edu/courses/8-04-quantum-physics-i-spring-2013/, as I believe it is one of the best resources for learning quantum mechanics.

Here's an unsettling story about something that can happen to electrons. The story is true, and the experiments described here have all been performed. The story concerns two particular properties of electrons: the precise physical definitios of these properties are irrelevant, here we take them to be some sort of "direction," either up or down, and "color," white or black. It happens to be an empirical fact that the electron can assume only one of the two possible values for each property. That is, we have never seen an electron in a state that is neither up nor down, or neither white nor black. We know this because we have devices that are able to measure these two properties with great accuracy.

Such an apparatus receives electrons through one aperture, then separates them into two beams, one for each possible value of the property, as seen in Fig. 14. Measurements with these devices are repeatable, in the sense that if we take the output beam corresponding to a particular value of the property and send it through the same apparatus again, we get the same value of the property with 100% certainty. For example, if we take the up beam from the direction apparatus and send it through the same apparatus again, we get only up electrons coming out:



Similarly, if we take the down beam from the direction apparatus and send it through the same apparatus again, we get only down electrons coming out. The same is true for the color apparatus.

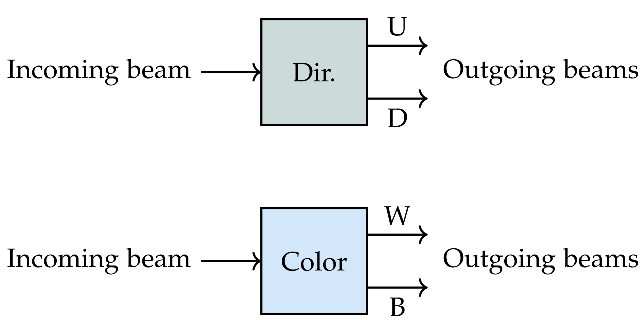
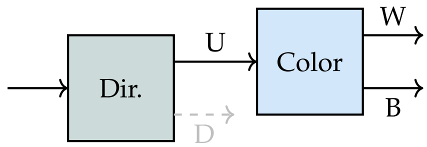


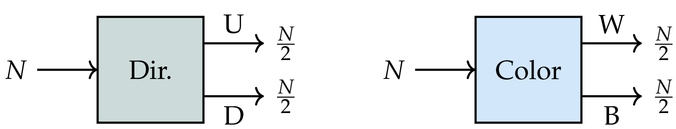
Figure 14: Showing the two devices which take in a beam of electrons and split them up into their property value. The label "Dir." stands for direction, and the label "Color" stands for color (who would've guessed). The first device splits the incoming beam into two outgoing beams, every electron in the upper beam has the direction property value of up (U), and every electron in the lower beam has the direction property value of down (D). The second device also splits the incoming beam into two outgoing beams, where every electron in the upper beam has the color property value of white (W), and every electron in the lower beam has the color property value of black (B).

Now, suppose that we want to investigate the possibility that the direction and color proprties of electrons are related in some way. For example, maybe all up electrons are white, and all down electrons are black? One way to look for such a relation might be to check for *correlations* between the values of the two properties. Using our devices, we can do this by taking the up beam from the direction apparatus and sending it through the color apparatus:

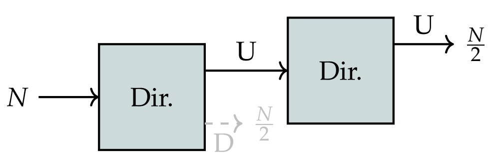


What we find is that the up beam from the direction apparatus is split into two beams of equal intensity by the color apparatus, one for white electrons and one for black electrons. If we were to repeat this experiment, but this time take the down beam from the direction apparatus and send it through the color apparatus, we would find the same thing: the down beam is also split into two beams of equal intensity by the color apparatus, one for white electrons and one for black electrons. Thus, so far there are no correlations between the direction and color properties of electrons, since both the up and down beams are split into equal parts by the color apparatus.

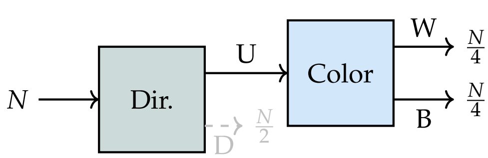
We want to make our observations more quantitave so we start tracking the number of electrons that go into/out of each apparatus. If the incoming beam into an apparatus contains $N \gg 1$ electrons, we find that the outgoing beams each contain $N/2$ electrons:



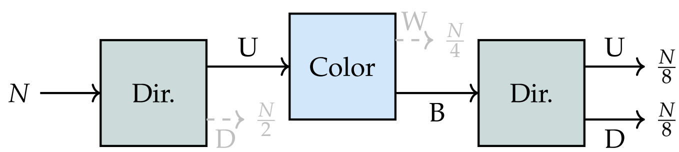
As before, we can repeat measurements to confirm that the number of electrons remains consistent:



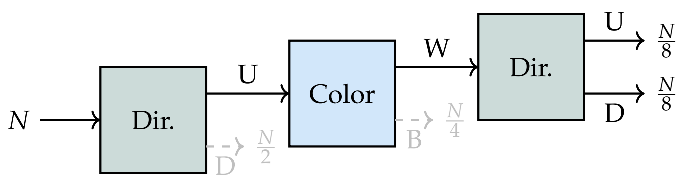
Now looking back at our experiment where we send the up beam from the direction apparatus through the color apparatus, we can also confirm that the number of electrons in each outgoing beam is consistent:



This clearly tells that knowing the direction property of an electron does not give us any information about its color property, and vice versa. As a sanity check, we can keep this same configuration and add another direction apparatus at the end. We know that we send $N/2$ electrons in the up beam into the color apparatus, and we get $N/4$ electrons out of each outgoing beam from the color apparatus. What one naively expects is that if we send the black beam from the color apparatus into the direction apparatus, we should get $N/4$ electrons in the up beam, meaning that we have $N/4$ electrons that are both up and black. What we see is as follows:



We've lost all information that we had about the direction property of the electrons, since we get $N/8$ electrons in the up beam and $N/8$ electrons in the down beam. This is a very strange result, since we started with $N/2$ electrons that were all up, and after sending them through the color apparatus, we end up with only $N/8$ electrons that are up. We test this again with the white beam from the color apparatus, and we get the same result:

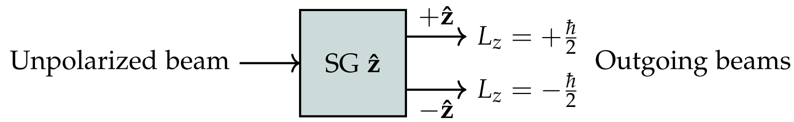


So it has nothing to with which color beam we chose, all it has to do with is the fact that we know which beam is being sent into the direction apparatus.

The Stern-Gerlach Experiment

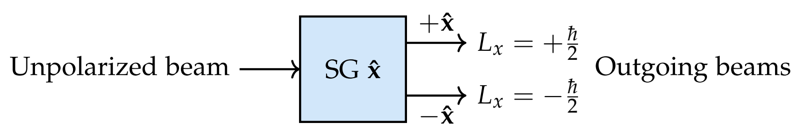
We take the experiments from before but make them a bit more physical in the Stern-Gerlach experiment, which is a real experiment that was performed in the early 20th century. In this experiment, a beam of silver atoms is sent through an inhomogeneous magnetic field, which splits the beam into two beams of equal intensity, one for each possible value of the spin property of the silver atoms. The spin property is a quantum property that can take on two possible values, which we can call up and down.

As before, we have a device which now takes an unpolarized beam of silver atoms and splits it into two beams of equal intensity, one for each possible value of the spin property. This initial device only considers the spin property of the silver atoms in the z-direction,



For reasons that we won't be going into here, the Stern-Gerlach experiment tells us that the silver atoms can only have two possible values of L_z, the angular momentum in the z-direction, namely $L_z = +\hbar/2$ and $L_z = -\hbar/2$. As seen above, we designate the "quantum state" of the silver atoms in the $L_z = +\hbar/2$ beam as $+\hat{\mathbf{z}}$ and the quantum state of the silver atoms in the $L_z = -\hbar/2$ beam as $-\hat{\mathbf{z}}$.

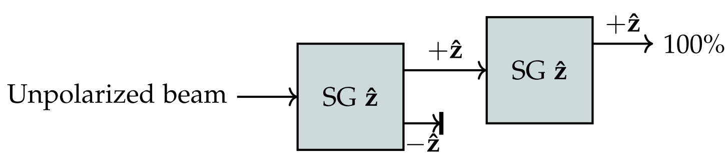
We can consider another Stern-Gerlach apparatus that splits the beam of silver atoms based on their spin property in the x-direction, as shown below:



This apparatus tells us that the silver atoms can also have two possible values of L_x, the angular momentum in the x-direction, namely $L_x = +\hbar/2$ and $L_x = -\hbar/2$. We designate the quantum state of the silver atoms in the $L_x = +\hbar/2$ beam as $+\hat{\mathbf{x}}$ and the quantum state of the silver atoms in the $L_x = -\hbar/2$ beam as $-\hat{\mathbf{x}}$.

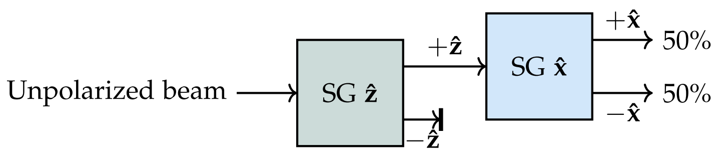
With these two Stern-Gerlach apparatuses in mind, we consider four different thought experiments:

1. **Thought Experiment 1:** As we did with the naive direction and color apparatuses, we want to verify that our measurements are consistent:



   We find that all of the atoms going into the second device come out in the $+\hat{\mathbf{z}}$ beam, consistent!

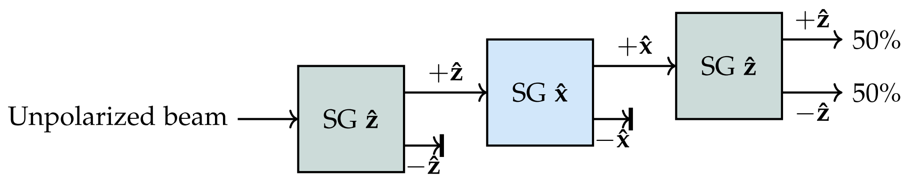
2. **Thought Experiment 2:** What if we chain together two different Stern-Gerlach apparatuses in different orientations? In particular, we direct the $+\hat{\mathbf{z}}$ beam from the first Stern-Gerlach apparatus into a second Stern-Gerlach apparatus that is oriented in the x-direction:



We find that half of the beam comes out in the $+\hat{\mathbf{x}}$ beam and half of the beam comes out in the $-\hat{\mathbf{x}}$ beam. Based on this outcome, we hypothesize that the atoms in the original, unpolarized beam can be divided into four equal groups, each with a different combination of the z and x spin properties:

$$(+\hat{\mathbf{z}}, +\hat{\mathbf{x}}), \quad (+\hat{\mathbf{z}}, -\hat{\mathbf{x}}), \quad (-\hat{\mathbf{z}}, +\hat{\mathbf{x}}), \quad (-\hat{\mathbf{z}}, -\hat{\mathbf{x}}).$$

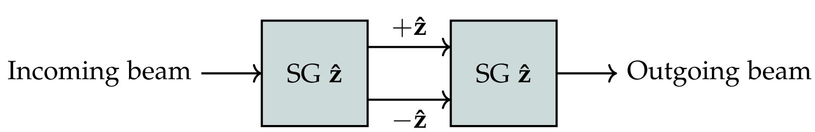
3. **Thought Experiment 3:** We test our hypothesis by adding a third Stern-Gerlach apparatus at the end of the second one, which is oriented in the z-direction:



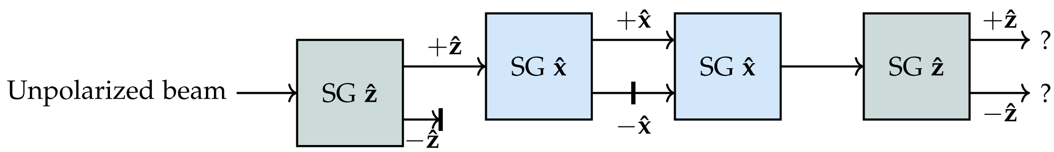
Had we not gone through the previous abstract thought experiments, we would have expected that all of the atoms in the $+\hat{\mathbf{x}}$ beam from the second Stern-Gerlach apparatus would come out in the $+\hat{\mathbf{z}}$ beam from the third Stern-Gerlach apparatus, since we hypothesized that there are atoms with the combination of properties $(+\hat{\mathbf{z}}, +\hat{\mathbf{x}})$. We therefore observe that the measurement of L_x 'erases' the information about L_z.

Thus, we learn that our hypothesis was misguided, we can either separate the beam into its z spin components or its x spin components, but we cannot separate the beam into its (z, x) spin components.

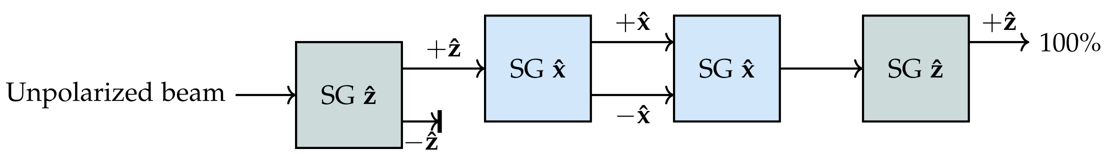
4. **Thought Experiment 4:** We introduce a new Stern-Gerlach apparatus that *combines* two beams into one beam, as shown below:



We then have the choice of whether we block one of the beams in the middle or not. If we block the $-\hat{\mathbf{z}}$ beam in the middle, then all of the atoms that go into the second Stern-Gerlach apparatus come out in the $+\hat{\mathbf{z}}$ beam, consistent with our previous observations. However, if we do not block either beam in the middle, then we find an unpolarized beam coming out of the second Stern-Gerlach apparatus. We insert such a device in the middle of the previous thought experiment,



where we optionally insert a block in the middle of the second Stern-Gerlach apparatus to block the $-\hat{\mathbf{x}}$ beam. If we do proceed with blocking the $-\hat{\mathbf{x}}$ beam, then the experimental setup is effectively the same as the one in Thought Experiment 3. If we instead block the $+\hat{\mathbf{x}}$ beam, then we again end up with the same experimental setup as in Thought Experiment 3, just with the $+\hat{\mathbf{x}}$ and $-\hat{\mathbf{x}}$ beams swapped. However, if both the $+\hat{\mathbf{x}}$ and $-\hat{\mathbf{x}}$ beams are allowed to go through, then we find that the final beam comes out entirely in the $+\hat{\mathbf{z}}$ beam,



These four thought experiments are the basis for the mathematical framework of quantum mechanics, which we will be developing in the rest of this chapter. The key takeaway from these thought experiments is that something we refer to as 'quantum interference' occurs, which is the phenomenon that the measurement of one property of a quantum system can affect the measurement of another property of the same quantum system, even if the two properties are not directly related to each other. In particular, we see that the measurement of L_x can affect the measurement of L_z, and vice versa. This is a fundamental feature of quantum mechanics that distinguishes it from classical mechanics, where measurements of different properties do not affect each other in this way.

This also lays the foundation for the concept of superposition, which is the idea that a quantum system can be in a combination of different states at the same time. Superposition then enables entanglement, the phenomenon where the state of one quantum system can be correlated with the state of another quantum system, even if the two systems are separated by a large distance. We've got a long way to go before we can get to superposition, and an even longer way to go before we can get to entanglement, but these thought experiments are the first step in that direction. To get us started on this journey, we first learn the language of quantum mechanics.

Finite-Dimensional Hilbert Spaces

Definition 94. A **(finite dimensional) Hilbert space** is a vector space $\mathcal{H}$ equipped with an inner product.

One may look at the definition on the left and question what the difference is between a Hilbert space and an inner product space. The difference is that a Hilbert space is an inner product space that is also complete, meaning that every Cauchy sequence in the space converges to a limit that is also in the space. In the finite-dimensional case, every inner product space is automatically complete, so there is no difference between the two concepts. However, in the infinite-dimensional case, there are inner product spaces that are not complete, and thus not Hilbert spaces.

In the infinite-dimensional case, the inner product must satisfy some additional properties, but in the finite-dimensional case, we can just take it to be the usual inner product.

Kets, Bras, and Operators

Following the conventions invented by Paul Dirac, we denote the elements of a Hilbert space $\mathcal{H}$ by **kets**:

$$|\psi\rangle\,, |\phi\rangle\,, |\chi\rangle \in \mathcal{H}.$$

By definition, a ket is just a vector in the Hilbert space, so we can add kets together and multiply them by scalars: for $|\psi\rangle$, $|\phi\rangle \in \mathcal{H}$ and $\lambda \in \mathbb{C}$,

$$|\psi\rangle + \lambda\,|\phi\rangle \in \mathcal{H}.$$

Question 95. *Using this new notation, list the properties of a vector space, seen in Defn. 45. Include the properties of an inner product space as well.*

Note that since the scalars we use here are complex numbers, we are dealing with a complex vector space.

We write covectors[21] as **bras**:

[21] Understood as linear maps from $\mathcal{H}$ to $\mathbb{C}$

$$\langle\omega|\,, \langle\xi|\,, \langle\gamma| \in \mathcal{H}^*,$$

where $\mathcal{H}^*$ is the dual space of $\mathcal{H}$, *i.e.*, the space of covectors. The **contraction** of a bra and a ket, *i.e.*, evaluating the bra on the ket, yields a complex number:

$$\langle\omega|\chi\rangle \equiv \langle\omega|\,(|\chi\rangle) \in \mathbb{C},$$

while also giving rise to the names bra and ket.

Since bras are linear maps from $\mathcal{H}$ to $\mathbb{C}$, they distribute over addition and scalar multiplication: for $\langle\omega|$, $\langle\xi| \in \mathcal{H}^*$, $|\chi\rangle \in \mathcal{H}$, and $\lambda \in \mathbb{C}$,

$$\langle\omega|\,(|\chi\rangle + \lambda\,|\xi\rangle) = \langle\omega|\chi\rangle + \lambda\,\langle\omega|\xi\rangle\,.$$

Likewise, addition and scalar multiplication of bras is defined by linearity: for $\langle\omega|$, $\langle\xi| \in \mathcal{H}^*$ and $\lambda \in \mathbb{C}$,

$$(\langle\omega| + \lambda\,\langle\xi|)(|\chi\rangle) = \langle\omega|\chi\rangle + \lambda\,\langle\xi|\chi\rangle\,.$$

Linear maps from $\mathcal{H}$ to itself are also important,

Definition 96. A **(linear) operator** on $\mathcal{H}$ is a linear map from $\mathcal{H}$ to itself. The result of applying an operator $\mathbf{O}$ to a state $|\psi\rangle$ is written as

$$\mathbf{O}\,|\psi\rangle \equiv \mathbf{O}(|\psi\rangle) \in \mathcal{H},$$

i.e., operators act on the left of kets.

Operators likewise act on the right of bras:

$$\langle\omega|\,\mathbf{O} \quad \text{is defined by} \quad (\langle\omega|\,\mathbf{O})(|\chi\rangle) = \langle\omega|\,(\mathbf{O}\,|\chi\rangle) = \langle\omega|\mathbf{O}|\chi\rangle\,,$$

where here the notation is chosen in such a way that is naturally associative.[22] We will see that such associativity is built into Dirac notation quite naturally, and it will be very useful for simplifying expressions involving operators, bras, and kets.

[22] We proceed by denoting operators with boldface letters, and reserving the use of regular letters for scalars. A common notation for operators is to use hats, *e.g.*, $\hat{O}$, but we will avoid this notation since it can be easily confused with the use of hats for unit vectors.

Example 97. Examples of such operators include the identity operator $\mathbf{I}$, which acts as the identity on all states $|\psi\rangle \in \mathcal{H}$, $\mathbf{I}\,|\psi\rangle = |\psi\rangle$, and the zero operator $\mathbf{0}$, which acts as the zero map on all states, $\mathbf{0}\,|\psi\rangle = 0$.

Definition 98. Two operators $\mathbf{X}$ and $\mathbf{Y}$ are **equal** if and only if, for all $\psi \in \mathcal{H}$,

$$\mathbf{X}\,|\psi\rangle = \mathbf{Y}\,|\psi\rangle\,.$$

As usual, operators can be added together and multiplied by scalars, and these operations are defined by linearity: for $\mathbf{X}$, $\mathbf{Y}$ operators on $\mathcal{H}$, $|\psi\rangle \in \mathcal{H}$, and $\lambda \in \mathbb{C}$,

$$(\mathbf{X} + \lambda\mathbf{Y})\,|\psi\rangle = \mathbf{X}\,|\psi\rangle + \lambda\mathbf{Y}\,|\psi\rangle = \mathbf{X}(|\psi\rangle) + \lambda\mathbf{Y}(|\psi\rangle).$$

Furthermore, operators can also be multiplied together, and this operation is defined by composition: for $\mathbf{X}$, $\mathbf{Y}$ operators on $\mathcal{H}$ and $|\psi\rangle \in \mathcal{H}$,

$$(\mathbf{XY})\,|\psi\rangle = \mathbf{X}(\mathbf{Y}\,|\psi\rangle) = \mathbf{X}(\mathbf{Y}(|\psi\rangle)).$$

Operator multiplication does not generally commute, $\mathbf{XY} \neq \mathbf{YX}$, so we have the following definition:

Question 99.

1. *Show that both the identity operator* $\mathbf{I}$ *and the zero operator* $\mathbf{0}$ *commute with all operators on* $\mathcal{H}$*, i.e., for any operator* $\mathbf{X}$ *on* $\mathcal{H}$*,* $[\mathbf{I}, \mathbf{X}] = [\mathbf{0}, \mathbf{X}] = 0$.
2. *Show that the commutator satisfies the following properties:*
   (a) $[\mathbf{X}, \mathbf{X}] = 0$,
   (b) $[\mathbf{X} + \lambda\mathbf{Y}, \mathbf{Z}] = [\mathbf{X}, \mathbf{Z}] + \lambda[\mathbf{Y}, \mathbf{Z}]$,
   (c) $[\mathbf{X}, \lambda\mathbf{Y}] = -\lambda[\mathbf{Y}, \mathbf{X}]$,
   (d) $[\mathbf{X}, [\mathbf{Y}, \mathbf{Z}]] + [\mathbf{Y}, [\mathbf{Z}, \mathbf{X}]] + [\mathbf{Z}, [\mathbf{X}, \mathbf{Y}]] = 0$.

Definition 100. The **commutator** of two operators $\mathbf{X}$ and $\mathbf{Y}$ is defined as

$$[\mathbf{X}, \mathbf{Y}] \equiv \mathbf{XY} - \mathbf{YX}.$$

Definition 101. The **outer product** of a ket $|\psi\rangle$ and a bra $\langle\phi|$ is the operator $|\psi\rangle\langle\phi|$ defined by

$$(|\psi\rangle\langle\phi|)(|\chi\rangle) = \langle\phi|\chi\rangle\, |\psi\rangle\, .$$

A nonzero operator that can be written as an outer product is called a **rank-1 operator**. More generally,

Definition 102. An operator $\mathbf{X}$ is a **rank-n operator** if it can be written as a sum of n rank-1 operators, but cannot be written as a sum of fewer than n rank-1 operators:

$$\mathbf{X} = \sum_{i=1}^{n} |\psi_i\rangle\langle\phi_i|\, ,$$

where $|\psi_i\rangle$, $|\phi_i\rangle \in \mathcal{H}$ for $i = 1, \ldots, n$.

One can show that if $\mathcal{H}$ is N-dimensional, then any operator on $\mathcal{H}$ is a rank-n operator for some $n \leqslant N$.

The Inner Product and the Conjugate Map

Thus far, all we have invoked are the properties of a complex vector space, but we have not yet used the inner product structure of the Hilbert space.

Definition 103. The **inner product** on a Hilbert space $\mathcal{H}$ is a map $\mathcal{H} \times \mathcal{H} \mapsto \mathbb{C}$ which takes a pair of kets as arguments and produces a complex number that satisfies the following properties for all $|\psi\rangle$, $|\chi\rangle$, $|\phi\rangle \in \mathcal{H}$ and $\lambda \in \mathbb{C}$:

1. $(|\psi\rangle\, , |\chi\rangle)^* = (|\chi\rangle\, , |\psi\rangle)$ (conjugate symmetry)
2. $(|\psi\rangle\, , |\chi\rangle + \lambda\, |\phi\rangle) = (|\psi\rangle\, , |\chi\rangle) + \lambda(|\psi\rangle\, , |\phi\rangle)$ (linearity in the second argument)

3. $(|\psi\rangle, |\psi\rangle) \geq 0$ with saturation if and only if $|\psi\rangle = 0$. (positive-definiteness)

Properties (1) and (2) together imply that the inner product is antilinear in the first argument, *i.e.*, for all $|\psi\rangle$, $|\chi\rangle$, $|\phi\rangle \in \mathcal{H}$ and $\lambda \in \mathbb{C}$,

$$(|\psi\rangle + \lambda |\chi\rangle, |\phi\rangle) = (|\psi\rangle, |\phi\rangle) + \lambda^*(|\chi\rangle, |\phi\rangle).$$

Definition 104. The **norm** of a ket $|\psi\rangle$ is defined as

$$\| |\psi\rangle \| \equiv \sqrt{(|\psi\rangle, |\psi\rangle)}.$$

The definition of the inner product ensures that the norm of any ket is a non-negative real number, and that the norm of a ket is zero if and only if the ket is the zero ket. It also allows us to reinterpret the bra:

Definition 105. The **conjugate** of a ket $|\psi\rangle$ is the bra $\langle\psi|$,

$$|\psi\rangle^\dagger \equiv \langle\psi|,$$

defined by

$$|\psi\rangle^\dagger (|\chi\rangle) \equiv (|\psi\rangle, |\chi\rangle).$$

This is a linear map from $\mathcal{H}$ to $\mathbb{C}$, since the inner product is linear in its second argument. Likewise, since the inner product is antilinear in its first argument, the conjugate map is antilinear: for all $|\psi\rangle$, $|\chi\rangle$, $|\phi\rangle \in \mathcal{H}$ and $\lambda \in \mathbb{C}$,

$$(|\psi\rangle + \lambda |\chi\rangle)^\dagger = |\psi\rangle^\dagger + \lambda^* |\chi\rangle^\dagger = \langle\psi| + \lambda^* \langle\chi|.$$

Thus, $\dagger : \mathcal{H} \to \mathcal{H}^*$ is an antilinear map from kets to bras.[23]

By the footnote on the right, and after showing that the map $\dagger : \mathcal{H} \to \mathcal{H}^*$ is injective, we can conclude that the conjugate map is a bijection between $\mathcal{H}$ and $\mathcal{H}^*$, and thus it is an *invertible* map. Abusing notation, we write the inverse of the conjugate map from bras to kets $\dagger^{-1} = \dagger : \mathcal{H}^* \to \mathcal{H}$ using the same symbol, such that

$$\langle\omega|^\dagger = |\omega\rangle.$$

Thus, $\left(|\psi\rangle^\dagger\right)^\dagger = |\psi\rangle$ and there is a direct, antilinear correspondence between kets and bras.

[23] The Riesz representation theorem states that for every bra $\langle\omega| \in \mathcal{H}^*$, there exists a unique ket $|\omega\rangle \in \mathcal{H}$ such that $\langle\omega| = |\omega\rangle^\dagger$. In other words, the map $\dagger : \mathcal{H} \to \mathcal{H}^*$ is surjective.

Since

$$(|\psi\rangle, |\chi\rangle) = |\psi\rangle^\dagger |\chi\rangle = \langle\psi|\chi\rangle$$

by the definition of $\dagger : \mathcal{H} \to \mathcal{H}^*$, we can write the inner product $(|\psi\rangle, |\chi\rangle)$ equivalently as

$$\langle\psi|\chi\rangle = \langle\chi|\psi\rangle^* .$$

This is the notation we will use for the inner product from now on, where the antilinearity of the inner product in its first argument manifests itself as the antilinearity of the map $\dagger : |\psi\rangle \to \langle\psi|$.

Question 106.

1. *For two kets $|\psi\rangle, |\chi\rangle \in \mathcal{H}$, with $\langle\psi|\chi\rangle = 0$, i.e., assuming they are orthogonal, show that the norm of their sum is given by $\||\psi\rangle + |\chi\rangle\|^2 = \||\psi\rangle\|^2 + \||\chi\rangle\|^2$.*
2. *For all $|\psi\rangle, |\chi\rangle \in \mathcal{H}$, show that $\||\psi\rangle + |\chi\rangle\|^2 + \||\psi\rangle - |\chi\rangle\|^2 = 2\||\psi\rangle\|^2 + 2\||\chi\rangle\|^2$.*

Self-adjoint Operators and Orthonormal Bases

Definition 107. The **adjoint** of an operator $\mathbf{X}$ is the operator $\mathbf{X}^\dagger$ defined by

$$\left(\mathbf{X}^\dagger\right)(|\psi\rangle) = \left(|\psi\rangle^\dagger \mathbf{X}\right)^\dagger = (\langle\psi| \mathbf{X})^\dagger.$$

Definition 108. An operator $\mathbf{X}$ is **self-adjoint** if it is equal to its adjoint, $\mathbf{X} = \mathbf{X}^\dagger$, **unitary** if its adjoint is equal to its inverse, $\mathbf{X}^\dagger = \mathbf{X}^{-1}$, and **normal** if it commutes with its adjoint, $\left[\mathbf{X}, \mathbf{X}^\dagger\right] = 0$.

Equivalently, an operator $\mathbf{X}$ is **unitary** if $\mathbf{X}\mathbf{X}^\dagger = \mathbf{X}^\dagger\mathbf{X} = \mathbf{I}$, and **normal** if $\mathbf{X}\mathbf{X}^\dagger = \mathbf{X}^\dagger\mathbf{X}$. Note that all self-adjoint and unitary operators are normal, but not all normal operators are self-adjoint or unitary.

Question 109. *Using the definitions of the adjoint and the properties of operators, show that for operators $\mathbf{X}$ and $\mathbf{Y}$ on $\mathcal{H}$,*

1. $\langle\psi|\mathbf{X}|\chi\rangle^* = \langle\chi|\mathbf{X}^\dagger|\psi\rangle$, *and*
2. $(\mathbf{X}\mathbf{Y})^\dagger = \mathbf{Y}^\dagger\mathbf{X}^\dagger$.

Definition 110. A ket $|\psi\rangle$ is **normalized** if its norm is 1, $\||\psi\rangle\| = 1$, and two kets $|\psi\rangle$ and $|\chi\rangle$ are **orthogonal** if their inner product is zero, $\langle\psi|\chi\rangle = 0$. A set of kets $\{|1\rangle, |2\rangle, \ldots, |N\rangle\}$, is **orthonormal** if each ket in the set is normalized and any two distinct kets in the set are orthogonal, *i.e.*,

$$\langle i|j\rangle = \delta_{ij}, \tag{11}$$

where δ_{ij} is the Kronecker delta, which is 1 if $i = j$ and 0 otherwise.

Any nonzero ket can be normalized by dividing it by its norm:

$$|\psi\rangle' = \frac{1}{\||\psi\rangle\|} |\psi\rangle .$$

Definition 111. An orthonormal set of kets $\{|1\rangle, |2\rangle, \ldots, |N\rangle\}$ is an **orthonormal basis** if any ket $|\psi\rangle \in \mathcal{H}$ can be written as a linear combination of the kets in the set, *i.e.*, for all $|\psi\rangle \in \mathcal{H}$, there exist complex numbers $\psi_1, \psi_2, \ldots, \psi_N$ such that

$$|\psi\rangle = \sum_{i=1}^{N} \psi_i |i\rangle , \tag{12}$$

where ψ_i are the **components** of $|\psi\rangle$ in the basis $\{|1\rangle, |2\rangle, \ldots, |N\rangle\}$.

When $\mathcal{H}$ is finite dimensional, the size of any orthonormal set is at most the dimension of $\mathcal{H}$, $N \leqslant \dim(\mathcal{H})$. Moreover, any orthonormal set of size equal to the dimension of $\mathcal{H}$ is an orthonormal basis for $\mathcal{H}$. In other words, if $\dim(\mathcal{H}) = N$, then any orthonormal set of N kets is an orthonormal basis for $\mathcal{H}$. Such an orthonormal set is said to be **complete**.

Given an orthonormal basis $\{|1\rangle, |2\rangle, \ldots, |N\rangle\}$ for $\mathcal{H}$, we can extract the components of any ket $|\psi\rangle \in \mathcal{H}$ in the basis by taking the inner product of $|\psi\rangle$ with each ket in the basis and using Eqn. (11):

$$\langle i|\psi\rangle = \sum_{j=1}^{N} \psi_j \langle i|j\rangle = \psi_i.$$

Substituting this into Eqn. (12), we obtain

$$\begin{aligned} |\psi\rangle &= \langle 1|\psi\rangle \cdot |1\rangle + \langle 2|\psi\rangle \cdot |2\rangle + \cdots + \langle N|\psi\rangle \cdot |N\rangle \\ &= |1\rangle\langle 1| \, |\psi\rangle + |2\rangle\langle 2| \, |\psi\rangle + \cdots + |N\rangle\langle N| \, |\psi\rangle \\ &= (|1\rangle\langle 1| + |2\rangle\langle 2| + \cdots + |N\rangle\langle N|) \, |\psi\rangle , \end{aligned}$$

where we first move the scalar factors to the right and then reinterpret them as outer products acting on $|\psi\rangle$. Since this is true for all

$|\psi\rangle \in \mathcal{H}$, we have

$$|1\rangle\langle 1| + |2\rangle\langle 2| + \cdots + |N\rangle\langle N| = \mathbf{I},$$

which is known as the **completeness relation** for the orthonormal basis $\{|1\rangle, |2\rangle, \ldots, |N\rangle\}$.

Definition 112. An orthonormal set is **complete** if and only if it satisfies such a completeness relation.

Question 113. *Let $\{|1\rangle, |2\rangle, \ldots, |k\rangle\}$ be an orthonormal set. Show that*

$$\mathbf{\Pi} = \sum_{i=1}^{k} |i\rangle\langle i|$$

is an orthogonal projector.

Definition 114. A **projector $\mathbf{\Pi}$** is an operator that is idempotent, $\mathbf{\Pi}^2 = \mathbf{\Pi}$. An **orthogonal projector** is a projector that is also self-adjoint, $\mathbf{\Pi}^\dagger = \mathbf{\Pi} = \mathbf{\Pi}^2$.

Definition 115. An **eigenket** of an operator $\mathbf{X}$ is a nonzero ket $|\psi\rangle$ such that

$$\mathbf{X}|\psi\rangle = \lambda|\psi\rangle,$$

where λ is a complex number called the **eigenvalue** corresponding to $|\psi\rangle$.

Proposition 116. *If $\mathbf{X}$ is self-adjoint, then*

1. *its eigenvalues are real, and*
2. *eigenkets corresponding to distinct eigenvalues are orthogonal.*

Proof.

1. Suppose $|\psi\rangle$ is an eigenket of $\mathbf{X}$ with eigenvalue λ, *i.e.*,

$$\mathbf{X}|\psi\rangle = \lambda|\psi\rangle. \tag{13}$$

Taking the adjoint of both sides, we have

$$(\mathbf{X}|\psi\rangle)^\dagger = (\lambda|\psi\rangle)^\dagger \quad \Longrightarrow \quad \langle\psi|\mathbf{X} = \lambda^*\langle\psi|. \tag{14}$$

Multiplying both sides of Eqn. (13) on the left by $\langle\psi|$ says

$$\langle\psi|\mathbf{X}|\psi\rangle = \lambda\langle\psi|\psi\rangle.$$

Similarly, multiplying both sides of Eqn. (14) on the right by $|\psi\rangle$ says

$$\langle\psi|\mathbf{X}|\psi\rangle = \lambda^* \langle\psi|\psi\rangle .$$

That is, $\lambda \langle\psi|\psi\rangle = \lambda^* \langle\psi|\psi\rangle$. Since $|\psi\rangle$ is an eigenket, it is nonzero, so $\langle\psi|\psi\rangle > 0$. Thus, we can divide both sides by $\langle\psi|\psi\rangle$ to conclude that $\lambda = \lambda^*$, so λ is real.

2. Suppose $|\psi_{1,2}\rangle$ are eigenkets of $\mathbf{X}$ with distinct eigenvalues $\lambda_{1,2}$, *i.e.*,

$$\mathbf{X}\,|\psi_1\rangle = \lambda_1\,|\psi_1\rangle\,, \quad \mathbf{X}\,|\psi_2\rangle = \lambda_2\,|\psi_2\rangle\,.$$

Multiplying the first equation on the left by $\langle\psi_2|$ and the second equation on the left by $\langle\psi_1|$ and then taking the adjoint of the second equation, we have

$$\langle\psi_2|\mathbf{X}|\psi_1\rangle = \lambda_1\,\langle\psi_2|\psi_1\rangle\,, \quad \langle\psi_1|\mathbf{X}|\psi_2\rangle = \lambda_2\,\langle\psi_1|\psi_2\rangle\,.$$

Since $\mathbf{X}$ is self-adjoint, $\langle\psi_2|\mathbf{X}|\psi_1\rangle = \langle\psi_1|\mathbf{X}|\psi_2\rangle^*$. Thus, $\lambda_1\,\langle\psi_2|\psi_1\rangle = \lambda_2\,\langle\psi_1|\psi_2\rangle^* = \lambda_2\,\langle\psi_2|\psi_1\rangle$. Since λ_1 and λ_2 are distinct, we must have $\langle\psi_2|\psi_1\rangle = 0$, so $|\psi_1\rangle$ and $|\psi_2\rangle$ are orthogonal.

□

More generally, we have the following the following result,

Theorem 117 (Spectral Theorem)**.** *For an self-adjoint operator* $\mathbf{X}$ *on a finite-dimensional Hilbert space* $\mathcal{H}$*, there exists an orthonormal basis* $\{|1\rangle, |2\rangle, \ldots, |N\rangle\}$ *and real numbers* $\lambda_1, \lambda_2, \ldots, \lambda_N$ *such that*

$$\mathbf{X} = \sum_{i=1}^{N} \lambda_i\,|i\rangle\langle i|\,.$$

Thus, in reference to Quest. 113, the spectral theorem implies that any orthogonal projector takes on the form $\mathbf{\Pi} = \sum_{i=1}^{k} |i\rangle\langle i|$ for some orthonormal set $\{|1\rangle, |2\rangle, \ldots, |k\rangle\}$.

Note that the orthonormal basis produced by the spectral theorem consists of eigenkets of the self-adjoint operator $\mathbf{X}$, and the real numbers $\lambda_1, \lambda_2, \ldots, \lambda_N$ are the corresponding eigenvalues. Thus, a self-adjoint operator has an orthonormal basis of eigenkets.

The spectral theorem has a few important generalizations:

1. The spectral theorem also holds for normal operators, except that the eigenvalues become complex numbers instead of real numbers. Thus, a normal operator with all real eigenvalues is necessarily self-adjoint.

2. The spectral theorem also holds for infinite-dimensional Hilbert spaces, but the statement is more complicated and involves the notion of a *spectral measure*.

3. If $\mathbf{X}$, $\mathbf{Y}$ are self-adjoint operators that commute, $[\mathbf{X}, \mathbf{Y}] = 0$, then there exists an orthonormal basis of kets that are simultaneous eigenkets of both $\mathbf{X}$ and $\mathbf{Y}$:

$$\mathbf{X} = \sum_{i=1}^{N} \lambda_i \left|i\right\rangle\!\left\langle i\right|, \quad \mathbf{Y} = \sum_{i=1}^{N} \mu_i \left|i\right\rangle\!\left\langle i\right|,$$

where

$$\langle i|j\rangle = \delta_{ij}, \quad \sum_{i=1}^{N} \left|i\right\rangle\!\left\langle i\right| = \mathbf{I},$$

and $|i\rangle$ is an eigenket of both $\mathbf{X}$ and $\mathbf{Y}$ with eigenvalues λ_i and μ_i, respectively.

Statements (1) and (3) are true in reverse as well, *i.e.*, any operators with an orthonormal basis of eigenkets are normal, and any operators sharing an orthonormal basis of eigenkets commute.

Unitary Transformations

Recall that an operator $\mathbf{U}$ is unitary if $\mathbf{U}^\dagger = \mathbf{U}^{-1}$, or equivalently, if $\mathbf{U}\mathbf{U}^\dagger = \mathbf{U}^\dagger\mathbf{U} = \mathbf{I}$. Alternatively, we can define a unitary operator in terms of its action on kets:

Definition 118. An operator $\mathbf{U}$ is **unitary** if it preserves the inner product, *i.e.*, for all $|\psi\rangle$, $|\chi\rangle \in \mathcal{H}$,

$$(\mathbf{U}\left|\psi\right\rangle, \mathbf{U}\left|\chi\right\rangle) = (\left|\psi\right\rangle, \left|\chi\right\rangle).$$

Suppose that $\{|1\rangle, |2\rangle, \ldots, |N\rangle\}$ is an orthonormal basis for $\mathcal{H}$. Then, the set of kets $\{|1'\rangle, |2'\rangle, \ldots, |N'\rangle\}$, where $|i'\rangle \equiv \mathbf{U}|i\rangle$, is also an orthonormal basis for $\mathcal{H}$, since

$$\langle i'|j'\rangle = \left\langle i\middle|\mathbf{U}^\dagger\mathbf{U}\middle|j\right\rangle = \langle i|j\rangle = \delta_{ij}.$$

By inserting the completeness relation for the original basis, we obtain a decomposition of the unitary operator $\mathbf{U}$ in terms of the original and transformed bases:

$$\mathbf{I} = \sum_{i=1}^{N} \left|i\right\rangle\!\left\langle i\right| \quad \implies \quad \mathbf{U} = \mathbf{U}\sum_{i=1}^{N} \left|i\right\rangle\!\left\langle i\right| = \sum_{i=1}^{N} \left|i'\right\rangle\!\left\langle i\right|.$$

For *any* two orthonormal bases $\{|1\rangle, |2\rangle, \ldots, |N\rangle\}$ and $\{|1'\rangle, |2'\rangle, \ldots, |N'\rangle\}$ for $\mathcal{H}$, there exists a unitary operator $\mathbf{U}$ such that $|i'\rangle = \mathbf{U}|i\rangle$ for all i. In other words, any two orthonormal bases for a Hilbert space are related by a unitary transformation. The transformation is indeed unitary, since

$$\mathbf{U}^\dagger\mathbf{U} = \sum_{i=1}^{N} |i\rangle\langle i'| \sum_{j}^{N} |j'\rangle\langle j| = \sum_{i,j=1}^{N} \underbrace{\langle i'|j'\rangle}_{\delta_{ij}} |i\rangle\langle j| = \sum_{i=1}^{N} |i\rangle\langle i| = \mathbf{I},$$

and likewise $\mathbf{U}\mathbf{U}^\dagger = \mathbf{I}$.[24]

[24] In a finite dimensional Hilbert space, it suffices to show that $\mathbf{U}^\dagger\mathbf{U} = \mathbf{I}$ to conclude that $\mathbf{U}$ is unitary, since this implies that $\mathbf{U}$ is invertible and that $\mathbf{U}^{-1} = \mathbf{U}^\dagger$. However, in an infinite dimensional Hilbert space, it is possible for an operator to be a left inverse but not a right inverse, so we must also check that $\mathbf{U}\mathbf{U}^\dagger = \mathbf{I}$ to conclude that $\mathbf{U}$ is unitary.

Definition 119. The **unitary transformation** of an operator $\mathbf{X}$ by a unitary operator $\mathbf{U}$ is the operator $\mathbf{X}_U$ defined by

$$\mathbf{X}_U = \mathbf{U}\mathbf{X}\mathbf{U}^\dagger.$$

If $[\mathbf{X}, \mathbf{U}] = 0$ then $\mathbf{X}_U = \mathbf{X}$, so $\mathbf{X}$ is invariant under the unitary transformation by $\mathbf{U}$.

Going into the details a bit, if $[\mathbf{X}, \mathbf{U}] = 0$, then

$$\mathbf{X}_U = \mathbf{U}\mathbf{X}\mathbf{U}^\dagger = \mathbf{X}\mathbf{U}\mathbf{U}^\dagger = \mathbf{X}.$$

Generally, $\mathbf{X}_U \neq \mathbf{X}$, but $\mathbf{X}_U$ has similar properties to $\mathbf{X}$, as will be seen in the question at the end of this paragraph. Operators related by a unitary transformation are said to be **unitarily equivalent**.

A unitary transformation is what a mathematician would call an **automorphism** on the Hilbert space. They are also close to what a physicist would call a **symmetry**, but a symmetry is usually defined as a unitary transformation that leaves the Hamiltonian invariant, *i.e.*, a unitary transformation by $\mathbf{U}$ such that $[\mathbf{H}, \mathbf{U}] = 0$.

Question 120. *Show that*

1. *if $\mathbf{X}$ is self-adjoint, then $\mathbf{X}_U$ is also self-adjoint,*
2. *if $\mathbf{X}$ is unitary, then $\mathbf{X}_U$ is also unitary,*
3. *$|\psi\rangle$ is an eigenket of $\mathbf{X}$ with eigenvalue λ if and only if $\mathbf{U}|\psi\rangle$ is an eigenket of $\mathbf{X}_U$ with the same eigenvalue λ, and*
4. $\mathbf{X}_U\mathbf{Y}_U = (\mathbf{X}\mathbf{Y})_U$.

Matrices

Given an orthonormal basis $\{|1\rangle, |2\rangle, \ldots, |N\rangle\}$ for $\mathcal{H}$, we can use the associated completeness relation to decompose bras and kets into their components in the basis:

$$|\psi\rangle = \sum_{i=1}^{N} |i\rangle\langle i| \, |\psi\rangle = \sum_{i=1}^{N} \langle i|\psi\rangle \, |i\rangle, \qquad \langle\phi| = \sum_{i=1}^{N} |i\rangle\langle i| \, \langle\phi| = \sum_{i=1}^{N} \langle\phi|i\rangle \, |i\rangle,$$

where $\langle i|\psi\rangle$ are the components of $|\psi\rangle$ in the basis $\{|1\rangle, |2\rangle, \ldots, |N\rangle\}$, and $\langle\phi|i\rangle$ are the components of $\langle\phi|$ in the dual basis $\{\langle 1|, \langle 2|, \ldots, \langle N|\}$. We can also decompose operators into their matrix elements in the basis:

$$\mathbf{X} = \sum_{i,j=1}^{N} |i\rangle\langle i| \, \mathbf{X} \, |j\rangle\langle j| = \sum_{i,j=1}^{N} \langle i|\mathbf{X}|j\rangle \, |i\rangle\langle j| \, ,$$

where $\langle i|\mathbf{X}|j\rangle$ are the matrix elements of $\mathbf{X}$ in the basis $\{|1\rangle, |2\rangle, \ldots, |N\rangle\}$. When we multiply two operators, $\mathbf{X}$ and $\mathbf{Y}$, together, the matrix elements of the product $\mathbf{XY}$ are given by

$$\langle i|\mathbf{XY}|k\rangle = \sum_{j=1}^{N} \langle i|\mathbf{X}|j\rangle \, \langle j|\mathbf{Y}|k\rangle \, ,$$

which is the familiar formula for matrix multiplication. Similarly, extracting the components of $\mathbf{X}\,|\psi\rangle$ in the basis, we have

$$\langle i|\mathbf{X}|\psi\rangle = \sum_{j=1}^{N} \langle i|\mathbf{X}|j\rangle \, \langle j|\psi\rangle \, ,$$

which is the matrix-vector product of the matrix $\mathbf{X}$ and the column vector of components of $|\psi\rangle$ in the basis. Dealing with bras analogously, we can realize the entire Hilbert space $\mathcal{H}$ concretely as the space of column vectors of components in the basis, and the dual space $\mathcal{H}^*$ as the space of row vectors of components in the dual basis, and operators as matrices of components in the basis:

$$|\psi\rangle \to \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_N \end{pmatrix}, \qquad \langle\phi| \to \begin{pmatrix} \phi_1 & \phi_2 & \cdots & \phi_N \end{pmatrix}, \qquad \mathbf{X} \to \begin{pmatrix} X_{11} & X_{12} & \cdots & X_{1N} \\ X_{21} & X_{22} & \cdots & X_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ X_{N1} & X_{N2} & \cdots & X_{NN} \end{pmatrix},$$

where $\psi_i = \langle i|\psi\rangle$, $\phi_i = \langle\phi|i\rangle$, and $X_{ij} = \langle i|\mathbf{X}|j\rangle$. Moreover, we now see that the conjugate map and the adjoint correspond to taking the conjugate transpose of the matrix in question, *i.e.*, $|\psi\rangle^*$ corresponds to the row vector of conjugate components, and $\mathbf{X}^\dagger$ corresponds to the conjugate transpose of the matrix of components of $\mathbf{X}$.

Despite the pleasing simplicity of this, it is important no to conflate the components of a ket, bra, or operator in a particular basis with the ket, bra, or operator itself. For instance, suppose that the values ψ_i, ϕ_i, and X_{ij}, etc., change by a unitary transformation. This could mean several things:

1. $|\psi\rangle \to \mathbf{U}\,|\psi\rangle$, $\langle\phi| \to \langle\phi|\,\mathbf{U}^\dagger$, $X \to \mathbf{U}X\mathbf{U}^\dagger$, etc., which is an **active transformation** of the kets, bras, and operators themselves, while the basis remains fixed, or

2. $|\psi\rangle$, $\langle\phi|$, X, etc. are unchanged, but the basis changes by $|i\rangle \to \mathbf{U}\,|i\rangle$, which is a **passive transformation** of the basis while the kets, bras, and operators remain fixed.

These two transformations are mathematically equivalent, since the components of the kets, bras, and operators in the new basis after the passive transformation are the same as the components of the transformed kets, bras, and operators in the original basis after the active transformation. However, they are physically distinct, since the active transformation corresponds to a physical change in the state of the system, while the passive transformation corresponds to a change in our description of the system without any physical change. Combining these two transformations, we can also consider a scenario where both the kets, bras, and operators and the basis change by a unitary transformation, while leaving the components of the kets, bras, and operators unchanged.

Working with matrices makes it easier to define certain operations. Particularly, we can define the determinant and trace of an operator $\mathbf{X}$ as the determinant and trace of the corresponding matrix of components in a particular basis, $\langle i|\mathbf{X}|j\rangle$.

The Postulates of Quantum Mechanics

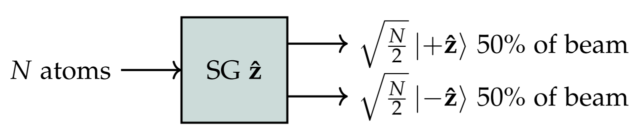
We explore how the postulates of quantum mechanics emerge from the Stern-Gerlach experiment and the mathematical framework of Hilbert spaces, kets, bras, and operators. In particular, the Stern-Gerlach experiment is described by a two-dimensional Hilbert space, the qubit.

Analysis of the Stern-Gerlach Experiment

We assign a quantum state vector $|\psi\rangle \in \mathcal{H}$ to each atomic beam in the Stern-Gerlach experiment, where the properties of the Hilber space $\mathcal{H}$ remain to be determined. We also normlize the state vectors so that the square of the norm of the state $|\psi\rangle$ is equal to the *number* of atoms in the beam,

$$N_{\text{beam}} = \langle\psi|\psi\rangle\,.$$

The original Stern-Gerlach experiment takes an unpolarized beam of N atoms and splits it into two polarized beams:

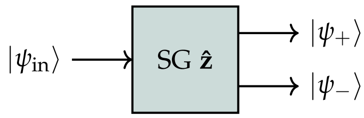


Here we've taken the quantum states of the two output beams (up to normalization) to define the state vectors $|+\hat{\mathbf{z}}\rangle$ and $|-\hat{\mathbf{z}}\rangle$, which we take to be normalized:

$$\langle+\hat{\mathbf{z}}|+\hat{\mathbf{z}}\rangle = \langle-\hat{\mathbf{z}}|-\hat{\mathbf{z}}\rangle = 1.$$

Thus, the prefactor $\sqrt{\frac{N}{2}}$ is chosen so that the square of the norm of each output state is equal to the number of atoms in the corresponding output beam, which is $N/2$.

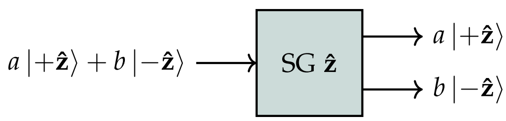
Now consider passing a polarized beam through the same apparatus:



where $|\psi_{\text{in}}\rangle$ is the state of the incoming beam, and $|\psi_+\rangle$ and $|\psi_-\rangle$ are the states of the two output beams. Our previous experience tells us that

1. if $|\psi_{\text{in}}\rangle = |+\hat{\mathbf{z}}\rangle$, then $|\psi_+\rangle = |+\hat{\mathbf{z}}\rangle$ and $|\psi_-\rangle = 0$, and
2. if $|\psi_{\text{in}}\rangle = |-\hat{\mathbf{z}}\rangle$, then $|\psi_+\rangle = 0$ and $|\psi_-\rangle = |-\hat{\mathbf{z}}\rangle$.

Since the number of atoms in the incoming beam is equal to the number of atoms in the outgoing beams, a **quantum superposition** of $|+\hat{\mathbf{z}}\rangle$ and $|-\hat{\mathbf{z}}\rangle$ must be preserved by the apparatus, so we can write



Thus, the number of atoms coming out up and down are

$$N_+ = (a^* \langle +\hat{\mathbf{z}}|)(a\,|+\hat{\mathbf{z}}\rangle) = |a|^2, \qquad N_- = (b^* \langle -\hat{\mathbf{z}}|)(b\,|-\hat{\mathbf{z}}\rangle) = |b|^2,$$

so the total number of atoms coming out is

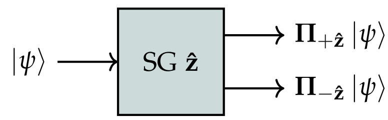
$$N_{\text{out}} = N_+ + N_- = |a|^2 + |b|^2.$$

Comparing this with the number of atoms entering the apparatus, we have

$$\begin{aligned} N &= (a^* \langle +\hat{\mathbf{z}}| + b^* \langle -\hat{\mathbf{z}}|)(a\,|+\hat{\mathbf{z}}\rangle + b\,|-\hat{\mathbf{z}}\rangle) \\ &= |a|^2 + |b|^2 + a^* b \langle +\hat{\mathbf{z}}|-\hat{\mathbf{z}}\rangle + b^* a \langle -\hat{\mathbf{z}}|+\hat{\mathbf{z}}\rangle \\ &= |a|^2 + |b|^2 + 2\,\mathrm{Re}\,[a^* b \langle +\hat{\mathbf{z}}|-\hat{\mathbf{z}}\rangle]. \end{aligned}$$

In order for this to be exactly the number of atoms coming out, we must have $\mathrm{Re}\,[a^* b \langle +\hat{\mathbf{z}}|-\hat{\mathbf{z}}\rangle] = 0$. Since this must hold for all a and b, we must have $\langle +\hat{\mathbf{z}}|-\hat{\mathbf{z}}\rangle = 0$. Thus, the two output states $|+\hat{\mathbf{z}}\rangle$ and $|-\hat{\mathbf{z}}\rangle$ are orthogonal to each other. This illustrates a general principle of quantum mechanics: *distinct measurement outcomes correspond to orthogonal quantum states.*

Therefore, $\{|+\hat{\mathbf{z}}\rangle, |-\hat{\mathbf{z}}\rangle\}$ forms an orthonormal set, implying that the Hilbert space $\mathcal{H}$ describing the Stern-Gerlach experiment is atleast two-dimensional. We limit ourselves to the minimal choice of a two-dimensional Hilbert space, so $\{|+\hat{\mathbf{z}}\rangle, |-\hat{\mathbf{z}}\rangle\}$ is an orthonormal basis for $\mathcal{H}$. Th effect of the SG device is then to split the beam into orthogonal components:



where

$$\mathbf{\Pi}_{+\hat{\mathbf{z}}} = |+\hat{\mathbf{z}}\rangle\langle+\hat{\mathbf{z}}|, \qquad \mathbf{\Pi}_{-\hat{\mathbf{z}}} = |-\hat{\mathbf{z}}\rangle\langle-\hat{\mathbf{z}}|,$$

is a complete set of orthogonal projectors. The two beams correspond to the atomic angular momenta being measured to be $\pm\hbar/2$ along the z-axis. We represent this fact in the Hilbert space language by defining the **spin operator** $\mathbf{S}_z$ as

$$\mathbf{S}_z = \frac{\hbar}{2}(|+\hat{\mathbf{z}}\rangle\langle+\hat{\mathbf{z}}| - |-\hat{\mathbf{z}}\rangle\langle-\hat{\mathbf{z}}|) = \frac{\hbar}{2}(\mathbf{\Pi}_{+\hat{\mathbf{z}}} - \mathbf{\Pi}_{-\hat{\mathbf{z}}}), \tag{15}$$

so that $|+\hat{\mathbf{z}}\rangle$ and $|-\hat{\mathbf{z}}\rangle$ are eigenkets of $\mathbf{S}_z$ with eigenvalues $+\hbar/2$ and $-\hbar/2$, respectively. By the following definition,

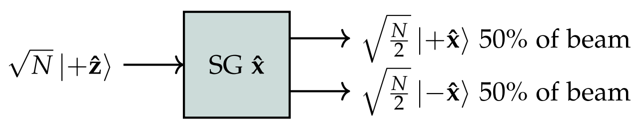
Definition 121. An **observable** is a self-adjoint operator representing a measurable physical quantity, whose eigenvalues correspond to the possible measurement outcomes of that quantity, and whose eigenkets correspond to the states in which the system will be found after a measurement yielding that outcome.

$\mathbf{S}_z$ is an observable, since it is self-adjoint, and its eigenvalues $\pm\hbar/2$ correspond to the possible measurement outcomes of the z-component of the atomic angular momentum, and its eigenkets $|+\hat{\mathbf{z}}\rangle$ and $|-\hat{\mathbf{z}}\rangle$ correspond to the states in which the system will be found after a measurement yielding those outcomes. Measuring many identically prepared atoms in this way, the average value of the z-component of the atomic angular momentum is given by

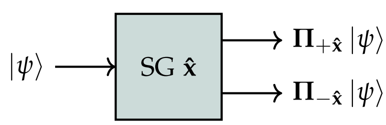
$$\begin{aligned}
\langle S_z\rangle &= \frac{1}{N}\left[\frac{\hbar}{2}N_+ - \frac{\hbar}{2}N_-\right] \\
&= \frac{1}{\langle\psi|\psi\rangle}\left[\frac{\hbar}{2}\langle\psi|+\hat{\mathbf{z}}\rangle\langle+\hat{\mathbf{z}}|\psi\rangle - \frac{\hbar}{2}\langle\psi|-\hat{\mathbf{z}}\rangle\langle-\hat{\mathbf{z}}|\psi\rangle\right] \\
&= \frac{1}{\langle\psi|\psi\rangle}\langle\psi|\left(\frac{\hbar}{2}|+\hat{\mathbf{z}}\rangle\langle+\hat{\mathbf{z}}| - \frac{\hbar}{2}|-\hat{\mathbf{z}}\rangle\langle-\hat{\mathbf{z}}|\right)|\psi\rangle \\
&= \frac{\langle\psi|\mathbf{S}_z|\psi\rangle}{\langle\psi|\psi\rangle} \equiv \langle\mathbf{S}_z\rangle,
\end{aligned}$$

known as the **expectation value** of $\mathbf{S}_z$ in the state $|\psi\rangle$.

Where before we had a direction and a color apparatus, we now have different polarizations of the atomic beam. Thus far we've only considered the Stern-Gerlach experiment with the magnetic field gradient oriented along the z-axis, but we can also orient the magnetic field gradient along the x-axis. For our next thought experiment, suppose we send a beam of atoms polarized in the z-direction through a Stern-Gerlach apparatus with the magnetic field gradient oriented along the x-axis:



As before, we take the quantum states of the two output beams to define the state vectors $|+\hat{\mathbf{x}}\rangle$ and $|-\hat{\mathbf{x}}\rangle$, which we take to be normalized. For the same reason as before, these must be orthogonal, so $\{|+\hat{\mathbf{x}}\rangle, |-\hat{\mathbf{x}}\rangle\}$ is also an orthonormal basis for $\mathcal{H}$, and the SG device with the magnetic field gradient oriented along the x-axis splits the beam into orthogonal components:



Therefore, with an initial state of $|\psi\rangle = \sqrt{N}\,|+\hat{\mathbf{z}}\rangle$, the number of atoms coming out up and down are

$$\sqrt{N}\,\mathbf{\Pi}_{+\hat{\mathbf{x}}}\,|+\hat{\mathbf{z}}\rangle = \sqrt{N}\,|+\hat{\mathbf{x}}\rangle\,\langle+\hat{\mathbf{x}}|+\hat{\mathbf{z}}\rangle\,, \qquad \sqrt{N}\,\mathbf{\Pi}_{-\hat{\mathbf{x}}}\,|+\hat{\mathbf{z}}\rangle = \sqrt{N}\,|-\hat{\mathbf{x}}\rangle\,\langle-\hat{\mathbf{x}}|+\hat{\mathbf{z}}\rangle\,.$$

Matching this with what we have observed, we must have

$$\langle+\hat{\mathbf{x}}|+\hat{\mathbf{z}}\rangle = \langle-\hat{\mathbf{x}}|+\hat{\mathbf{z}}\rangle = \frac{1}{\sqrt{2}}.$$

Therefore, essentially by reverse-engineering,

$$|+\hat{\mathbf{z}}\rangle = \frac{1}{\sqrt{2}}(|+\hat{\mathbf{x}}\rangle + |-\hat{\mathbf{x}}\rangle). \tag{16}$$

Since $|-\hat{\mathbf{z}}\rangle$ is orthogonal to $|+\hat{\mathbf{z}}\rangle$, it must be of the form

$$|-\hat{\mathbf{z}}\rangle = \frac{1}{\sqrt{2}}\,e^{i\phi}\,(|+\hat{\mathbf{x}}\rangle - |-\hat{\mathbf{x}}\rangle),$$

up to an *a priori* unknown phase factor $e^{i\phi}$. So far, we have not fixed the phase of $|-\hat{\mathbf{z}}\rangle$ relative to $|+\hat{\mathbf{z}}\rangle$, so we can choose $\phi = 0$ without loss of generality, giving us

$$|-\hat{\mathbf{z}}\rangle = \frac{1}{\sqrt{2}}(|+\hat{\mathbf{x}}\rangle - |-\hat{\mathbf{x}}\rangle). \tag{17}$$

Question 122. *Find expressions for $|+\hat{\mathbf{x}}\rangle$ and $|-\hat{\mathbf{x}}\rangle$ in terms of $|+\hat{\mathbf{z}}\rangle$ and $|-\hat{\mathbf{z}}\rangle$ using Eqns.* (16) *&* (17)*.*

As before, we define an observable $\mathbf{S}_x$ corresponding to the x-component of the atomic angular momentum as

$$\mathbf{S}_x = \frac{\hbar}{2}(|{+\hat{\mathbf{x}}}\rangle\langle{+\hat{\mathbf{x}}}| - |{-\hat{\mathbf{x}}}\rangle\langle{-\hat{\mathbf{x}}}|) = \frac{\hbar}{2}(\mathbf{\Pi}_{+\hat{\mathbf{x}}} - \mathbf{\Pi}_{-\hat{\mathbf{x}}}),$$

so that $|{+\hat{\mathbf{x}}}\rangle$ and $|{-\hat{\mathbf{x}}}\rangle$ are eigenkets of $\mathbf{S}_x$ with eigenvalues $+\hbar/2$ and $-\hbar/2$, respectively.

Example 123. We now have two different observables, $\mathbf{S}_z$ and $\mathbf{S}_x$, corresponding to the z- and x-components of the atomic angular momentum, respectively. In terms of the basis $\{|{+\hat{\mathbf{z}}}\rangle, |{-\hat{\mathbf{z}}}\rangle\}$, we have

$$|{\pm\hat{\mathbf{x}}}\rangle = \frac{1}{\sqrt{2}}(|{+\hat{\mathbf{z}}}\rangle + |{-\hat{\mathbf{z}}}\rangle),$$

as you should've gotten from the previous question. Therefore, the observable $\mathbf{S}_x$ can be expressed in terms of the basis $\{|{+\hat{\mathbf{z}}}\rangle, |{-\hat{\mathbf{z}}}\rangle\}$ as

$$\begin{aligned}
\mathbf{S}_x &= \frac{\hbar}{2}(|{+\hat{\mathbf{x}}}\rangle\langle{+\hat{\mathbf{x}}}| - |{-\hat{\mathbf{x}}}\rangle\langle{-\hat{\mathbf{x}}}|) \\
&= \frac{\hbar}{2}\left[\frac{1}{\sqrt{2}}(|{+\hat{\mathbf{z}}}\rangle + |{-\hat{\mathbf{z}}}\rangle)\frac{1}{\sqrt{2}}(\langle{+\hat{\mathbf{z}}}| + \langle{-\hat{\mathbf{z}}}|)\right. \\
&\qquad\qquad \left. - \frac{1}{\sqrt{2}}(|{+\hat{\mathbf{z}}}\rangle - |{-\hat{\mathbf{z}}}\rangle)\frac{1}{\sqrt{2}}(\langle{+\hat{\mathbf{z}}}| - \langle{-\hat{\mathbf{z}}}|)\right] \\
&= \frac{\hbar}{4}\left[(|{+\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}| + |{+\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}| + |{-\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}| + |{-\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}|)\right. \\
&\qquad\qquad \left. - (|{+\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}| - |{+\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}| - |{-\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}| + |{-\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}|)\right] \\
&= \frac{\hbar}{4}\left[\cancel{|{+\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}|} + |{+\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}| + |{-\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}| + \cancel{|{-\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}|}\right. \\
&\qquad\qquad \left. - \cancel{|{+\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}|} + |{+\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}| + |{-\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}| - \cancel{|{-\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}|}\right] \\
&= \frac{\hbar}{2}(|{+\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}| + |{-\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}|),
\end{aligned}$$

which is not diagonal in the basis $\{|{+\hat{\mathbf{z}}}\rangle, |{-\hat{\mathbf{z}}}\rangle\}$. As matrices, then, we have

$$\mathbf{S}_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \mathbf{S}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{18}$$

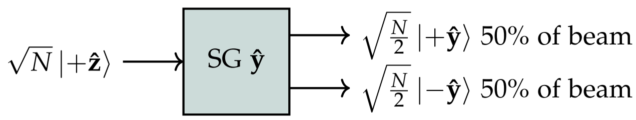
Note that $\mathbf{S}_z$ and $\mathbf{S}_x$ do not commute with each other, since

$$\begin{aligned}
[\mathbf{S}_z, \mathbf{S}_x] &= \frac{\hbar^2}{4}[(|{+\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}| - |{-\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}|)(|{+\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}| + |{-\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}|) - (|{+\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}| + |{-\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}|)(|{+\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}| - |{-\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}|)] \\
&= \frac{\hbar^2}{4}(|{+\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}| - |{-\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}| + |{+\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}| - |{-\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}|) \\
&= \frac{\hbar^2}{2}(|{+\hat{\mathbf{z}}}\rangle\langle{-\hat{\mathbf{z}}}| - |{-\hat{\mathbf{z}}}\rangle\langle{+\hat{\mathbf{z}}}|) \neq 0.
\end{aligned} \tag{19}$$

Mathematically, this reflects the fact that their eigenspace projectors do not commute with each other; *i.e.*, when measuring both $\mathbf{S}_z$ and $\mathbf{S}_x$, the outcome depends on the order in which the measurements are performed. For this reason, we say that $\mathbf{S}_z$ and $\mathbf{S}_x$ are **incompatible observables**.[25]

[25] Alternatively, we can say that $\mathbf{S}_z$ and $\mathbf{S}_x$ are **not simulaneously observable**.

We now consider what happens when we instead measure the y-component of the atomic angular momentum. With the entering atoms in the state $|+\hat{\mathbf{z}}\rangle$, this is a repetition of the previous experiment, except with the SG device oriented along the y-axis instead of the x-axis:



We proceed just as before, defining $|+\hat{\mathbf{y}}\rangle$ and $|-\hat{\mathbf{y}}\rangle$ as the states of the two output beams, which we take to be normalized. We arrive at the final result:

$$|+\hat{\mathbf{z}}\rangle = \frac{1}{\sqrt{2}}(|+\hat{\mathbf{y}}\rangle + |-\hat{\mathbf{y}}\rangle), \qquad |-\hat{\mathbf{z}}\rangle = \frac{1}{\sqrt{2}}\,\mathrm{e}^{i\delta}\,(|+\hat{\mathbf{y}}\rangle - |-\hat{\mathbf{y}}\rangle).$$

Here δ is another *a priori* unknown phase factor. However, we no longer have the freedom to choose δ to be zero, since we have already fixed the relative phase of $|-\hat{\mathbf{z}}\rangle$ and $|+\hat{\mathbf{z}}\rangle$ to be zero. Thus, for the time being, we obtain

$$|\pm\hat{\mathbf{y}}\rangle = \frac{1}{\sqrt{2}}\Big(|+\hat{\mathbf{z}}\rangle \pm \mathrm{e}^{-i\delta}\,|-\hat{\mathbf{z}}\rangle\Big).$$

Question 124. *Verify the expression for $|\pm\hat{\mathbf{y}}\rangle$ in terms of $|+\hat{\mathbf{z}}\rangle$ and $|-\hat{\mathbf{z}}\rangle$.*

To fix δ, we note that by analogy with what we've done before, we must have $|\langle +\hat{\mathbf{x}}|+\hat{\mathbf{y}}\rangle| = 1/\sqrt{2}$.[26] We find:

[26] Equivalently, this can be deduced by analyzing yet another rotated version of the second thought experiment.

$$\begin{aligned}\langle +\hat{\mathbf{x}}|+\hat{\mathbf{y}}\rangle &= \frac{1}{2}\Big(\langle +\hat{\mathbf{z}}|+\hat{\mathbf{z}}\rangle + \mathrm{e}^{-i\delta}\,\langle +\hat{\mathbf{z}}|-\hat{\mathbf{z}}\rangle + \langle -\hat{\mathbf{z}}|+\hat{\mathbf{z}}\rangle + \mathrm{e}^{-i\delta}\,\langle -\hat{\mathbf{z}}|-\hat{\mathbf{z}}\rangle\Big)\\ &= \frac{1}{2}\Big(1+0+0+\mathrm{e}^{-i\delta}\Big) = \frac{1}{2}\Big(1+\mathrm{e}^{-i\delta}\Big),\end{aligned}$$

recalling the definition of cosine in terms of the complex exponential,

$$\cos(\theta) = \frac{1}{2}\Big(\mathrm{e}^{i\theta} + \mathrm{e}^{-i\theta}\Big),$$

we have

$$\langle +\hat{\mathbf{x}}|+\hat{\mathbf{y}}\rangle = \frac{1}{2}\Big(1+\mathrm{e}^{-i\delta}\Big) = \frac{1}{2}\Big(1+\mathrm{e}^{-i\delta}\Big)\cdot\frac{\mathrm{e}^{i\delta/2}}{\mathrm{e}^{i\delta/2}} = \frac{1}{2}\Big(\mathrm{e}^{i\delta/2}+\mathrm{e}^{-i\delta/2}\Big)\,\mathrm{e}^{-i\delta/2} = \mathrm{e}^{-i\delta/2}\cos\Big(\frac{\delta}{2}\Big).$$

Taking the norm squared of both sides, we have

$$|\langle +\hat{\mathbf{x}}|+\hat{\mathbf{y}}\rangle|^2 = \cos^2(\frac{\delta}{2}) = \frac{1}{2},$$

which has the solution $\delta = \pm\pi/2$, corresponding to $e^{i\delta} = \pm i$. Conventionally, we take $\delta = -\pi/2$, so that

$$|\pm\hat{\mathbf{y}}\rangle = \frac{1}{\sqrt{2}}(|+\hat{\mathbf{z}}\rangle \pm i\,|-\hat{\mathbf{z}}\rangle).$$

Thus, the observable associated to a measurement of the y-component of the atomic angular momentum is

$$\mathbf{S}_y = \frac{\hbar}{2}(|+\hat{\mathbf{y}}\rangle\langle+\hat{\mathbf{y}}| - |-\hat{\mathbf{y}}\rangle\langle-\hat{\mathbf{y}}|) = \frac{\hbar}{2}\left(\mathbf{\Pi}_{+\hat{\mathbf{y}}} - \mathbf{\Pi}_{-\hat{\mathbf{y}}}\right) = \frac{\hbar}{2i}(|+\hat{\mathbf{z}}\rangle\langle-\hat{\mathbf{z}}| - |-\hat{\mathbf{z}}\rangle\langle+\hat{\mathbf{z}}|), \tag{20}$$

where

$$\mathbf{\Pi}_{+\hat{\mathbf{y}}} = |+\hat{\mathbf{y}}\rangle\langle+\hat{\mathbf{y}}| \quad \text{and} \quad \mathbf{\Pi}_{-\hat{\mathbf{y}}} = |-\hat{\mathbf{y}}\rangle\langle-\hat{\mathbf{y}}|\,,$$

and the right-hand side can be verified by direct substitution as we did in an earlier example. We notice that

$$\mathbf{S}_y = \frac{1}{i\hbar}[\mathbf{S}_z, \mathbf{S}_x]$$

from Eqn. (19). In fact, you will show that the commutator of any two of the three observables $\mathbf{S}_x$, $\mathbf{S}_y$, and $\mathbf{S}_z$ is $\pm i\hbar$ times the third observable. Thus, no two components of the spin can be measured simultaneously.

Question 125.

1. *Verify the second equality in Eqn. (20) by direct substitution.*

2. *Use Eqns. (18) & (20) to derive the* ***spin algebra*** *in three dimensions,* $\mathfrak{spin}(3)$*:*

$$\left[\mathbf{S}_x, \mathbf{S}_y\right] = i\hbar\mathbf{S}_z, \qquad \left[\mathbf{S}_y, \mathbf{S}_z\right] = i\hbar\mathbf{S}_x, \qquad \left[\mathbf{S}_z, \mathbf{S}_x\right] = i\hbar\mathbf{S}_y.$$

Working in the basis $\{|+\hat{\mathbf{z}}\rangle\,, |-\hat{\mathbf{z}}\rangle\}$, the matrix representations of $\mathbf{S}_x$, $\mathbf{S}_y$, and $\mathbf{S}_z$ are

$$\mathbf{S}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \mathbf{S}_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \mathbf{S}_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Stripping off the factor of $\hbar/2$, we have the **Pauli matrices**:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Any real 2×2 self-adjoint matrix is a real linear combination of $\mathcal{I}$, σ_x, σ_y, and σ_z, where $\mathcal{I}$ is the 2×2 identity matrix. One can show that

$$\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = \mathcal{I}, \qquad \sigma_x\sigma_y = -\sigma_y\sigma_z = i\sigma_z, \qquad \sigma_y\sigma_z = -\sigma_z\sigma_x = i\sigma_x, \qquad \sigma_z\sigma_x = -\sigma_x\sigma_y = i\sigma_y,$$

which can be fully captured by

$$\sigma_i \sigma_j = \delta_{ij} \mathcal{I} + i \epsilon_{ijk} \sigma_k,$$

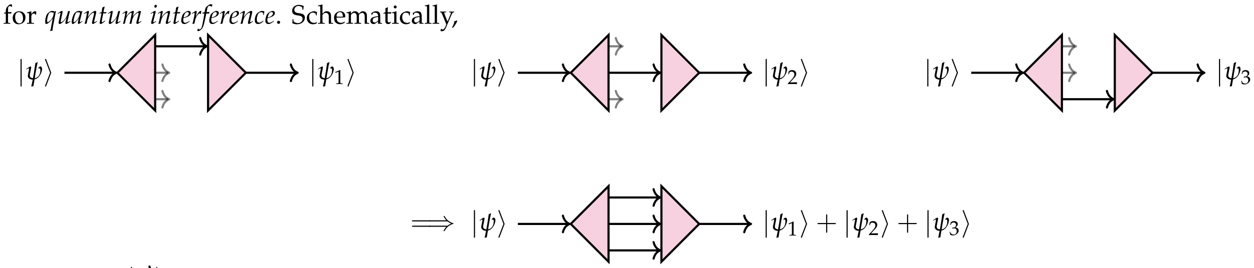
where ϵ_{ijk} is the Levi-Civita symbol, defined as

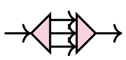
$$\epsilon_{ijk} = \begin{cases} +1 & \text{if } (i, j, k) \text{ is an even permutation of } (x, y, z), \\ -1 & \text{if } (i, j, k) \text{ is an odd permutation of } (x, y, z), \\ 0 & \text{otherwise,} \end{cases}$$

and repeated indices are summed over.

Postulate 1: Quantum States

As we've been arguing, quantum states are represented by vectors in a complex Hilbert space. When a given outcome can be reached in several different ways, the final state vector is the sum of the state vectors arising from restricting to each individual scenario, allowing for *quantum interference*. Schematically,



where  denotes a device similar to the modified SG device we used previously.

The overall phase of a state vector is not physically meaningful; only the relative phase between interfering components can be measured. Likewise, the overall normalization is not meaningful, though we may sometimes fix it by convention, but the relative normalization between interfering components is physically meaningful.

The essential nature of quantum states is encapsulated in the following postulate:

Postulate 1: [P1] Associated to a quantum system, there is a complex, projective Hilbert space $\mathcal{H}$ of possible **quantum states** $|\psi\rangle$.

Here "projective" means that the state $\lambda |\psi\rangle$ is physically equivalent to $|\psi\rangle$ for any nonzero complex number λ and that the state $|\psi\rangle = 0$ is not a valid quantum state.[27]

[27] In other words, quantum states are not individual vectors in $\mathcal{H}$, but rays through the origin of $\mathcal{H}$.

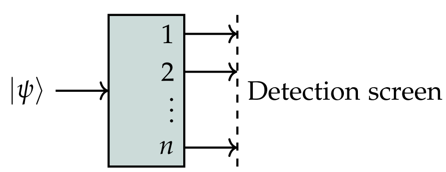
Note that P1 does not tell us how to idenitfy the appropriate Hilbert space for a given physical system. In physics, we must have some information about the 'classical limit' of the system in order to construct the appropriate Hilbert space. This process of finding the appropriate Hilbert space for a given physical system is called **quantization**. This process often involves a large degree of intuition and guesswork, so we will not consider it to be on equal footing with the postulates. However, it is useful to formalize it as much as possible. Summarizing our prior observations about quanutm interference, we observe:

Quantization Rule 1: [QR1] If a quantum system is allowed to take multiple classical paths to the same final configuration, then the final quantum state is the sum of the quantum states arising from restricting to each individual path.

This rule plays a central role in an alternative formulation of quantum mechanics called the **path integral formulation**. Suitably interpreted, it can be used to construct the Hilbert space of a quantum system from the classical limit of the system. Note, however, that while P1 is a purely mathematical statement, QR1 requires significant interpretation to make it precise.

Postulate 2: Measurements

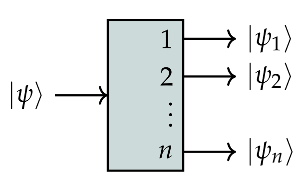
A **measurement** is something like performing a generalized SG experiment which produces n possible outcomes, each with probability p_i that depends on the input state:



Here we labeled the possible outcomes of the measurement by 1, 2, . . ., n for simplicity. To analyze such an experiment, suppose that we perform $N \gg 1$ trials of the experiment with the same incoming state $|\psi\rangle$, which we choose to normalize as

$$\langle\psi|\psi\rangle = N.$$

Let $|\psi_i\rangle$ denote the final state when the measurement outocome is i, normalized so that $\langle\psi_i|\psi_i\rangle = N_i$, where N_i is the **expected** (average) number of times that the outcome i is observed. We illustrate this as follows:



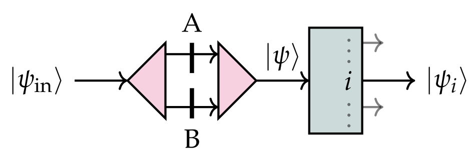
With these normalizations, the final states for each measurement outcome must be a linear function of the initial state, *i.e.*, there exist linear operators $\mathbf{O}_i$, independent of $|\psi\rangle$, such that

$$|\psi_i\rangle = \mathbf{O}_i\,|\psi\rangle\,. \tag{21}$$

To see why this must be the case, consider the more general ansatz $|\psi_i\rangle = f_i(|\psi\rangle)$ for some unknown, potentially nonlinear function $f_i : \mathcal{H} \to \mathcal{H}$. We make the following observations:

1. Recalling $|\psi\rangle \to \lambda\,|\psi\rangle$ for real $\lambda > 0$ corresponds to increasing the number of iterations of the experiment, $N \to \lambda^2 N$, which changes the expected number of occurences of each output proportionally, $N_i \to \lambda^2 N_i$. Thus, we must have $f_i(\lambda\,|\psi\rangle) = \lambda f_i(|\psi\rangle)$ for all $\lambda > 0$.

2. Let the output of a quantum interference experiment be directed into our measurement device:



   With only one shutter open, we get certain states (depending on the setup) going into and coming out of the measurement device:

$$\begin{aligned} &\text{Shutter A open}: \qquad |\psi\rangle = |\psi_A\rangle\,, \qquad |\psi_i\rangle = \left|\psi_i^{(A)}\right\rangle, \\ &\text{Shutter B open}: \qquad |\psi\rangle = |\psi_B\rangle\,, \qquad |\psi_i\rangle = \left|\psi_i^{(B)}\right\rangle. \end{aligned}$$

   Therefore, with both shutters open the quantum interference principle, QR1, says that

$$|\psi\rangle = |\psi_A\rangle + |\psi_B\rangle\,, \qquad |\psi_i\rangle = \left|\psi_i^{(A)}\right\rangle + \left|\psi_i^{(B)}\right\rangle.$$

   Thus, we have

$$f_i(|\psi_A\rangle + |\psi_B\rangle) = f_i(|\psi_A\rangle) + f_i(|\psi_B\rangle),$$

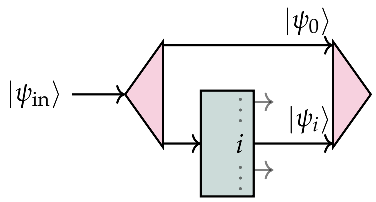
3. All that remains is to understand how f_i is affected by phase factors. Because the overall phase is not physically meaningful, we must have

$$f_i\left(\mathrm{e}^{i\phi}\,|\psi\rangle\right) = \mathrm{e}^{i\phi'}\,f_i(|\psi\rangle)$$

   where the relationship between ϕ and ϕ' is yet to be determined. One can use properties (1) and (2), in addition to the equation just above, to show that $\phi' = \pm\phi$, *i.e.*, the output states are either linear or antilinear functions of the input state.

This is what we wanted to show, but it raises an interesting question: could the output states be antilinear functions of the input state?

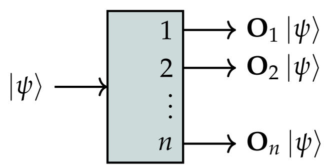
Consider the following interference experiment:



Here we interfere paths that do and do not pass through the measurement device. If the measurement device produces antilinear output states, then

$$|\psi_{\text{in}}\rangle \to e^{i\phi}\,|\psi_{\text{in}}\rangle \quad \text{would lead to} \quad |\psi_i\rangle \to e^{-i\phi}\,|\psi_i\rangle\,.$$

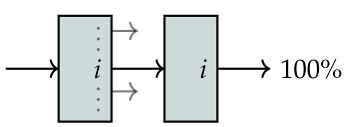
However, since the other path does not pass through the antilinear device, we would have $|\psi_0\rangle \to e^{i\phi}\,|\psi_0\rangle$, which would produce a measurable interference effect. This contradicts the fact that the overall phase is not physically meaningful, so we conclude that the output states must be linear functions of the input state:



Per Eqn. (21), the expected number of times that the outcome i is observed is

$$N_i = \langle\psi_i|\psi_i\rangle = \left\langle\psi\middle|\mathbf{O}_i^\dagger\mathbf{O}_i\middle|\psi\right\rangle. \tag{22}$$

Since each measurement results in one outcome, these must all sum to N. To say more, we enforce some requirements from what we observed in our previous thought experiments. First, we insist that the measurement be repeatable, meaning that if we perform the same measurement again immediately after the first measurement, we get the same outcome:



We therefore argue that

$$\mathbf{O}_i = \mathbf{U}\mathbf{\Pi}_i,$$

where $\{\mathbf{\Pi}_1, \mathbf{\Pi}_2, \ldots, \mathbf{\Pi}_n\}$ is a complete set of orthogonal projectors, and $\mathbf{U}$ is a unitary operator that commutes with them.[28]

[28] This argument should really come after performing some work. In particular, the fact that each measurement produces one outcome and is repeatable is equivalent to saying

$$\sum_{i=1}^{n} \mathbf{O}_i^\dagger\mathbf{O}_i = \mathbf{I} \quad \text{and} \quad \mathbf{O}_i\mathbf{O}_j = \delta_{ij}\mathbf{O}_i.$$

It can then be shown that $\mathbf{O}_j^\dagger\mathbf{O}_i = \delta_{ij}\mathbf{\Pi}_i$, and hence $\mathbf{U} \equiv \sum_{i=1}^{n}\mathbf{O}_i$ is a unitary operator that commutes with the $\mathbf{\Pi}_i$.

We can now compute

$$\mathbf{O}_i^\dagger \mathbf{O}_i = \mathbf{\Pi}_i \mathbf{U}^\dagger \mathbf{U} \mathbf{\Pi}_i = \mathbf{\Pi}_i^2 = \mathbf{\Pi}_i$$

which, by Eqn. (22),

$$\implies N_i = \langle \psi_i | \psi_i \rangle = \left\langle \psi \middle| \mathbf{O}_i^\dagger \mathbf{O}_i \middle| \psi \right\rangle = \langle \psi | \mathbf{\Pi}_i | \psi \rangle \,.$$

Thus, the probability of measuring the outcome i is

$$p_i = \frac{N_i}{N} = \frac{\langle \psi | \mathbf{\Pi}_i | \psi \rangle}{\langle \psi | \psi \rangle},$$

known as the **Born rule**.

After a measurement results in the outcome i, the state of the system is

$$|\psi_i\rangle = \mathbf{O}_i |\psi\rangle = \mathbf{U}\mathbf{\Pi}_i |\psi\rangle \,.$$

What is the effect of the unitary operator $\mathbf{U}$? If we perform the measurement again, as we did in our previous figure, we get

$$|\psi_i'\rangle = \mathbf{O}_i |\psi_i\rangle = \mathbf{U}\mathbf{\Pi}_i |\psi_i\rangle = \mathbf{U}\mathbf{\Pi}_i \mathbf{U}\mathbf{\Pi}_i |\psi\rangle = \mathbf{U}^2 \mathbf{\Pi}_i^2 |\psi\rangle = \mathbf{U}^2 \mathbf{\Pi}_i |\psi\rangle \,,$$

where we've used the fact that $\mathbf{U}$ commutes with $\mathbf{\Pi}_i$ and that projectors are idempotent. Thus, each subsequent measurement after the first one has the effect of applying $\mathbf{U}$ an additional time.

If $\mathbf{\Pi}_i$ is of rank-1, say $\mathbf{\Pi}_i = |\xi\rangle\langle\xi|$, then because $\mathbf{U}$ commutes with $\mathbf{\Pi}_i$, $|\chi\rangle$ must be an eigenvector of $\mathbf{U}$:

$$\mathbf{U} |\chi\rangle = \mathrm{e}^{i\theta} |\chi\rangle \qquad \implies \qquad \mathbf{U}\mathbf{\Pi}_i |\psi\rangle = \mathrm{e}^{i\theta}\, \mathbf{\Pi}_i |\psi\rangle \,.$$

Thus, the unitary operator $\mathbf{U}$ only contributes an overall phase factor to the post-measurement state, which is not physically meaningful. If, instead, $\mathbf{\Pi}_i$ is of higher rank, then $\mathbf{U}$ can have a nontrivial effect on the post-measurement state, changing the state with each repetition of the measurement.

For simplicity, let us consider only ideal measuring devices with $\mathbf{U} = \mathbf{I}$. This can be though of as a strong version of the repeatability requirement. For an ideal measuring device of this kind, each output state $|\psi_i\rangle = \mathbf{\Pi} |\psi\rangle$ is a projection of the input state $|\psi\rangle$ onto the subspace defined by $\mathbf{\Pi}_i$. Often, we assign distinct real numbers λ_i to each measurement outcome i so that

$$\mathbf{L} = \sum_{i=1}^{n} \lambda_i \mathbf{\Pi}_i,$$

is the spectral decomposition of a self-adjoint operator $\mathbf{L}$, λ_i are its eigenvalues, and $\mathbf{\Pi}_i$ are the projectors onto the corresponding eigenspaces. Every such observable defines a possible idealized measurement.

We can summarize what we've learned with the following postulate:

Postulate 2: [P2] Possible measurements are described by self-adjoint operators (**observables**) $\mathbf{L}$ on $\mathcal{H}$, where

1. the outcome of each measurement is an eigenvalue of $\mathbf{L}$,
2. the probability of measuring the eigenvalue λ_i when the state is $|\psi\rangle$ is given by the Born rule,
$$p_i = \frac{\langle\psi|\mathbf{\Pi}_i|\psi\rangle}{\langle\psi|\psi\rangle},$$
where $\mathbf{\Pi}_i$ is the projector onto the eigenspace of $\mathbf{L}$ with eigenvalue λ_i, and
3. after measuring the eigenvalue λ_i, the state of the system is projected onto the eigenspace of $\mathbf{L}$ with eigenvalue λ_i,
$$|\psi\rangle \to |\psi_i\rangle = \mathbf{\Pi}_i\,|\psi\rangle\,.$$

The last part of P2 can be interpreted as the "wavefunction collapse," coming from the Coppenhagen interpretation of quantum mechanics. Here we've chosen to describe measurement in the conventional way, using observables. Alternatively, P2 could be formulated using only a complete set of orthogonal projectors to describe possible measurements. However, observables come closer to classical intuition. Explicitly,

Quantization Rule 2: [QR2] Classical observables, which are real-valued functions on the classical phase space, correspond to quantum observables.

Like QR1, this is a less precise statement than any of the postulates, but it helps specify how quantum mechanics relates to classical mechanics, and it is a useful heuristic for constructing quantum theories from classical theories.

Postulate 3: Unitary Evolution

There are many ways to manipulate a quantum system without performing a measurement. We represent such a manipulation with the following diagram:

$$|\psi\rangle \longrightarrow \boxed{} \longrightarrow |\psi'\rangle$$

Such a process is said to **closed**.

By exactly the same reasoning as before, we argue that the output state must be linearly-related to the input state, *i.e.*, there exists a linear operator $\mathbf{U}$ such that

$$\left|\psi'\right\rangle = \mathbf{U}\left|\psi\right\rangle.$$

Moreover, since every input produces an output with 100% certainty, we must have $\langle\psi|\psi\rangle = \langle\psi'|\psi'\rangle$ for all $|\psi\rangle$, which implies that $\mathbf{U}$ is a unitary operator:

$$\left\langle\psi'\middle|\psi'\right\rangle = \left\langle\psi\middle|\mathbf{U}^\dagger\mathbf{U}\middle|\psi\right\rangle = \langle\psi|\psi\rangle\,.$$

This is then our third postulate:

Postulate 3: [P3] In any closed process, the state vector of a quantum system changes by a unitary transformation:

$$|\psi\rangle \to \left|\psi'\right\rangle = \mathbf{U}\left|\psi\right\rangle,$$

where $\mathbf{U}$ is a unitary operator on $\mathcal{H}$ that depends only on the process and not on the state $|\psi\rangle$.

To truly interpret this postulate, we need to understand what kinds of processes are closed. So far, we defined it to be a process in which no measurement takes place. More generally, a closed process is one in which the system of interest, call it $\mathcal{H}$, does not interact with any other system, say $\mathcal{H}'$. The result of such a mixing would be similar to a measurement, as the other system would thereby gain information about the initial state $|\psi\rangle$ of the system of interest. This phenomenon is known as entanglement. To account for processes that not closed, we can either

1. enlarge the Hilbert space to include the other system, so that the process is closed with respect to the enlarged Hilbert space, or

2. introduce "mixed states." These are not state vectors, but rather density operators (which are positive semidefinite, trace-1 operators on $\mathcal{H}$).

A simple closed process is simply waiting for some time to pass. We can also manipulate a quantum system by applying some external field to it, such as an electric or magnetic field. It is important to note that it does not matter who or what a measurement, any outside access to $|\psi\rangle$ will create entanglement between $\mathcal{H}$ and the rest of the universe, in which case the process is no longer closed.

Remarkably, the three postulates discussed above (with suitable modifications to accomodate continuous observables in an infinite-dimensional Hilbert space) apply to all closed quantum systems, and in some sense define what quantum mechanics is; much like Newton's laws define what classical mechanics is.

Quantum Information and Entanglement

Quantum States and Classical States

Now that we have an understanding of what a quantum system is, let us examine more carefully how it differs from a classical system. Here, we emphasize the following points:

1. Quantum states are *fuzzy*, since observables that do not commute, *i.e.*, incompatible observables, cannot be simultaneously measured with arbitrary precision.

We have already observed this phenomenon in the Stern-Gerlach experiment, where the spin along the z-axis and the spin along the x-axis are incompatible observables, but this observation can be made more precise as follows:

Theorem 126 (Generalized uncertainty principle)**.** *For any two self-adjoint operators* $\mathbf{A}$ *and* $\mathbf{B}$ *on* $\mathcal{H}$ *and any state* $|\psi\rangle \in \mathcal{H}$*, we have*

$$\left\langle (\mathbf{\Delta A})^2 \right\rangle \left\langle (\mathbf{\Delta B})^2 \right\rangle \geqslant \frac{1}{4} |\langle [\mathbf{A}, \mathbf{B}] \rangle|^2,$$

where the expectation values are all evaluated in the state $|\psi\rangle$*.*

Here, for any operator $\mathbf{X}$, the state-dependent operator $\mathbf{\Delta X}$ is defined as

$$\mathbf{\Delta X} \equiv \mathbf{X} - \langle \mathbf{X} \rangle \mathbf{I} = \mathbf{X} - \frac{\langle \psi | \mathbf{X} | \psi \rangle}{\langle \psi | \psi \rangle} \mathbf{I}.$$

It therefore follows that

$$\langle \mathbf{\Delta X} \rangle = \langle \mathbf{X} \rangle - \langle \mathbf{X} \rangle \langle \mathbf{I} \rangle = 0$$

since

$$\langle \mathbf{I} \rangle = \frac{\langle \psi | \mathbf{I} | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\langle \psi | \psi \rangle}{\langle \psi | \psi \rangle} = 1.$$

Moreover, the stastical **variance** of the observable $\mathbf{X}$ in the state $|\psi\rangle$ is

$$\sigma_X^2 = \left\langle (\mathbf{\Delta X})^2 \right\rangle = \left\langle \mathbf{X}^2 \right\rangle - \langle \mathbf{X} \rangle^2 .$$

Thus, the generalized uncertainty principle states that the product of the variances of two incompatible observables $\mathbf{A}$ and $\mathbf{B}$ satisfies a

Question 127.

1. *Show that* $\mathbf{\Delta X}$ *is self-adjoint if* $\mathbf{X}$ *is self-adjoint.*
2. *Show that* $\left\langle (\mathbf{\Delta X})^2 \right\rangle = \langle \mathbf{X}^2 \rangle - \langle \mathbf{X} \rangle^2$.

sharp, nonzero lower bound in states where the expectation value of their commutator is nonzero, $|\langle[\mathbf{A},\mathbf{B}]\rangle| \neq 0$. Equivalently,

$$\sqrt{\left\langle(\mathbf{\Delta A})^2\right\rangle}\sqrt{\left\langle(\mathbf{\Delta B})^2\right\rangle} \geqslant \frac{1}{2}|\langle[\mathbf{A},\mathbf{B}]\rangle|,$$

where $\sqrt{\left\langle(\mathbf{\Delta X})^2\right\rangle}$ is the **standard deviation** of the observable $\mathbf{X}$ in the state $|\psi\rangle$. Therefore, when $|\langle[\mathbf{A},\mathbf{B}]\rangle| \neq 0$, measurements of $\mathbf{A}$ and $\mathbf{B}$ in the state $|\psi\rangle$ cannot both be sharp at once. We now proceed with proving the generalized uncertainty principle.

Proof. We start with an obvious statement: for all $|\alpha\rangle$, $|\beta\rangle \in \mathcal{H}$ and $\lambda \in \mathbb{C}$,

$$\||\alpha\rangle + \lambda|\beta\rangle\|^2 \geqslant 0.$$

Choosing

$$\lambda = -\frac{\langle\beta|\alpha\rangle}{\langle\beta|\beta\rangle},$$

the above expression can be expanded as

$$\begin{aligned}
\||\alpha\rangle + \lambda|\beta\rangle\|^2 &= (\langle\alpha| + \lambda^*\langle\beta|)(|\alpha\rangle + \lambda|\beta\rangle) \\
&= \langle\alpha|\alpha\rangle + \lambda\langle\alpha|\beta\rangle + \lambda^*\langle\beta|\alpha\rangle + |\lambda|^2\langle\beta|\beta\rangle \\
&= \langle\alpha|\alpha\rangle - \frac{\langle\beta|\alpha\rangle}{\langle\beta|\beta\rangle}\langle\alpha|\beta\rangle - \frac{\langle\alpha|\beta\rangle}{\langle\beta|\beta\rangle}\langle\beta|\alpha\rangle + \left|\frac{\langle\beta|\alpha\rangle}{\langle\beta|\beta\rangle}\right|^2\langle\beta|\beta\rangle \\
&= \langle\alpha|\alpha\rangle\langle\beta|\beta\rangle - \langle\beta|\alpha\rangle\langle\alpha|\beta\rangle - \langle\alpha|\beta\rangle\langle\beta|\alpha\rangle + \left|\frac{\langle\beta|\alpha\rangle}{\cancel{\langle\beta|\beta\rangle}}\right|^2\cancel{\langle\beta|\beta\rangle^2} \\
&= \langle\alpha|\alpha\rangle\langle\beta|\beta\rangle - |\langle\beta|\alpha\rangle|^2 - \cancel{|\langle\beta|\alpha\rangle|^2} + \cancel{|\langle\beta|\alpha\rangle|^2} \\
&= \langle\alpha|\alpha\rangle\langle\beta|\beta\rangle - |\langle\beta|\alpha\rangle|^2 \geqslant 0,
\end{aligned}$$

$$\implies \langle\alpha|\alpha\rangle\langle\beta|\beta\rangle \geqslant |\langle\alpha|\beta\rangle|^2. \tag{23}$$

This is known as the **Cauchy-Schwarz inequality**, with saturation condition $|\alpha\rangle = \lambda|\beta\rangle$ for some $\lambda \in \mathbb{C}$.

Substituting $|\alpha\rangle = \mathbf{\Delta A}|\psi\rangle$ and $|\beta\rangle = \mathbf{\Delta B}|\psi\rangle$ into Eqn. (23), we get

$$\left\langle(\mathbf{\Delta A})^2\right\rangle\left\langle(\mathbf{\Delta B})^2\right\rangle \geqslant |\langle\mathbf{\Delta A\Delta B}\rangle|^2, \tag{24}$$

where we used the fact that $\mathbf{\Delta A}$ and $\mathbf{\Delta B}$ are self-adjoint to write $\langle\alpha|\alpha\rangle = \left\langle(\mathbf{\Delta A})^2\right\rangle$ and $\langle\beta|\beta\rangle = \left\langle(\mathbf{\Delta B})^2\right\rangle$. We can decompose the right-hand side as

$$\mathbf{\Delta A\Delta B} = \frac{1}{2}(\mathbf{\Delta A\Delta B} + \mathbf{\Delta B\Delta A} + \mathbf{\Delta A\Delta B} - \mathbf{\Delta B\Delta A}) = \frac{1}{2}([\mathbf{\Delta A},\mathbf{\Delta B}] + \{\mathbf{\Delta A},\mathbf{\Delta B}\}),$$

where we recognize the commutator, but we may not have recognized the **anticommutator**:

$$\{\mathbf{X},\mathbf{Y}\} \equiv \mathbf{XY} + \mathbf{YX}.$$

We also notice that the commutator reduces to

$$[\mathbf{\Delta A},\mathbf{\Delta B}] = [\mathbf{A} - \langle\mathbf{A}\rangle\mathbf{I}, \mathbf{B} - \langle\mathbf{B}\rangle\mathbf{I}] = [\mathbf{A},\mathbf{B}] - \cancelto{0}{[\langle\mathbf{A}\rangle\mathbf{I},\mathbf{B}]} - \cancelto{0}{[\mathbf{A},\langle\mathbf{B}\rangle\mathbf{I}]} + \cancelto{0}{[\langle\mathbf{A}\rangle\mathbf{I},\langle\mathbf{B}\rangle\mathbf{I}]} = [\mathbf{A},\mathbf{B}].$$

Note that for any two self-adjoint operators $\mathbf{X}$ and $\mathbf{Y}$,

$$[\mathbf{X}, \mathbf{Y}]^\dagger = (\mathbf{XY} - \mathbf{YX})^\dagger = \mathbf{Y}^\dagger\mathbf{X}^\dagger - \mathbf{X}^\dagger\mathbf{Y}^\dagger = \mathbf{YX} - \mathbf{XY} = -[\mathbf{X}, \mathbf{Y}]$$

since the adjoint reverses the order of the products. Thus, the commutator of two self-adjoint operators is anti-self-adjoint, and hence its expectation value is purely imaginary. On the other hand,

$$\{\mathbf{X}, \mathbf{Y}\}^\dagger = (\mathbf{XY} + \mathbf{YX})^\dagger = \mathbf{Y}^\dagger\mathbf{X}^\dagger + \mathbf{X}^\dagger\mathbf{Y}^\dagger = \mathbf{YX} + \mathbf{XY} = \{\mathbf{X}, \mathbf{Y}\},$$

so the anticommutator of two self-adjoint operators is self-adjoint, and hence its expectation value is purely real. Therefore,

$$\langle \mathbf{\Delta A \Delta B} \rangle = \frac{1}{2}(\underbrace{\langle [\mathbf{\Delta A}, \mathbf{\Delta B}] \rangle}_{\text{imaginary}} + \underbrace{\langle \{\mathbf{\Delta A}, \mathbf{\Delta B}\} \rangle}_{\text{real}}),$$

so taking the modulus squared of both sides gives

$$|\langle \mathbf{\Delta A \Delta B} \rangle|^2 = \frac{1}{4}\Big(|\langle [\mathbf{A}, \mathbf{B}] \rangle|^2 + |\langle \{\mathbf{\Delta A}, \mathbf{\Delta B}\} \rangle|^2\Big) \geqslant \frac{1}{4}|\langle [\mathbf{A}, \mathbf{B}] \rangle|^2.$$

Substituting this back into Eqn. (24), we get the desired result:

$$\Big\langle (\mathbf{\Delta A})^2 \Big\rangle \Big\langle (\mathbf{\Delta B})^2 \Big\rangle \geqslant \frac{1}{4}|\langle [\mathbf{A}, \mathbf{B}] \rangle|^2.$$

□

In this sense, quantum states have less information than classical states, since they cannot be simultaneously sharp in all observables. However,

2. Quantum states be in **superpositions** of different classical configurations, which is a phenomenon that has no classical analogue.

For example, the states $|+\hat{\mathbf{z}}\rangle$ and $|-\hat{\mathbf{z}}\rangle$ are both classically and quantum mechanically valid states of a spin-$\frac{1}{2}$ system, but the state $|\psi\rangle = \frac{1}{\sqrt{2}}(|+\hat{\mathbf{z}}\rangle + |-\hat{\mathbf{z}}\rangle)$ is a superposition of the two classical configurations with no classical analog.

In general, distinct classical states correspond to elements of an orthonormal basis of the Hilbert space, and quantum states are linear combinations of these basis elements. In this way, quantum states have more information than classical states.

There is an interesting interplay between quantum fuzziness and quantum superposition in the qubit Hilbert space. We started with a classical model system with a continuum of states. However, as shown by the SG experiment, there are actually only two classically distinct states, which we can represent with the orthonormal basis $|+\hat{\mathbf{z}}\rangle$ and $|-\hat{\mathbf{z}}\rangle$. However, by taking quantum superpositions of these two "classical" states, we can generate infinitely many quantum states, *e.g.*,

$$|+\hat{\mathbf{x}}\rangle = \frac{1}{\sqrt{2}}(|+\hat{\mathbf{z}}\rangle + |-\hat{\mathbf{z}}\rangle),$$

which can be represented as points on the Bloch sphere.

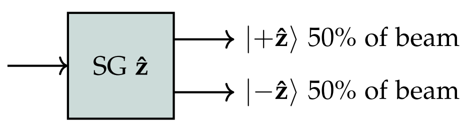
Of course, the quantum state of a single qubit can never be fully measured, since any attempt to measure it erases the input state while only providing one bit of (probabilistic) information about what it was. However, if we have many identically prepared qubits, we can perform measurements on them to determine the state with good accuracy.

In more complicated quantum systems, the space of quantum states is typically much larger than the classical state space, meaning that the quantum state vector contains a lot of **quantum information**. Again, we can only access a very small fraction of this information through measurements, unless we have many identically prepared systems to perform measurements on. Even then, the information we access depends on the measurements we choose to perform.

Mixed States

To further illustrate the difference between classical and quantum notions, let us compare classical uncertainty with quantum superposition.

In the original SG experiment, a beam of silver atoms split equally into two sub-beams, which we labeled as $|{+\hat{\mathbf{z}}}\rangle$ and $|{-\hat{\mathbf{z}}}\rangle$:



In our previous discussions, we never specified the state of the incoming beam, or even attempted to work it out. We can tell that it must contain *an equal mixture of up and down*, but this could mean at least two different things:

1. **Classical uncertainty**

   The incoming beam is a classical mixture of two sub-beams, one in the state $|{+\hat{\mathbf{z}}}\rangle$ and the other in the state $|{-\hat{\mathbf{z}}}\rangle$. Each atom has a definite $S_z = \pm\hbar/2$ value, but the spin of an atom is picked randomly from the two possibilities with equal probability.

2. **Quantum superposition**

   All the atoms in the incoming beam are in the same quantum state

   $$|\psi\rangle = \frac{1}{\sqrt{2}}(|{+\hat{\mathbf{z}}}\rangle + |{-\hat{\mathbf{z}}}\rangle),$$

   which is a superposition of the two classical configurations. Now the randomness of the outcome is not because we pick an atom to measure at random, but rather because the state of each atom is intrinsically uncertain in the S_z observable.

Since both of these scenarios produce the same measurement statistics, we cannot distinguish between them by performing these measurements on the incoming beam alone.[29]

[29] By “these measurements” I am referring to measurements of S_z on the incoming beam.

However, we are free to perform a different measurement on some of the atoms in the incoming beam, say a measurement of the $\hat{\mathbf{x}}$ component of the spin, S_x. Let’s analyze the outcomes of this measurement in the two scenarios:

1. **Classical uncertainty**

   Half of the time, the atom is in the state $|\psi\rangle = |+\hat{\mathbf{z}}\rangle$. The probability of each outcome of the S_x measurement is given by the Born rule:

$$p^{+}_{\text{up}} = |\langle +\hat{\mathbf{x}}|\psi\rangle|^2 = |\langle +\hat{\mathbf{x}}|+\hat{\mathbf{z}}\rangle|^2 = \frac{1}{2}, \qquad p^{+}_{\text{down}} = |\langle -\hat{\mathbf{x}}|\psi\rangle|^2 = |\langle -\hat{\mathbf{x}}|+\hat{\mathbf{z}}\rangle|^2 = \frac{1}{2}.$$

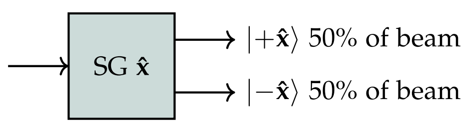
   The other half of the time, the atom is in the state $|\psi\rangle = |-\hat{\mathbf{z}}\rangle$. The probability of each outcome of the S_x measurement is

$$p^{-}_{\text{up}} = |\langle +\hat{\mathbf{x}}|\psi\rangle|^2 = |\langle +\hat{\mathbf{x}}|-\hat{\mathbf{z}}\rangle|^2 = \frac{1}{2}, \qquad p^{-}_{\text{down}} = |\langle -\hat{\mathbf{x}}|\psi\rangle|^2 = |\langle -\hat{\mathbf{x}}|-\hat{\mathbf{z}}\rangle|^2 = \frac{1}{2}.$$

   Thus, in net, the probability of each outcome of the S_x measurement is

$$p_{\text{up}} = \frac{1}{2}p^{+}_{\text{up}} + \frac{1}{2}p^{-}_{\text{up}} = \frac{1}{2}, \qquad p_{\text{down}} = \frac{1}{2}p^{+}_{\text{down}} + \frac{1}{2}p^{-}_{\text{down}} = \frac{1}{2}.$$

   Therefore, there is an equal probability of each measurement outcome, just like the S_z measurement.



2. **Quantum superposition**

   Here all of the atoms in the incoming beam are in the same state $|\psi\rangle = \frac{1}{\sqrt{2}}(|+\hat{\mathbf{z}}\rangle + |-\hat{\mathbf{z}}\rangle) = |+\hat{\mathbf{x}}\rangle$. The probability of each outcome of the S_x measurement is

$$p_{\text{up}} = |\langle +\hat{\mathbf{x}}|\psi\rangle|^2 = |\langle +\hat{\mathbf{x}}|+\hat{\mathbf{x}}\rangle|^2 = 1, \qquad p_{\text{down}} = |\langle -\hat{\mathbf{x}}|\psi\rangle|^2 = |\langle -\hat{\mathbf{x}}|+\hat{\mathbf{x}}\rangle|^2 = 0.$$

This is a very different measurement statistic from the previous scenario! In the real SG experiment, the outcome is the same regardless of how the apparatus is set up. Thus, the first scenario (or something like it) is the correct one. Classical uncertainty is therefore a distinct phenomena from quantum superposition, both which can arise in various situations.

More generally, suppose that each atom in the beam is in one of the normalized states $|\psi_1\rangle$, $|\psi_2\rangle$, …, $|\psi_k\rangle$ with probabilities p_1, p_2, …, p_k respectively, where $p_i > 0$ for all i and $\sum_{i=1}^{k} p_i = 1$. In this case, the probability of measuring $S_z = +\hbar/2$ is given by

$$p_{\text{up}} = \sum_{i=1}^{k} p_i |\langle +\hat{\mathbf{z}}|\psi_i\rangle|^2 = \sum_{i=1}^{k} p_i \langle +\hat{\mathbf{z}}|\psi_i\rangle \langle\psi_i|+\hat{\mathbf{z}}\rangle = \langle +\hat{\mathbf{z}}| \left(\sum_{i=1}^{k} p_i |\psi_i\rangle\langle\psi_i| \right) |+\hat{\mathbf{z}}\rangle = \langle +\hat{\mathbf{z}}|\boldsymbol{\rho}|+\hat{\mathbf{z}}\rangle,$$

where we define the **density operator**

$$\rho \equiv \sum_{i=1}^{k} p_i \, |\psi_i\rangle\langle\psi_i| \,.$$

While we have motivated the density operator using the qubit, this definition applies to any quantum system whose state vector is randomly drawn from some statistical ensemble. It can then be shown that

Theorem 128 (Density operator). *An operator ρ can be written in the form*

$$\rho = \sum_{i=1}^{k} p_i \, |\psi_i\rangle\langle\psi_i|$$

for some ensemble of normalized states $|\psi_1\rangle, |\psi_2\rangle, \ldots, |\psi_k\rangle$ and probabilities $p_1, p_2, \ldots, p_k$ if and only if

1. *ρ is self-adjoint, $\rho^\dagger = \rho$,*
2. *ρ is positive semidefinite, $\langle\rho\rangle \geqslant 0$ for all states $|\psi\rangle$,*
3. *ρ has unit trace, $\mathrm{Tr}(\rho) = 1$.*

Question 129. *Prove the above theorem (solution below).*

Proof.

$(\implies)$ If $\rho = \sum_{i=1}^{k} p_i \, |\psi_i\rangle\langle\psi_i|$, then

$$\rho^\dagger = \left(\sum_{i=1}^{k} p_i \, |\psi_i\rangle\langle\psi_i|\right)^\dagger = \sum_{i=1}^{k} p_i \, |\psi_i\rangle\langle\psi_i| = \rho,$$

$$\langle\rho\rangle = \langle\psi| \left(\sum_{i=1}^{k} p_i \, |\psi_i\rangle\langle\psi_i|\right) |\psi\rangle = \sum_{i=1}^{k} p_i |\langle\psi_i|\psi\rangle|^2 \geqslant 0,$$

$$\mathrm{Tr}(\rho) = \sum_{i=1}^{k} p_i \, \mathrm{Tr}(|\psi_i\rangle\langle\psi_i|) = \sum_{i=1}^{k} p_i \, \langle\psi_i|\psi_i\rangle = \sum_{i=1}^{k} p_i = 1.$$

$(\impliedby)$ If ρ is self-adjoint, positive semidefinite, and has unit trace, then by the spectral theorem, we can write

$$\rho = \sum_{i=1}^{k} \lambda_i \, |\psi_i\rangle\langle\psi_i| \,,$$

where $|\psi_1\rangle, |\psi_2\rangle, \ldots, |\psi_k\rangle$ are the eigenvectors of ρ and $\lambda_1, \lambda_2, \ldots, \lambda_k$ are the corresponding eigenvalues. Since ρ is positive semidefinite, we have $\lambda_i = \langle\rho\rangle_{|\psi_i\rangle} \geqslant 0$ for all i. Since ρ has unit trace, we have

$$1 = \mathrm{Tr}(\rho) = \sum_{i=1}^{k} \lambda_i \, \mathrm{Tr}(|\psi_i\rangle\langle\psi_i|) = \sum_{i=1}^{k} \lambda_i \, \langle\psi_i|\psi_i\rangle = \sum_{i=1}^{k} \lambda_i.$$

Thus, if we define $p_i = \lambda_i$ for all i, then we have $p_i \geqslant 0$ for all i and $\sum_{i=1}^{k} p_i = 1$, so

$$\rho = \sum_{i=1}^{k} p_i \left|\psi_i\right\rangle\!\left\langle\psi_i\right| ,$$

as desired.

$\square$

For example, in the classical uncertainty scenario from above, the density operator of the incoming beam is

$$\rho_{\text{cl}} = \frac{1}{2}\left|+\hat{\mathbf{z}}\right\rangle\!\left\langle+\hat{\mathbf{z}}\right| + \frac{1}{2}\left|-\hat{\mathbf{z}}\right\rangle\!\left\langle-\hat{\mathbf{z}}\right| = \frac{1}{2}\mathbf{I},$$

whereas in the quantum superposition scenario, the density operator of the incoming beam is

$$\rho_{\text{qm}} = \left|+\hat{\mathbf{x}}\right\rangle\!\left\langle+\hat{\mathbf{x}}\right| = \frac{1}{2}\left(\left|+\hat{\mathbf{z}}\right\rangle\!\left\langle+\hat{\mathbf{z}}\right| + \left|-\hat{\mathbf{z}}\right\rangle\!\left\langle-\hat{\mathbf{z}}\right| + \left|+\hat{\mathbf{z}}\right\rangle\!\left\langle-\hat{\mathbf{z}}\right| + \left|-\hat{\mathbf{z}}\right\rangle\!\left\langle+\hat{\mathbf{z}}\right|\right).$$

Clearly, $\rho_{\text{cl}} \neq \rho_{\text{qm}}$, so the density operator can distinguish between the two scenarios. The probabilities for measuring $S_x = +\hbar/2$ and $S_x = -\hbar/2$ in the two scenarios can be computed using the density operator as follows:

$$p_{\text{up}}^{\text{cl}} = \langle+\hat{\mathbf{x}}|\rho_{\text{cl}}|+\hat{\mathbf{x}}\rangle = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}, \qquad p_{\text{down}}^{\text{cl}} = \langle-\hat{\mathbf{x}}|\rho_{\text{cl}}|-\hat{\mathbf{x}}\rangle = \frac{1}{4} + \frac{1}{4} = \frac{1}{2},$$

in the classical uncertainty scenario, and

$$p_{\text{up}}^{\text{qm}} = \left\langle+\hat{\mathbf{x}}\middle|\rho_{\text{qm}}\middle|+\hat{\mathbf{x}}\right\rangle = 1, \qquad p_{\text{down}}^{\text{qm}} = \left\langle-\hat{\mathbf{x}}\middle|\rho_{\text{qm}}\middle|-\hat{\mathbf{x}}\right\rangle = 0,$$

in the quantum superposition scenario, as expected.

Note that the density operator contains less information than the underlying statistical ensemble of states. For example, one can consider a third scenario where 50% of the atoms are in the state $|+\hat{\mathbf{x}}\rangle$ and the other 50% are in the state $|-\hat{\mathbf{x}}\rangle$. This is a different statistical ensemble of states from the classical uncertainty scenario, but it has the same density operator

$$\rho'_{\text{cl}} = \frac{1}{2}\left|+\hat{\mathbf{x}}\right\rangle\!\left\langle+\hat{\mathbf{x}}\right| + \frac{1}{2}\left|-\hat{\mathbf{x}}\right\rangle\!\left\langle-\hat{\mathbf{x}}\right| = \frac{1}{2}\left(\left|+\hat{\mathbf{x}}\right\rangle\!\left\langle+\hat{\mathbf{x}}\right| + \left|-\hat{\mathbf{x}}\right\rangle\!\left\langle-\hat{\mathbf{x}}\right|\right) = \frac{1}{2}\mathbf{I} = \rho_{\text{cl}}.$$

In general, there are many different statistical ensembles of states that correspond to the same density operator. Nevertheless, one can show that the density operator contains enough information about the ensemble to determine the outcome of any measurement performed on the system, so it is a complete description of the state of the system *for all practical purposes*.

Theorem 130 (Density operator and measurement outcomes)**.** *For an observable* $\mathbf{L}$ *measured on a quantum state randomly drawn from an ensemble with density operator* $\boldsymbol{\rho}$*,*

1. *the probability of measuring an eigenvalue* λ *of* $\mathbf{L}$ *is given by*

$$p(\lambda) = \mathrm{Tr}(\boldsymbol{\rho} \, |\Pi\rangle\langle\Pi|_\lambda),$$

   where $|\Pi\rangle\langle\Pi|_\lambda$ *is the projection operator onto the eigenspace of* $\mathbf{L}$ *with eigenvalue* λ*,*

2. *the ensemble of states resulting from the measurement outcome* λ *is described by the density operator*

$$\boldsymbol{\rho}_\lambda = \frac{|\Pi\rangle\langle\Pi|_\lambda \, \boldsymbol{\rho} \, |\Pi\rangle\langle\Pi|_\lambda}{\mathrm{Tr}(\boldsymbol{\rho} \, |\Pi\rangle\langle\Pi|_\lambda)}.$$

Question 131. *Prove the above theorem (solution below).*

Proof.

1. The probability of measuring an eigenvalue λ of $\mathbf{L}$ is given by

$$p(\lambda) = \sum_{i=1}^{k} p_i |\langle\lambda|\psi_i\rangle|^2 = \sum_{i=1}^{k} p_i \, \langle\lambda|\psi_i\rangle \, \langle\psi_i|\lambda\rangle = \langle\lambda| \left(\sum_{i=1}^{k} p_i \, |\psi_i\rangle\langle\psi_i| \right) |\lambda\rangle = \mathrm{Tr}(\boldsymbol{\rho} \, |\Pi\rangle\langle\Pi|_\lambda),$$

   where we used the fact that $|\lambda\rangle$ is an eigenvector of $\mathbf{L}$ with eigenvalue λ to write $|\Pi\rangle\langle\Pi|_\lambda = |\lambda\rangle\langle\lambda|$.

2. The ensemble of states resulting from the measurement outcome λ is described by the density operator

$$\boldsymbol{\rho}_\lambda = \sum_{i=1}^{k} p'_i \, |\psi'_i\rangle\langle\psi'_i| \, ,$$

   where $|\psi'_i\rangle = \frac{|\Pi\rangle\langle\Pi|_\lambda |\psi_i\rangle}{\sqrt{\langle\psi_i||\Pi\rangle\langle\Pi|_\lambda|\psi_i\rangle}}$ is the normalized post-measurement state of the system if the pre-measurement state was $|\psi_i\rangle$, and $p'_i = \frac{p_i\langle\psi_i||\Pi\rangle\langle\Pi|_\lambda|\psi_i\rangle}{p(\lambda)}$ is the probability of the post-measurement state being $|\psi'_i\rangle$. Substituting these expressions into the definition of $\boldsymbol{\rho}_\lambda$, we get

$$\begin{aligned}
\boldsymbol{\rho}_\lambda &= \sum_{i=1}^{k} \frac{p_i \, \langle\psi_i| \, |\Pi\rangle\langle\Pi|_\lambda \, |\psi_i\rangle}{p(\lambda)} \left| \frac{|\Pi\rangle\langle\Pi|_\lambda \, |\psi_i\rangle}{\sqrt{\langle\psi_i| \, |\Pi\rangle\langle\Pi|_\lambda \, |\psi_i\rangle}} \right\rangle\!\left\langle \frac{|\Pi\rangle\langle\Pi|_\lambda \, |\psi_i\rangle}{\sqrt{\langle\psi_i| \, |\Pi\rangle\langle\Pi|_\lambda \, |\psi_i\rangle}} \right| \\
&= \frac{1}{p(\lambda)} \sum_{i=1}^{k} p_i \, |\Pi\rangle\langle\Pi|_\lambda \, |\psi_i\rangle \, \langle\psi_i| \, |\Pi\rangle\langle\Pi|_\lambda \\
&= \frac{1}{p(\lambda)} \, |\Pi\rangle\langle\Pi|_\lambda \left(\sum_{i=1}^{k} p_i \, |\psi_i\rangle\langle\psi_i| \right) |\Pi\rangle\langle\Pi|_\lambda \\
&= \frac{|\Pi\rangle\langle\Pi|_\lambda \, \boldsymbol{\rho} \, |\Pi\rangle\langle\Pi|_\lambda}{\mathrm{Tr}(\boldsymbol{\rho} \, |\Pi\rangle\langle\Pi|_\lambda)},
\end{aligned}$$

   as desired.

□

As a result of this theorem, we see that the statistical ensemble of states that generate $\boldsymbol{\rho}$ cannot be measured directly, since any measurement performed on the system can be described entirely in terms of $\boldsymbol{\rho}$. Likewise, the time evolution of the density operator can be described entirely in terms of $\boldsymbol{\rho}$ without reference to the underlying statistical ensemble of states:

Theorem 132 (Density operator and time evolution)**.** *In any closed process, the density operator also changes by a unitary transformation:*

$$\boldsymbol{\rho} \to \mathbf{U}\boldsymbol{\rho}\mathbf{U}^\dagger.$$

where we recall that a closed process is a process in which, by P3, the state vector changes by a unitary transformation.

Question 133. *Prove the above theorem (solution below).*

Proof. If the state vector of the system changes by a unitary transformation $\mathbf{U}$, then each state $|\psi_i\rangle$ in the statistical ensemble of states that generate $\boldsymbol{\rho}$ changes as $|\psi_i\rangle \to \mathbf{U}\,|\psi_i\rangle$. Substituting this into the definition of $\boldsymbol{\rho}$, we get

$$\boldsymbol{\rho} = \sum_{i=1}^{k} p_i \, |\psi_i\rangle\langle\psi_i| \to \sum_{i=1}^{k} p_i \Big(\mathbf{U}\,|\psi_i\rangle\langle\psi_i|\,\mathbf{U}^\dagger\Big) = \mathbf{U}\left(\sum_{i=1}^{k} p_i \, |\psi_i\rangle\langle\psi_i|\right)\mathbf{U}^\dagger = \mathbf{U}\boldsymbol{\rho}\mathbf{U}^\dagger,$$

as desired. □

Notice that the last three theorems strongly resemble the three postulates of quantum mechanics, except that they are stated in terms of the density operator instead of the state vector. In fact, some people choose to define quantum states as density operators acting on a Hilbert space, rather than as projective vectors on a Hilbert space. To see how these two definitions are related, note that in the case where the normalized state vector of the system is $|\psi\rangle$, the density operator of the system is given by

$$\boldsymbol{\rho} = |\psi\rangle\langle\psi|\,,$$

a rank-1 projection operator, so $|\psi\rangle$ is the unique eigenvector of $\boldsymbol{\rho}$ with eigenvalue 1. Thus, the projective Hilbert space can be identified with the set of rank-1 projection operators on the Hilbert space. These are called **pure states**, whereas density operators that are not rank-1 projection operators are called **mixed states**. In the special case of $\boldsymbol{\rho} \propto \mathbf{I}$, we say that the system is in a **maximally mixed state**, which is the state of maximum uncertainty about the system. Mixed states are especially useful in quantum statistical mechanics. However, for our purposes, we follow a different interpretation: mixed states describe open quantum systems. To understand what this means, we need to study something we've been avoiding so far:

Everything we have done so far using pure states can be rephrased in terms of mixed states. For example, the expectation value of an observable $\mathbf{L}$ in the state $|\psi\rangle$ can be written as

$$\begin{aligned}\langle\mathbf{L}\rangle &= \sum_i p_i \,\langle\psi_i|\mathbf{A}|\psi_i\rangle \\ &= \sum_i p_i \,\mathrm{Tr}(|\psi_i\rangle\langle\psi_i|\,\mathbf{A}) \\ &= \mathrm{Tr}\left(\sum_i p_i \,|\psi_i\rangle\langle\psi_i|\,\mathbf{A}\right) \\ &= \mathrm{Tr}(\boldsymbol{\rho}\mathbf{A}).\end{aligned}$$

Entanglement

Consider a quantum system consisting of two spin-1/2 particles. By individually sending these atoms through two SG $\hat{\mathbf{z}}$ apparatuses, we can measure both $S_z^(1)$ and $S_z^(2)$, with four possible outcomes:

$$|\uparrow\uparrow\rangle\,, \quad |\uparrow\downarrow\rangle\,, \quad |\downarrow\uparrow\rangle\,, \quad |\downarrow\downarrow\rangle\,,$$

where for ease of notation we have replaced $|+\hat{\mathbf{z}}\rangle \to |\uparrow\rangle$ and $|-\hat{\mathbf{z}}\rangle \to |\downarrow\rangle$, and the first and second arrows refer to the first and second particles respectively. The dimension of the Hilbert space of this system is 4, so any state of the system can be written as a linear combination of these four basis states. Additionally,

$$\begin{aligned}
\mathbf{S}_z^{(1)} &= \frac{\hbar}{2}(|\uparrow\uparrow\rangle\langle\uparrow\uparrow| + |\uparrow\downarrow\rangle\langle\uparrow\downarrow|) - \frac{\hbar}{2}(|\downarrow\uparrow\rangle\langle\downarrow\uparrow| + |\downarrow\downarrow\rangle\langle\downarrow\downarrow|),\\
\mathbf{S}_z^{(2)} &= \frac{\hbar}{2}(|\uparrow\uparrow\rangle\langle\uparrow\uparrow| + |\downarrow\uparrow\rangle\langle\downarrow\uparrow|) - \frac{\hbar}{2}(|\uparrow\downarrow\rangle\langle\uparrow\downarrow| + |\downarrow\downarrow\rangle\langle\downarrow\downarrow|).
\end{aligned}$$

This Hilbert space is a **tensor product** of the Hilbert spaces of the two particles, so we can write

$$|\uparrow\uparrow\rangle = |\uparrow\rangle \otimes |\uparrow\rangle\,, \quad |\uparrow\downarrow\rangle = |\uparrow\rangle \otimes |\downarrow\rangle\,, \quad |\downarrow\uparrow\rangle = |\downarrow\rangle \otimes |\uparrow\rangle\,, \quad |\downarrow\downarrow\rangle = |\downarrow\rangle \otimes |\downarrow\rangle\,,$$

as in general, $\mathcal{H}_1 \otimes \mathcal{H}_2$ is defined as the Hilbert space spanned by the **decomposable** kets $|\psi\rangle \in \mathcal{H}_1$ and $|\phi\rangle \in \mathcal{H}_2$ of the form $|\psi\rangle \otimes |\phi\rangle$. The function $\otimes$ is called the **tensor product**, and it is bilinear,

$$(|\psi\rangle + \lambda\,|\chi\rangle) \otimes |\phi\rangle = |\psi\rangle \otimes |\phi\rangle + \lambda\,|\chi\rangle \otimes |\phi\rangle\,, \qquad |\psi\rangle \otimes (|\phi\rangle + \lambda\,|\xi\rangle) = |\psi\rangle \otimes |\phi\rangle + \lambda\,|\psi\rangle \otimes |\xi\rangle\,,$$

where $|\psi\rangle, |\chi\rangle \in \mathcal{H}_1$, $|\phi\rangle, |\xi\rangle \in \mathcal{H}_2$. and $\lambda \in \mathbb{C}$. The inner product on $\mathcal{H}_1 \otimes \mathcal{H}_2$ is defined by

$$(|\psi\rangle \otimes |\phi\rangle\,, |\chi\rangle \otimes |\xi\rangle) \equiv \langle\psi|\chi\rangle\,\langle\phi|\xi\rangle\,,$$

and extended to all of $\mathcal{H}_1 \otimes \mathcal{H}_2$ by antilinearity.

Typically, we will drop the $\otimes$ symbol and write $|\psi\rangle_1\,|\phi\rangle_2$ (sometimes even $|\psi\phi\rangle$) instead of $|\psi\rangle_1 \otimes |\phi\rangle_2$ for decomposable kets in $\mathcal{H}_1 \otimes \mathcal{H}_2$, where the subscripts indicate which Hilbert space the ket belongs to. When writing the associated bra, it is natural to reverse the order:

$$(|\psi\rangle_1\,|\phi\rangle_2)^\dagger = \langle\phi|_2\,\langle\psi|_1\,.$$

Then,

$$(|\psi\rangle_1\,|\phi\rangle_2\,, |\chi\rangle_1\,|\xi\rangle_2) = (|\psi\rangle_1\,|\phi\rangle_2)^\dagger(|\chi\rangle_1\,|\xi\rangle_2) = (\langle\phi|_2\,\langle\psi|_1)(|\chi\rangle_1\,|\xi\rangle_2) = \langle\phi|_2\,\langle\psi|\chi\rangle_1\,|\xi\rangle_2 = \langle\psi|\chi\rangle_1\,\langle\phi|\xi\rangle_2\,,$$

as expected. If one maintains the order of the kets throughout a computation there is no ambiguity, so the subscripts can be dropped as well. However, it is recommended to keep the subscripts when working with more than two Hilbert spaces, or when working with operators that act on only one of the Hilbert spaces, to avoid confusion.

If $\mathcal{H}_A$ has orthonormal basis $\{|1\rangle_A, |2\rangle_A, \ldots, |M\rangle_A\}$ and $\mathcal{H}_B$ has orthonormal basis $\{|1\rangle_B, |2\rangle_B, \ldots, |N\rangle_B\}$, then $\mathcal{H}_A \otimes \mathcal{H}_B$ has orthonormal basis

$$\{|1\rangle_A |1\rangle_B, |1\rangle_A |2\rangle_B, \ldots, |1\rangle_A |N\rangle_B, |2\rangle_A |1\rangle_B, |2\rangle_A |2\rangle_B, \ldots, |2\rangle_A |N\rangle_B, \ldots, |M\rangle_A |1\rangle_B, |M\rangle_A |2\rangle_B, \ldots, |M\rangle_A |N\rangle_B\}.$$

It therefore follows that $\dim(\mathcal{H}_A \otimes \mathcal{H}_B) = \dim(\mathcal{H}_A) \cdot \dim(\mathcal{H}_B) = M \cdot N$.

Operators on $\mathcal{H}_A \otimes \mathcal{H}_B$ can be constructed from operators on $\mathcal{H}_A$ and $\mathcal{H}_B$ as follows. If $\mathbf{X}_A$ is an operator on $\mathcal{H}_A$ and $\mathbf{Y}_B$ is an operator on $\mathcal{H}_B$, then we can define the operator $\mathbf{X}_A \otimes \mathbf{Y}_B$ on $\mathcal{H}_A \otimes \mathcal{H}_B$ by

$$(\mathbf{X}_A \otimes \mathbf{Y}_B) |\psi\rangle_A |\phi\rangle_B = (\mathbf{X}_A |\psi\rangle_A) \otimes (\mathbf{Y}_B |\phi\rangle_B).$$

All operators on $\mathcal{H}_A \otimes \mathcal{H}_B$ can be written as linear combinations of decomposable operators.

Bibliography